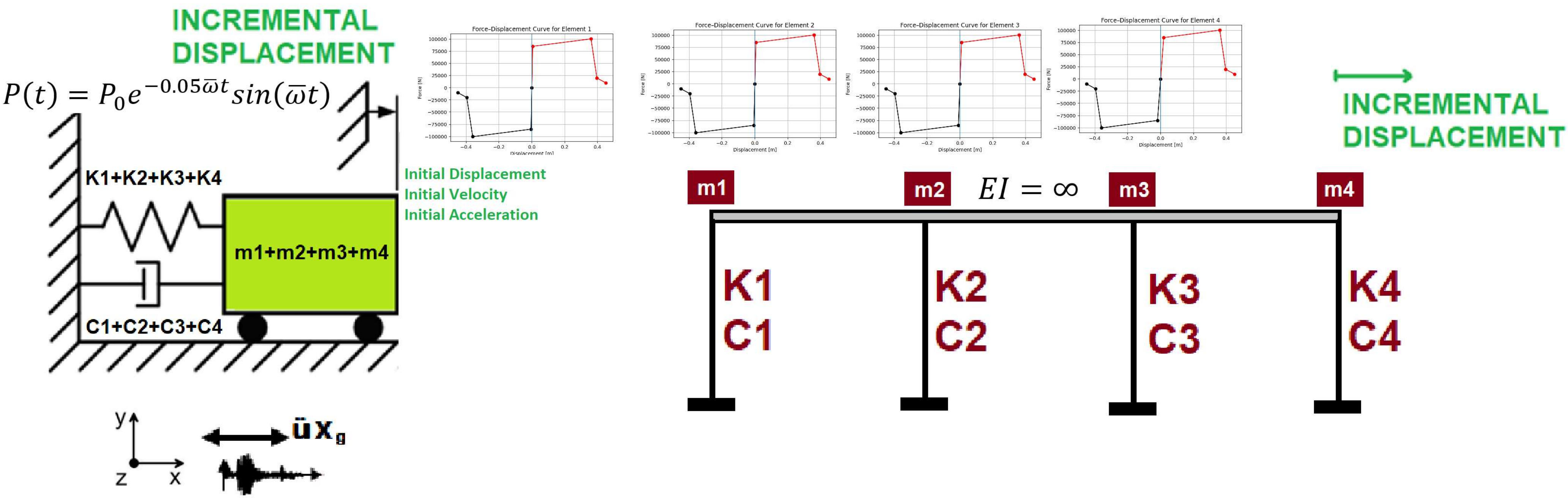
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF SINGLE-DEGREE-FREEDOM WITH FOUR ELEMENTS STRUCTURE (SDOF) : AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)


INCREMENTAL DISPLACEMENT
$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$
K1+K2+K3+K4
m1+m2+m3+m4
C1+C2+C3+C4
Initial Displacement
Initial Velocity
Initial Acceleration
y
z
x
$\ddot{u}\,x_g$
Force-Displacement Curve for Element 1
Force-Displacement Curve for Element 2
Force-Displacement Curve for Element 3
Force-Displacement Curve for Element 4
Displacement [m]
Force [N]
m1
m2
$EI = \infty$
m3
m4
K1
C1
K2
C2
K3
C3
K4
C4
INCREMENTAL DISPLACEMENT

$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

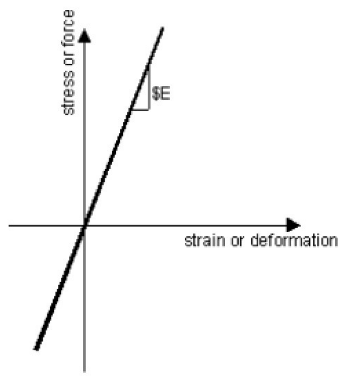
$$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$$

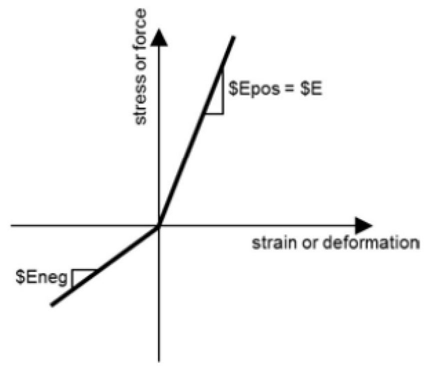
$$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$$

$$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$$

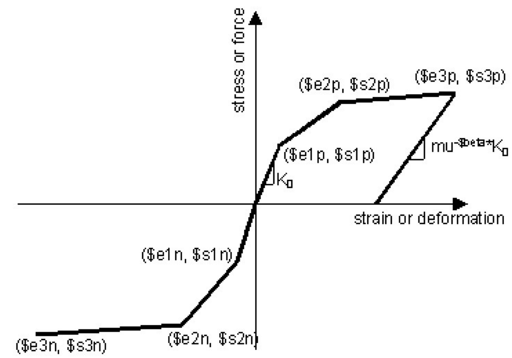
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

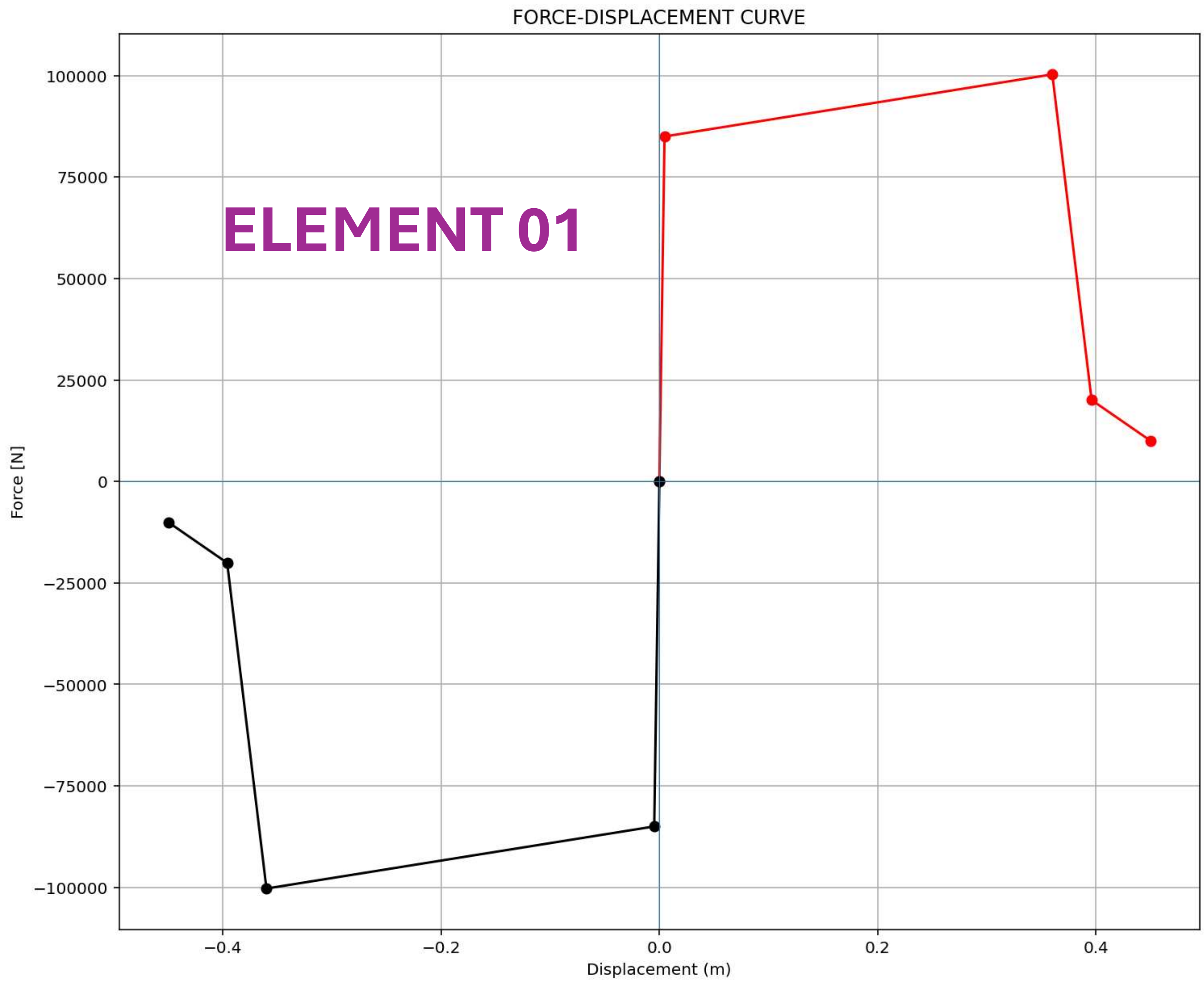




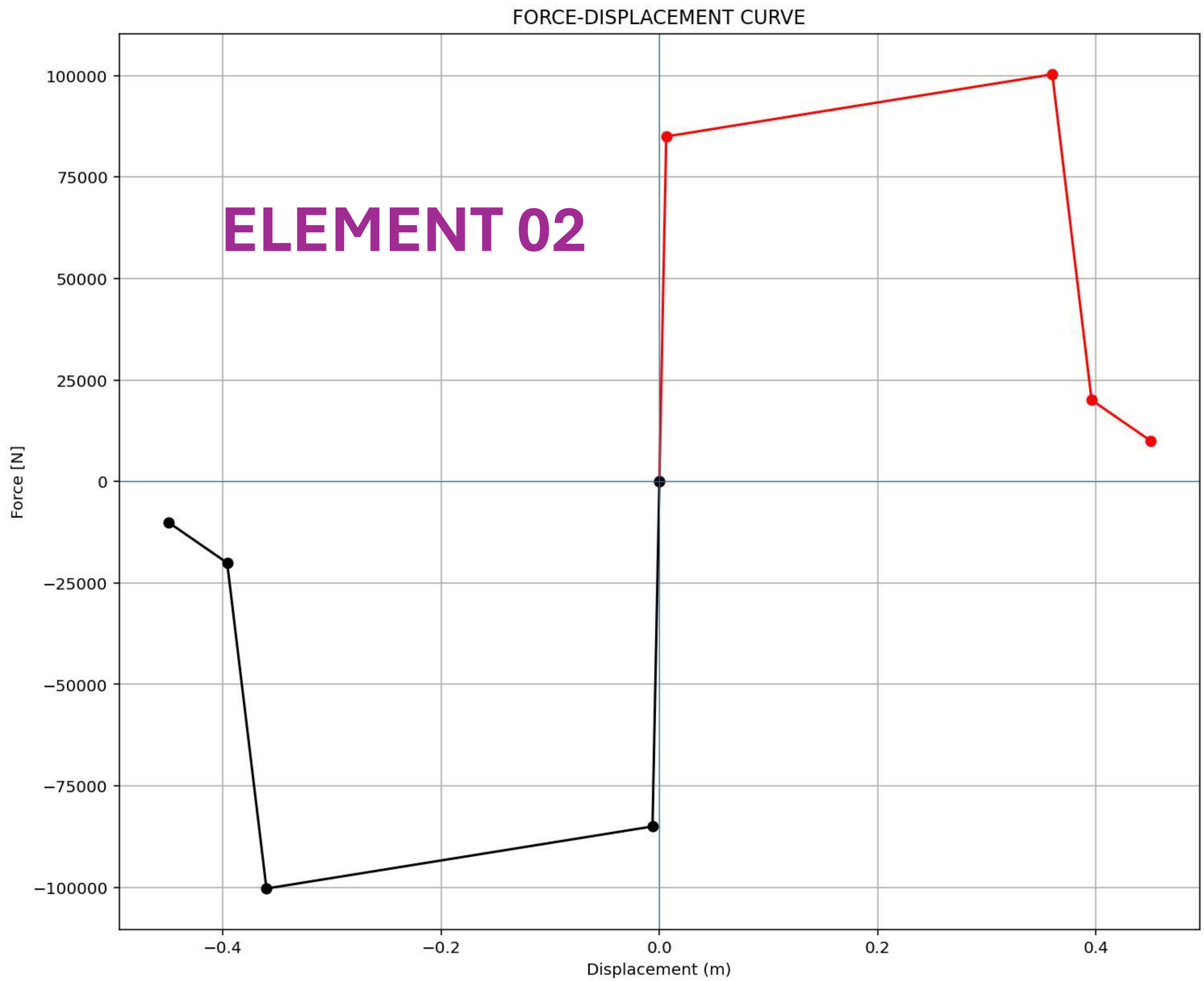
ELASTIC MATERIAL

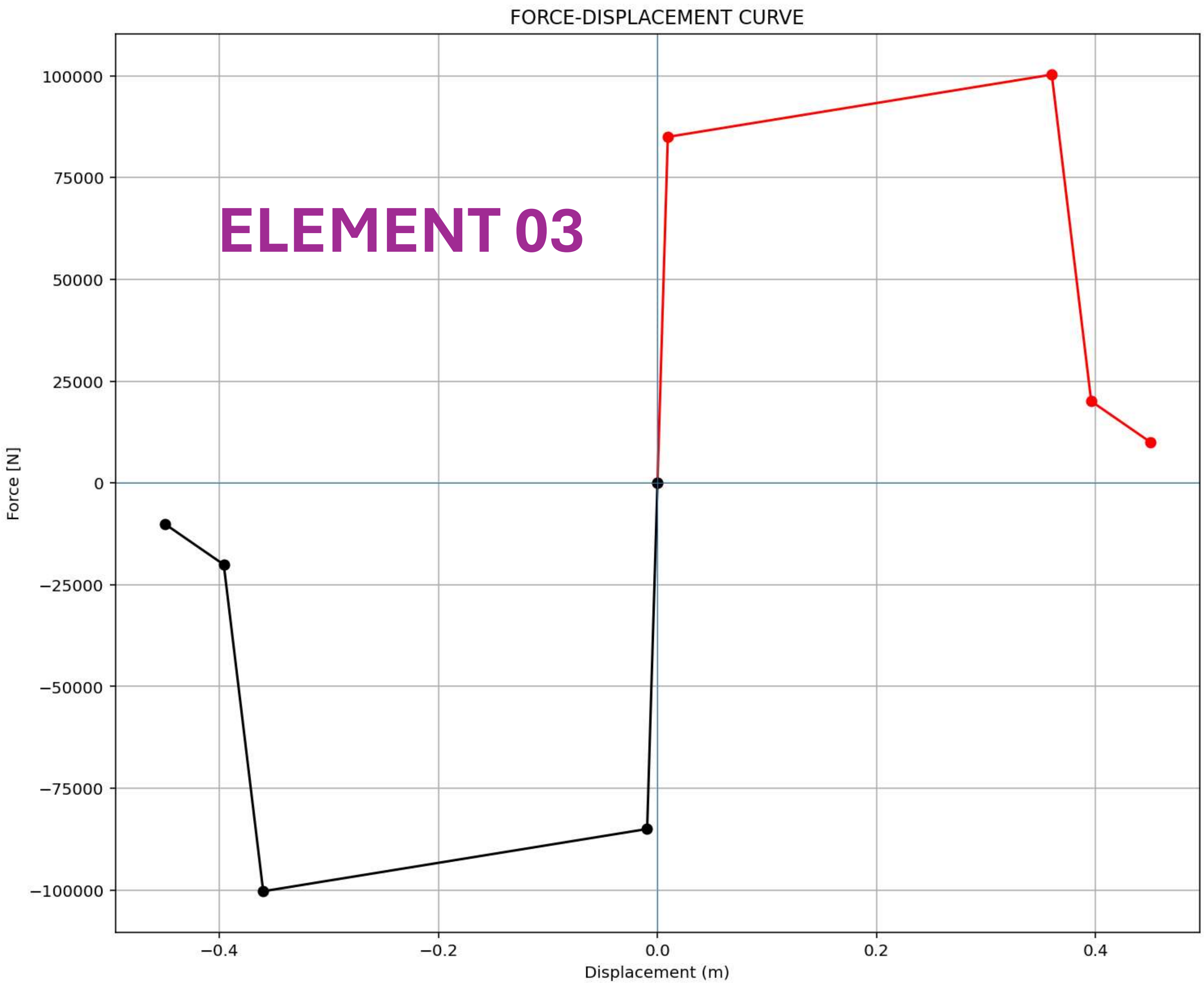


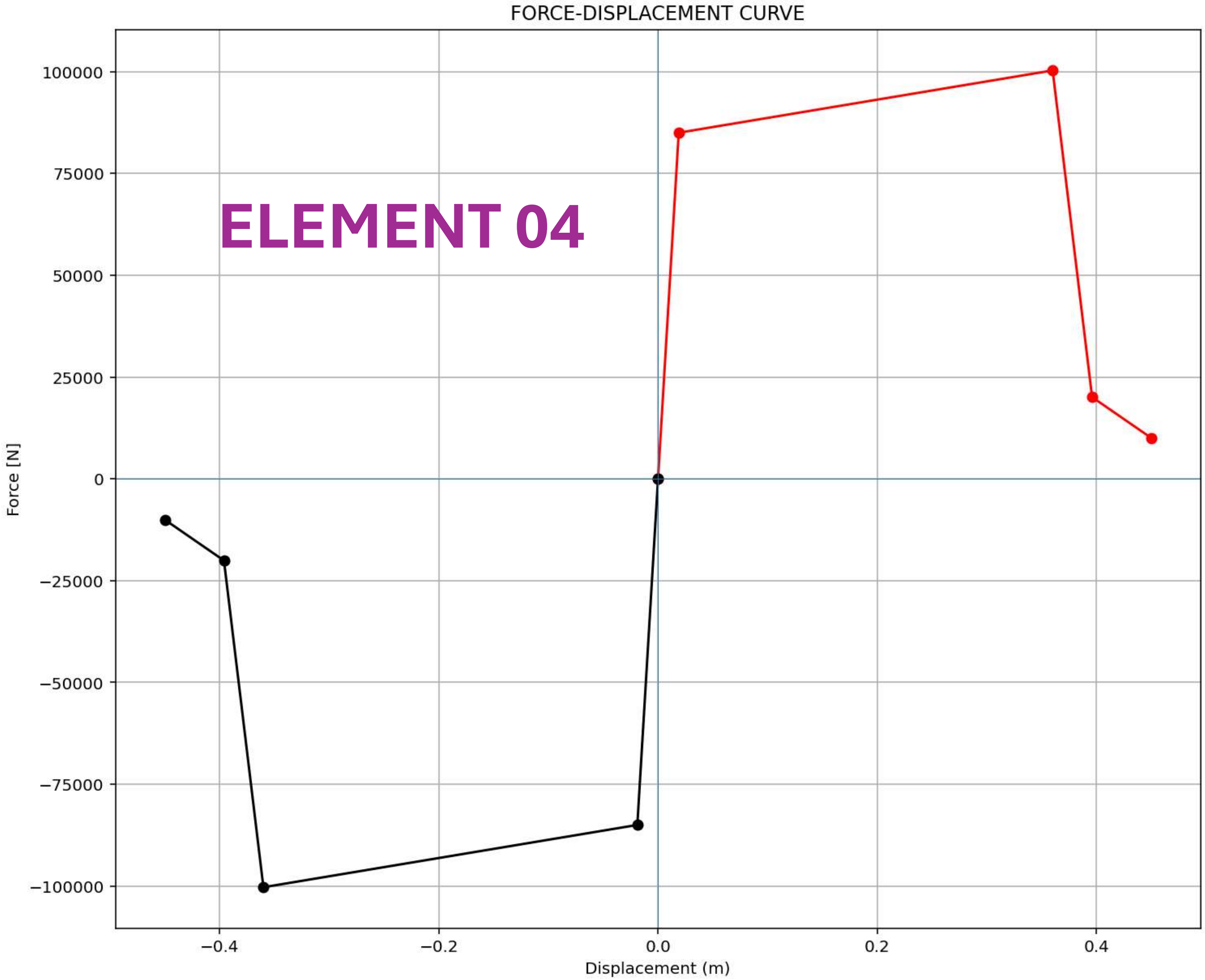
INELASTIC MATERIAL



FORCE-DISPLACEMENT CURVE
ELEMENT 01
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)

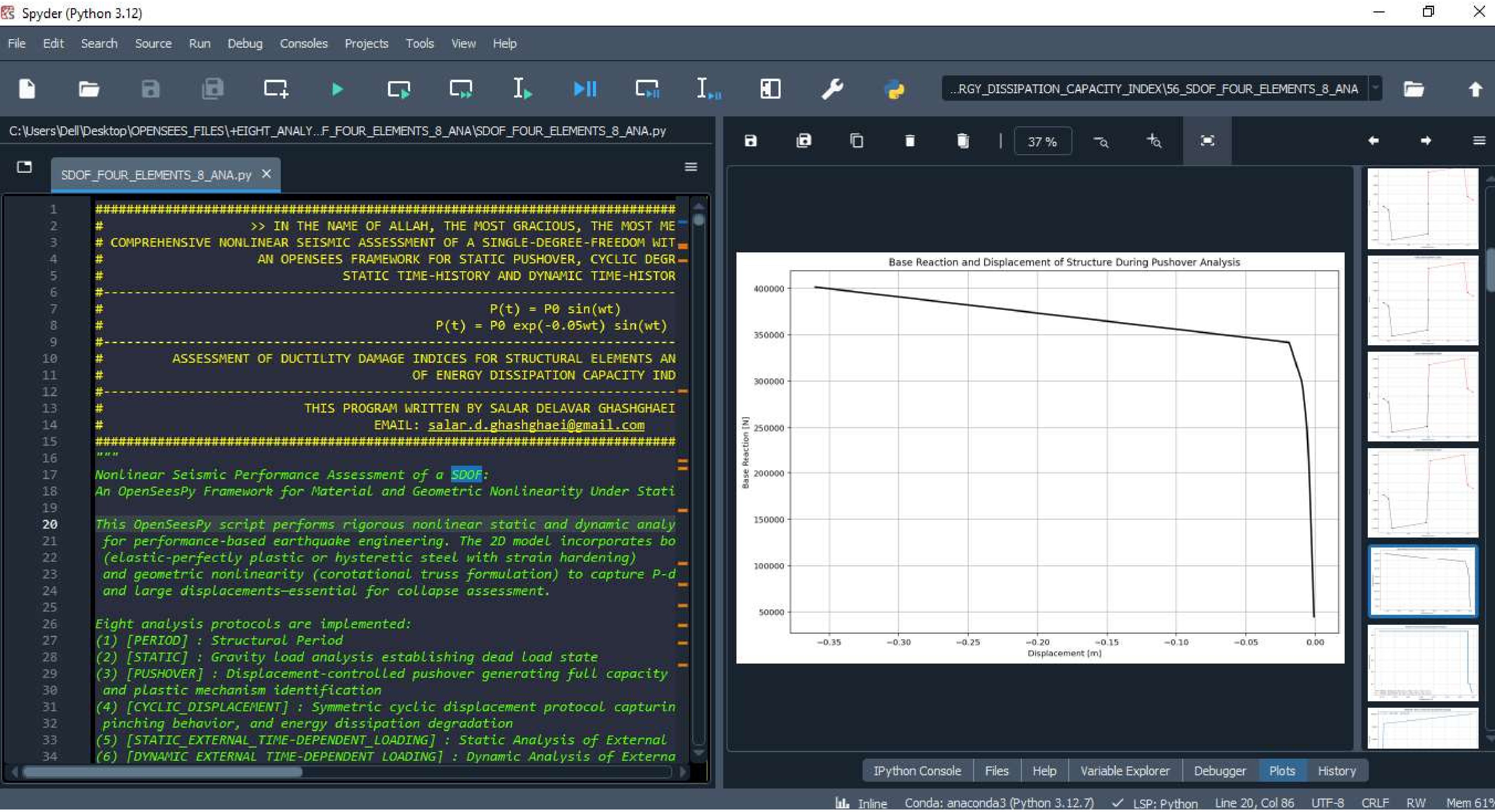

FORCE-DISPLACEMENT CURVE
ELEMENT 02
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE
ELEMENT 03
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)

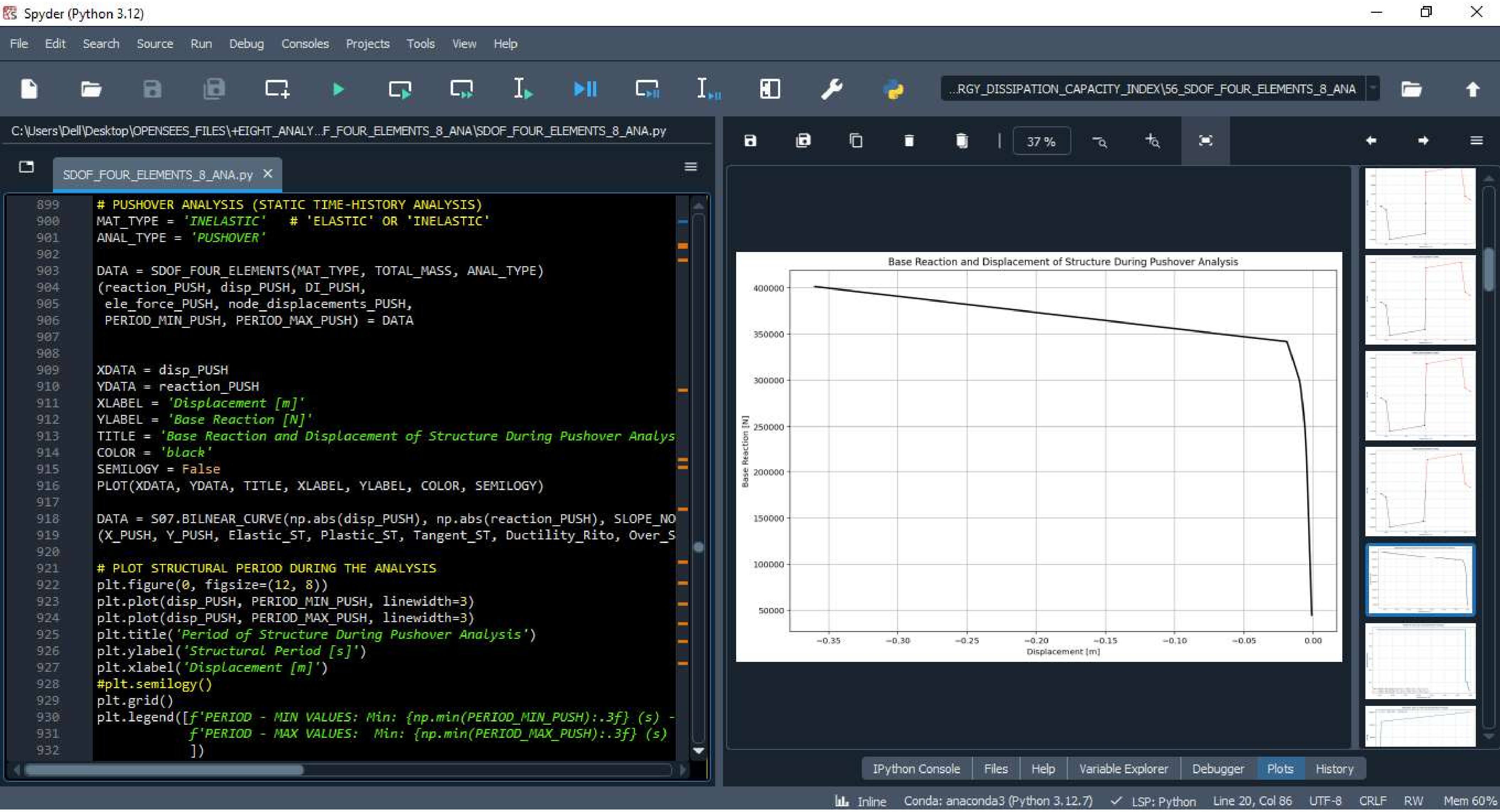


FORCE-DISPLACEMENT CURVE
ELEMENT 04
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)

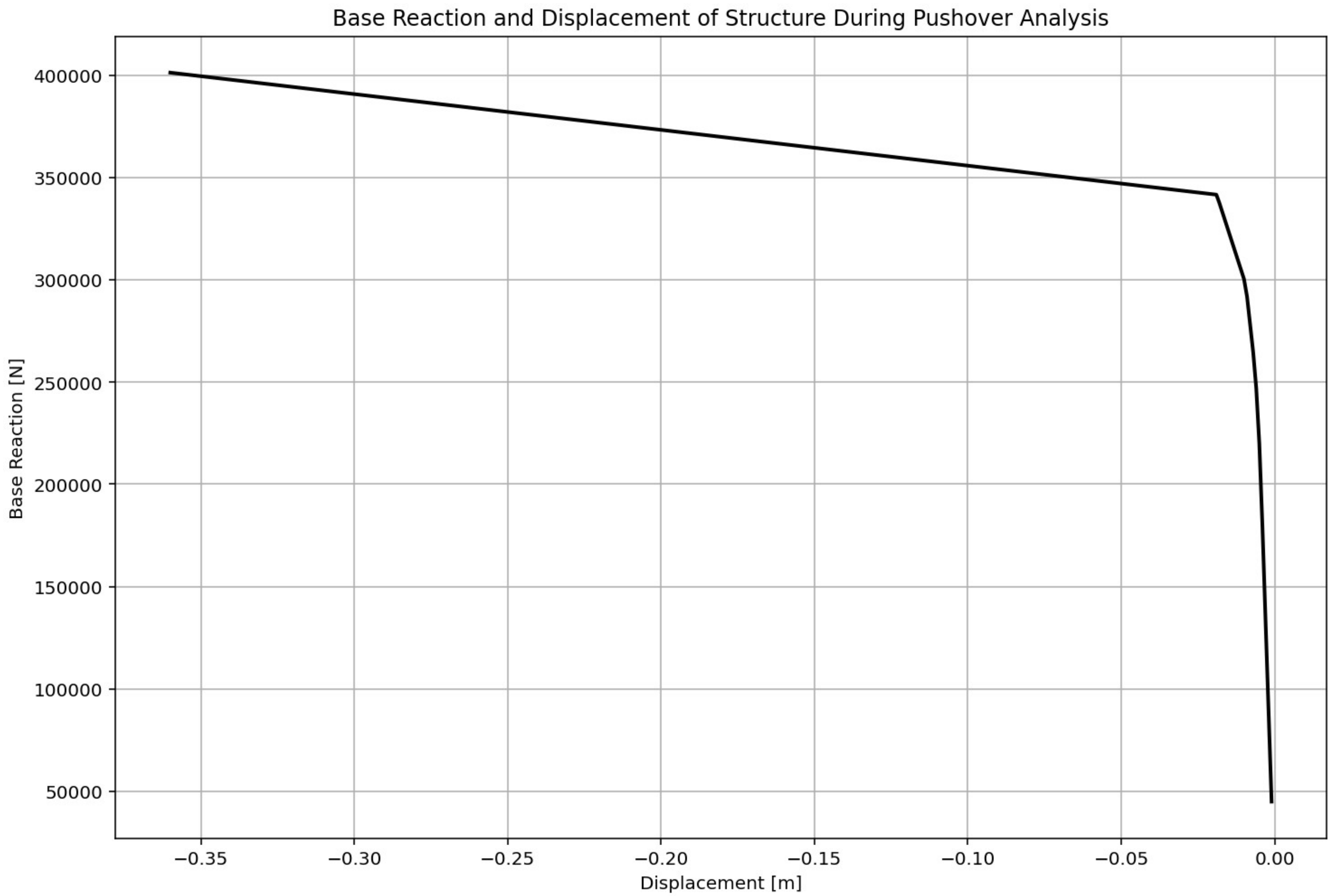
STATIC ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
..RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
#>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM WIT
AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGR
STATIC TIME-HISTORY AND DYNAMIC TIME-HISTOR
P(t) = P0 sin(wt)
P(t) = P0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
OF ENERGY DISSIPATION CAPACITY IND
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a SDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
for performance-based earthquake engineering. The 2D model incorporates bo
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity (corotational truss formulation) to capture P-d
and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
pinching behavior, and energy dissipation degradation
(5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External
(6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of Externa
37 %
Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
Displacement [m]
400000
350000
300000
250000
200000
150000
100000
50000
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 20, Col 86 UTF-8 CRLF RW Mem 61%

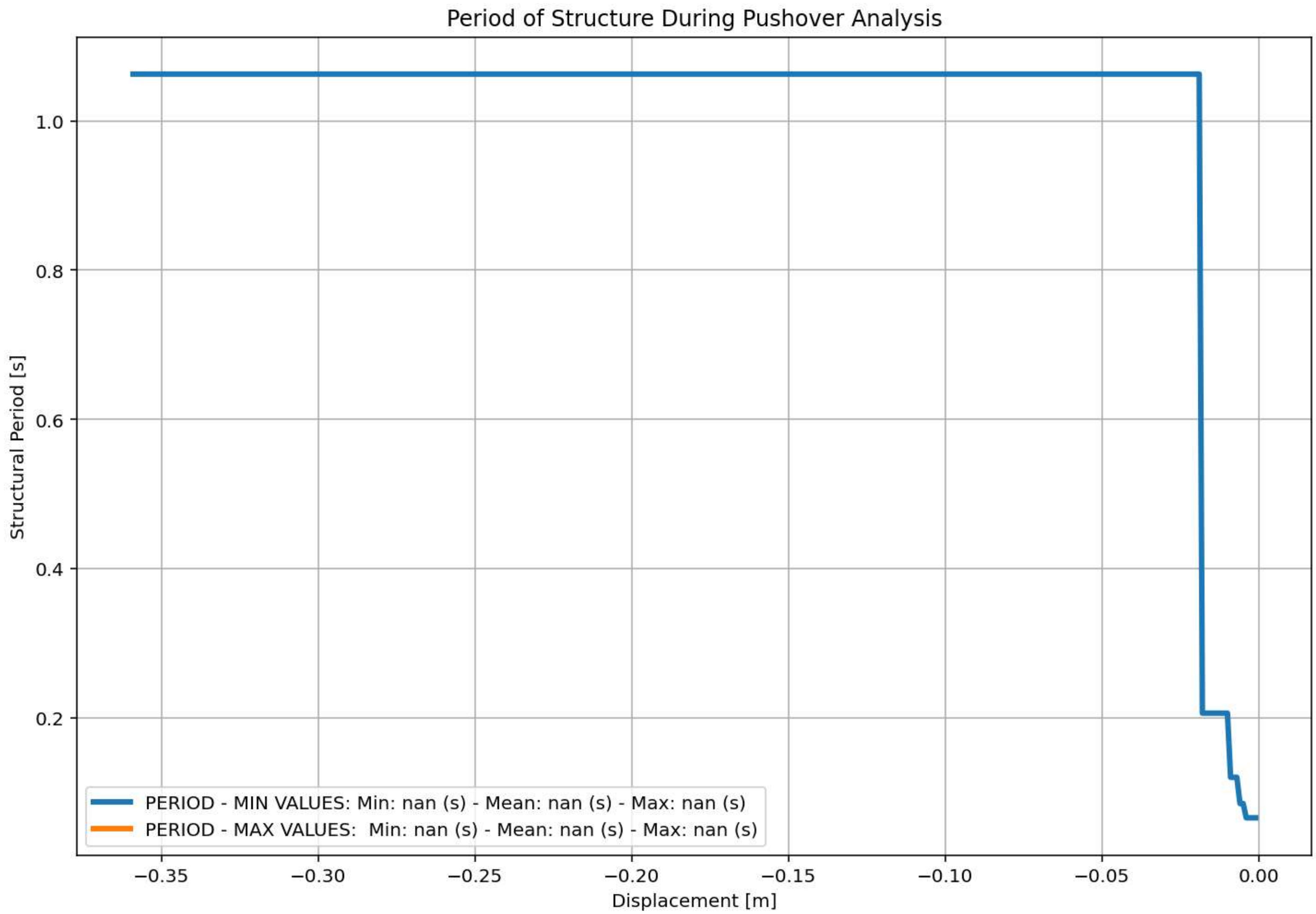
PUSHOVER ANALYSIS

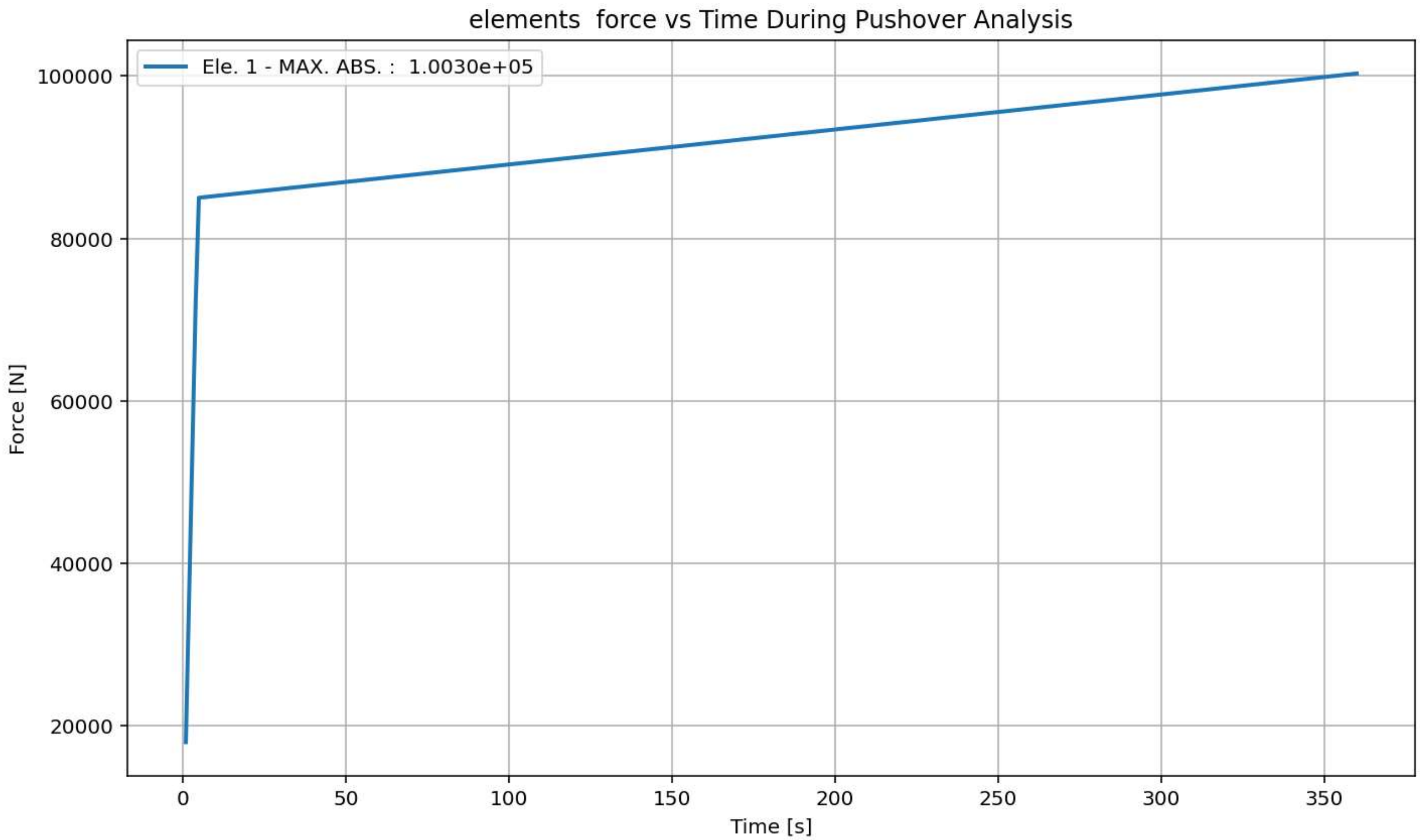


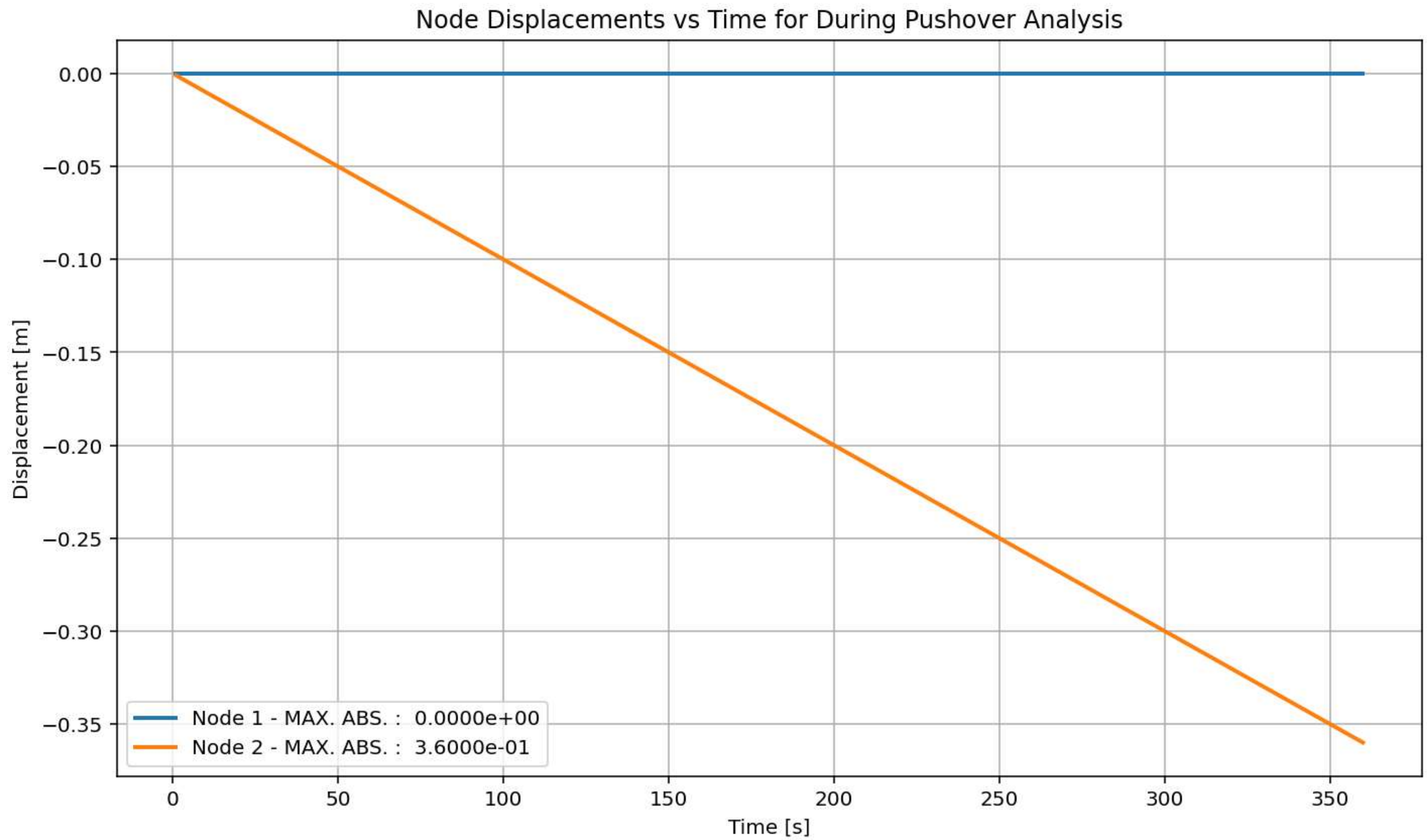
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
# PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PUSHOVER'
DATA = SDOF_FOUR_ELEMENTS(MAT_TYPE, TOTAL_MASS, ANAL_TYPE)
(reaction_PUSH, disp_PUSH, DI_PUSH,
 ele_force_PUSH, node_displacements_PUSH,
 PERIOD_MIN_PUSH, PERIOD_MAX_PUSH) = DATA
XDATA = disp_PUSH
YDATA = reaction_PUSH
XLABEL = 'Displacement [m]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Pushover Analys
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
DATA = S07.BILNEAR_CURVE(np.abs(disp_PUSH), np.abs(reaction_PUSH), SLOPE_NO
(X_PUSH, Y_PUSH, Elastic_ST, Plastic_ST, Tangent_ST, Ductility_Rito, Over_S
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_PUSH, PERIOD_MIN_PUSH, linewidth=3)
plt.plot(disp_PUSH, PERIOD_MAX_PUSH, linewidth=3)
plt.title('Period of Structure During Pushover Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [m]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_PUSH):.3f} (s) -
            f'PERIOD - MAX VALUES: Min: {np.min(PERIOD_MAX_PUSH):.3f} (s)
            ])
37 %
Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
Displacement [m]
400000
350000
300000
250000
200000
150000
100000
50000
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 20, Col 86 UTF-8 CRLF RW Mem 60%

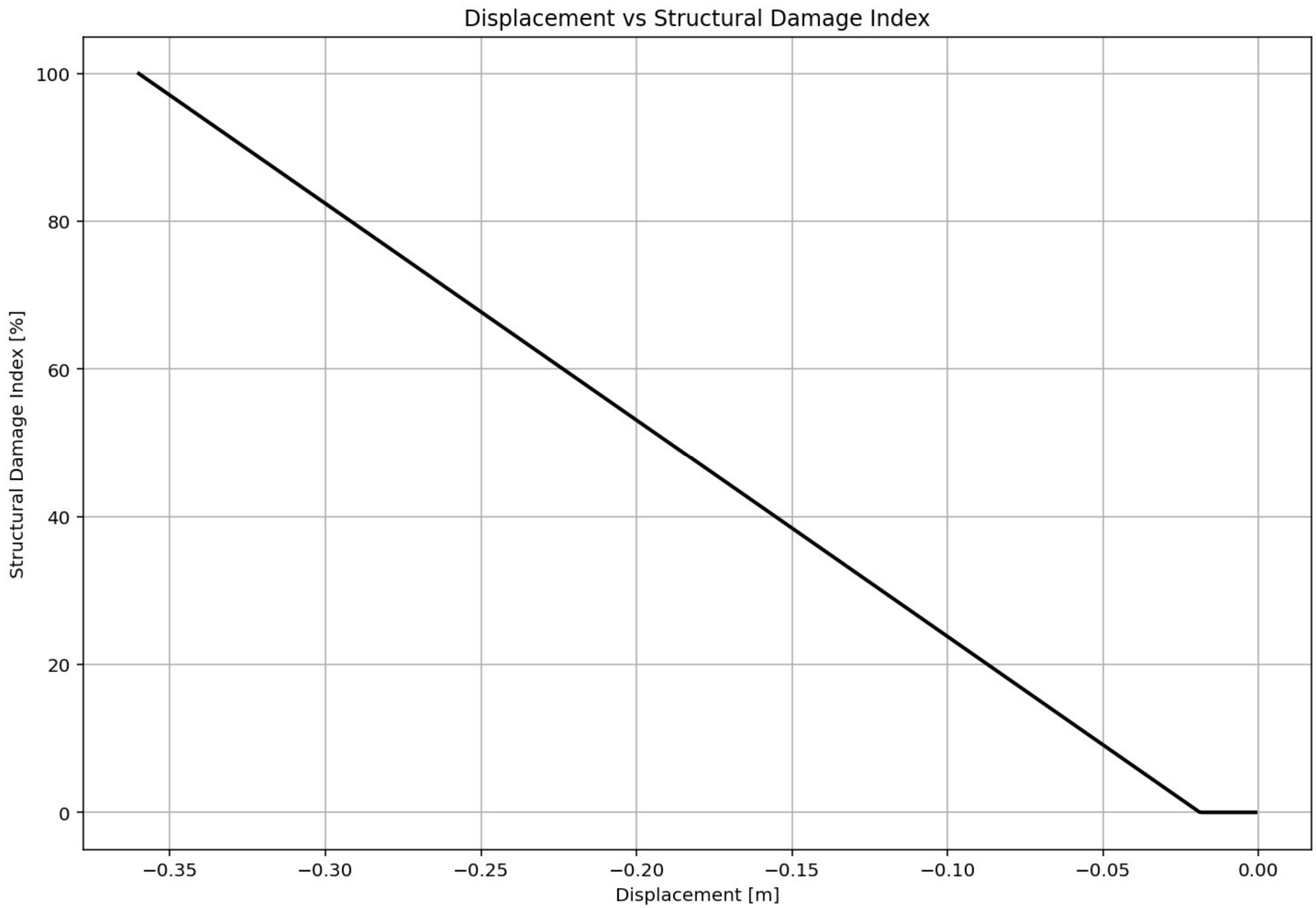


Base Reaction and Displacement of Structure During Pushover Analysis
Base Reaction [N]
400000
350000
300000
250000
200000
150000
100000
50000
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]


Period of Structure During Pushover Analysis
Structural Period [s]
1.0
0.8
0.6
0.4
0.2
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)

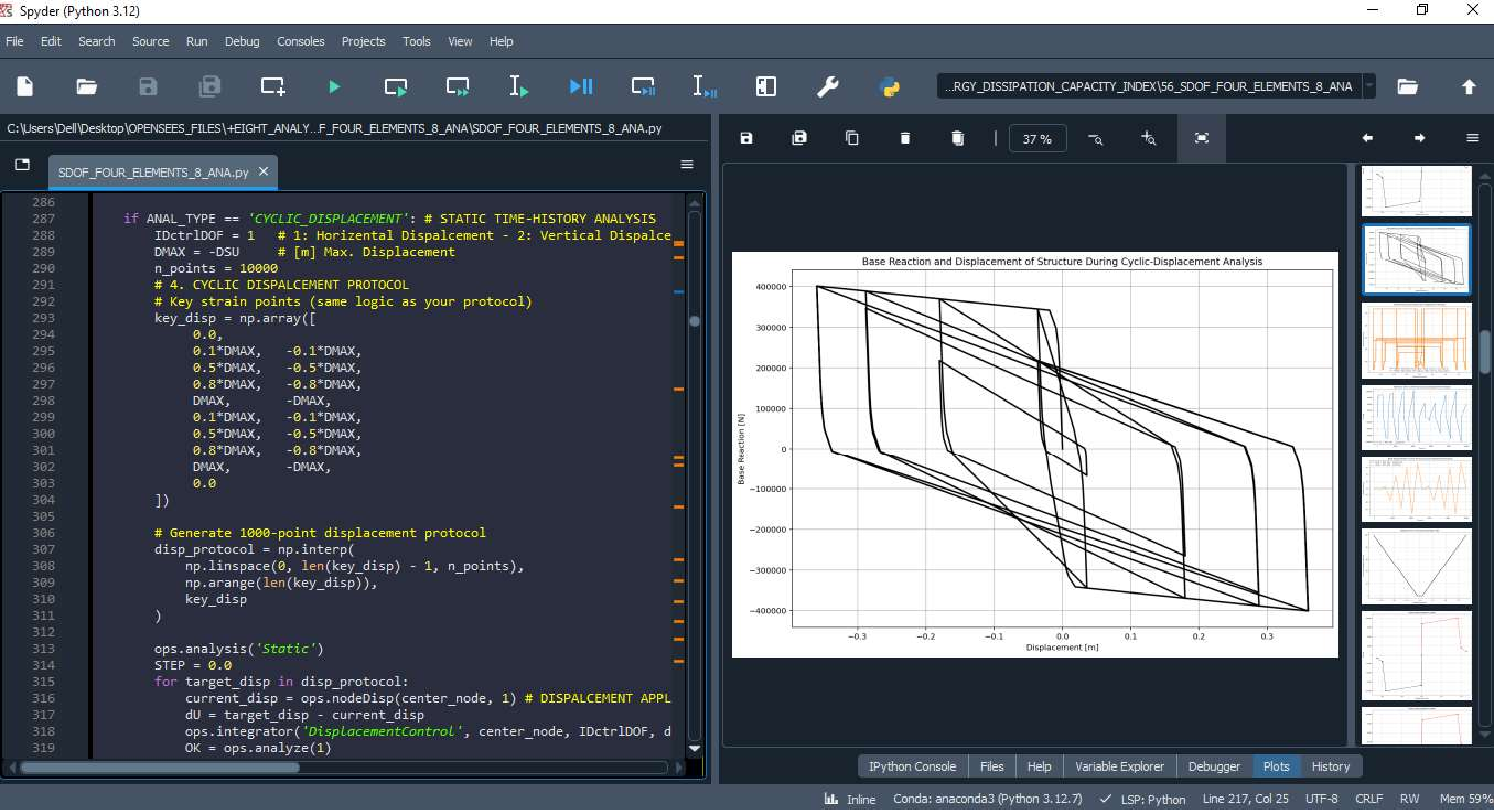

elements force vs Time During Pushover Analysis
Ele. 1 - MAX. ABS. : 1.0030e+05
Force [N]
100000
80000
60000
40000
20000
0
50
100
150
200
250
300
350
Time [s]


Node Displacements vs Time for During Pushover Analysis
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
−0.30
−0.35
Displacement [m]
0
50
100
150
200
250
300
350
Time [s]
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.6000e-01

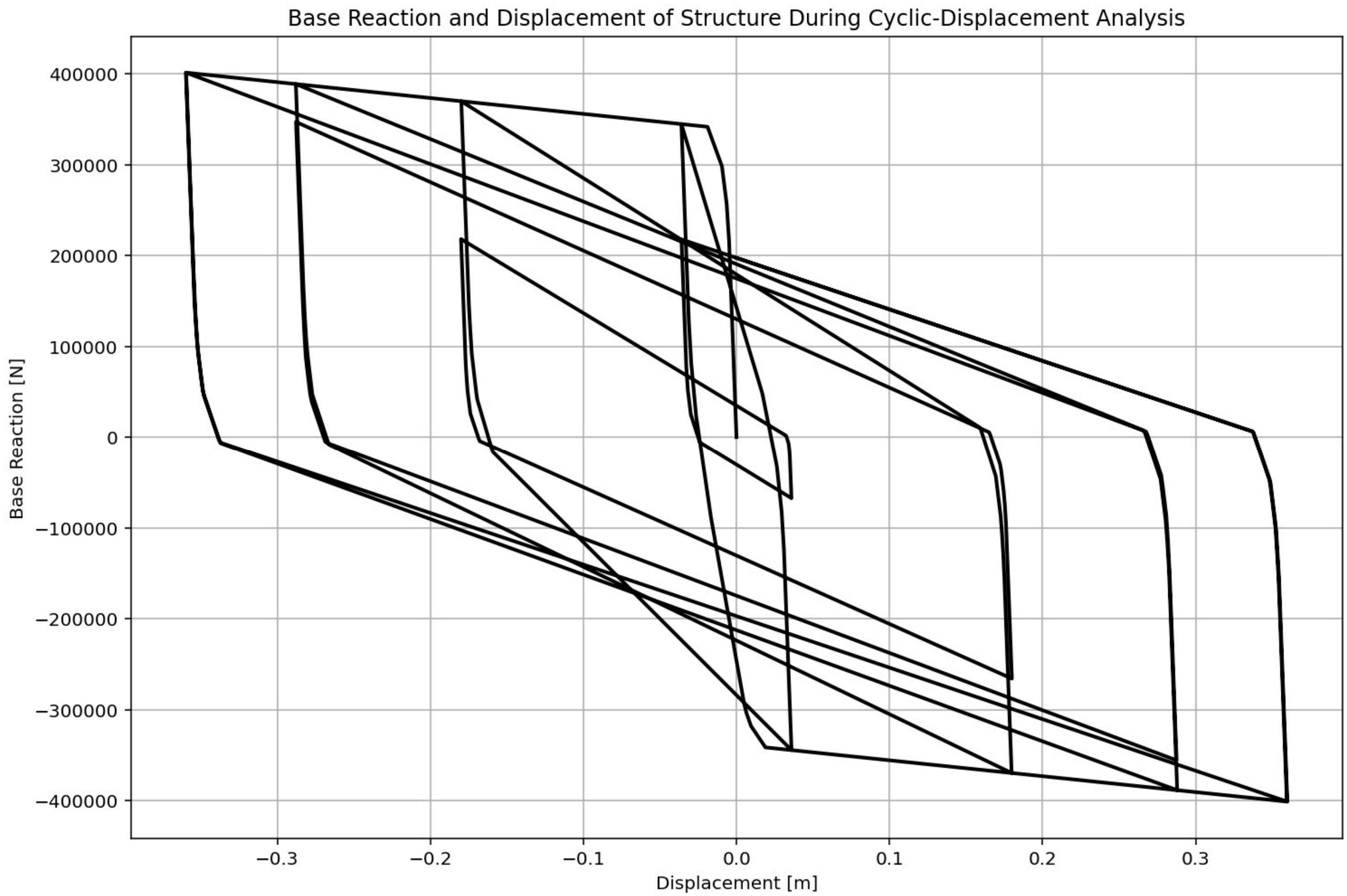


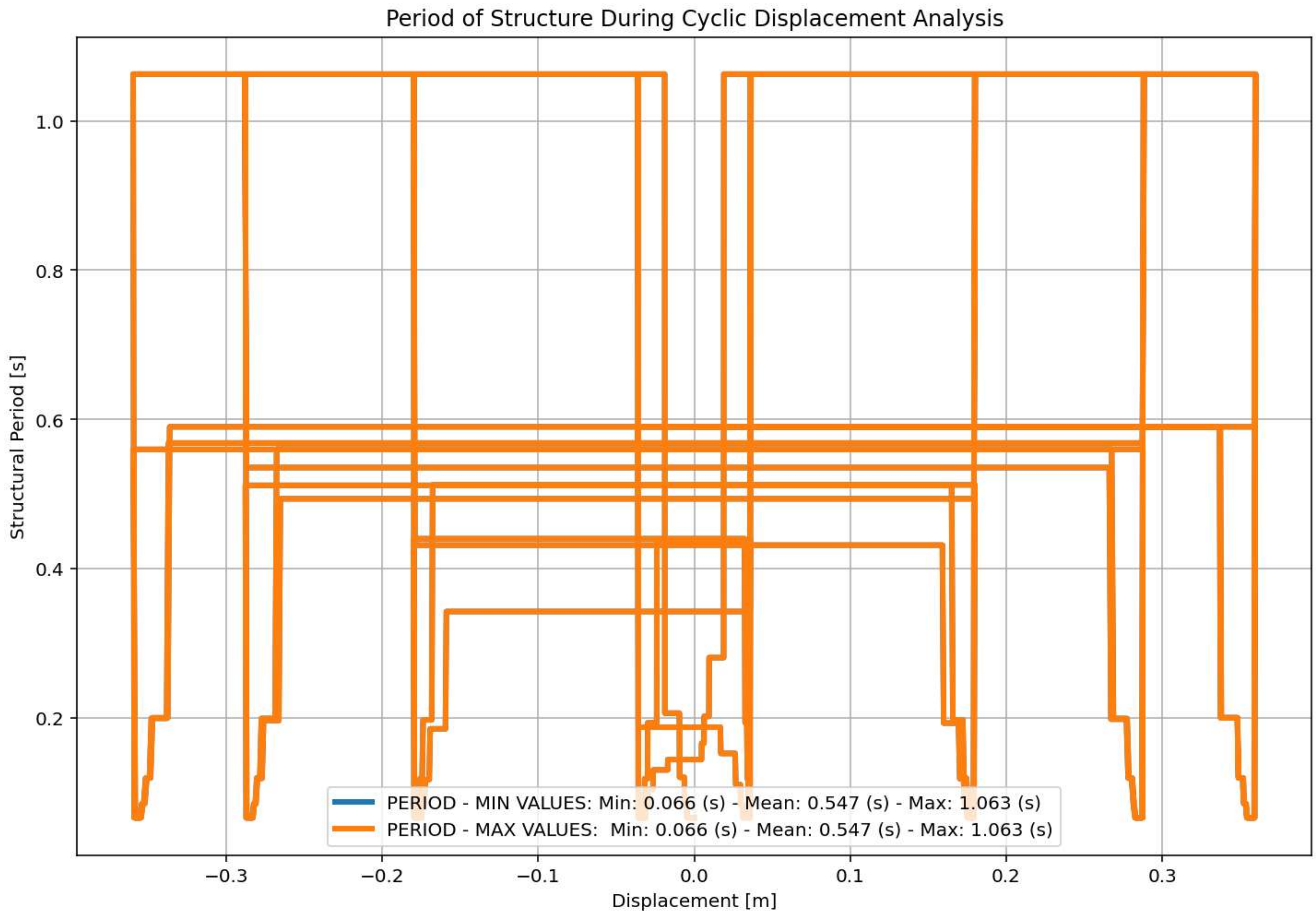
Displacement vs Structural Damage Index
Structural Damage Index [%]
100
80
60
40
20
0
−0.35
−0.30
−0.25
−0.20
−0.15
−0.10
−0.05
0.00
Displacement [m]

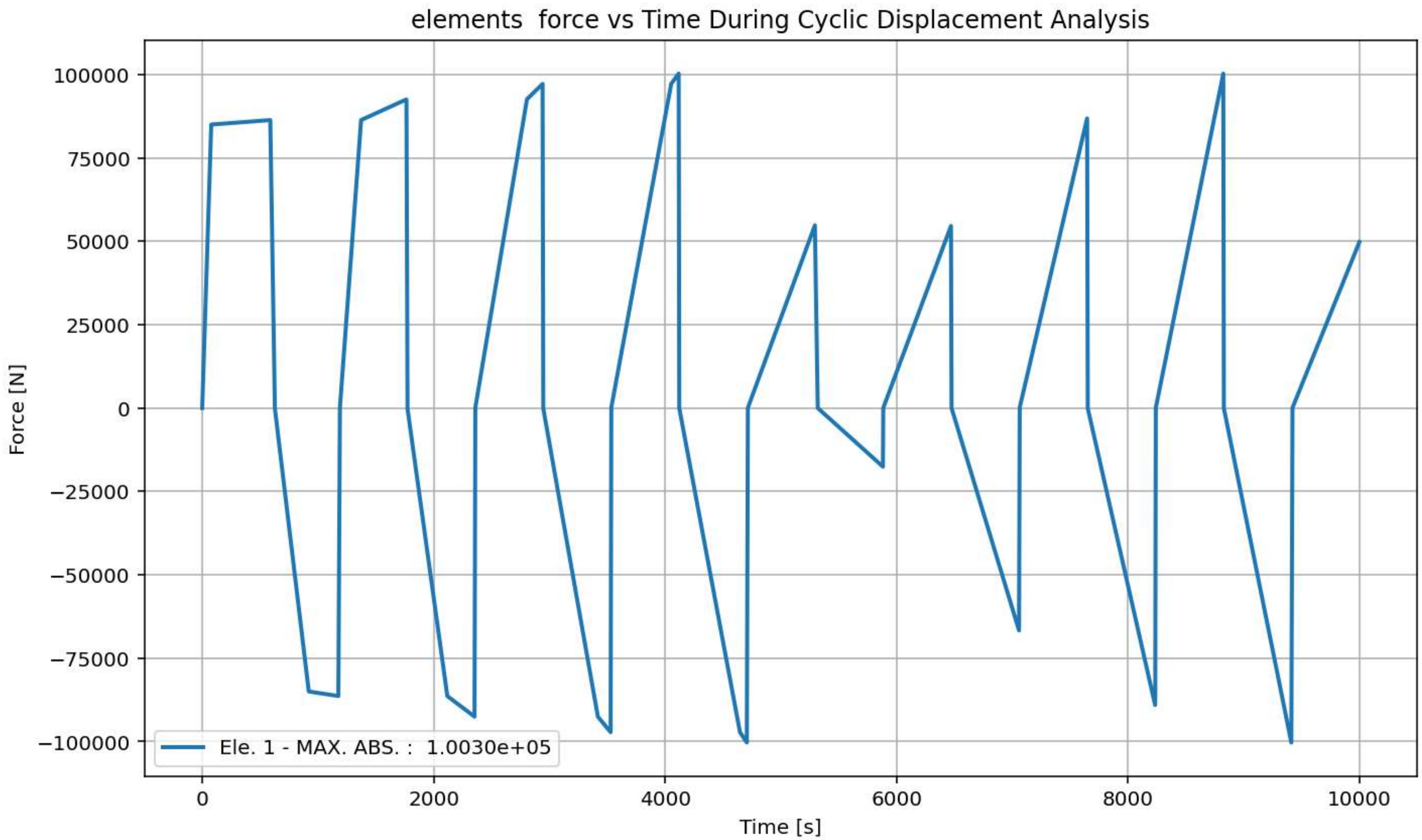
CYCLIC DISPLACEMENT ANALYSIS

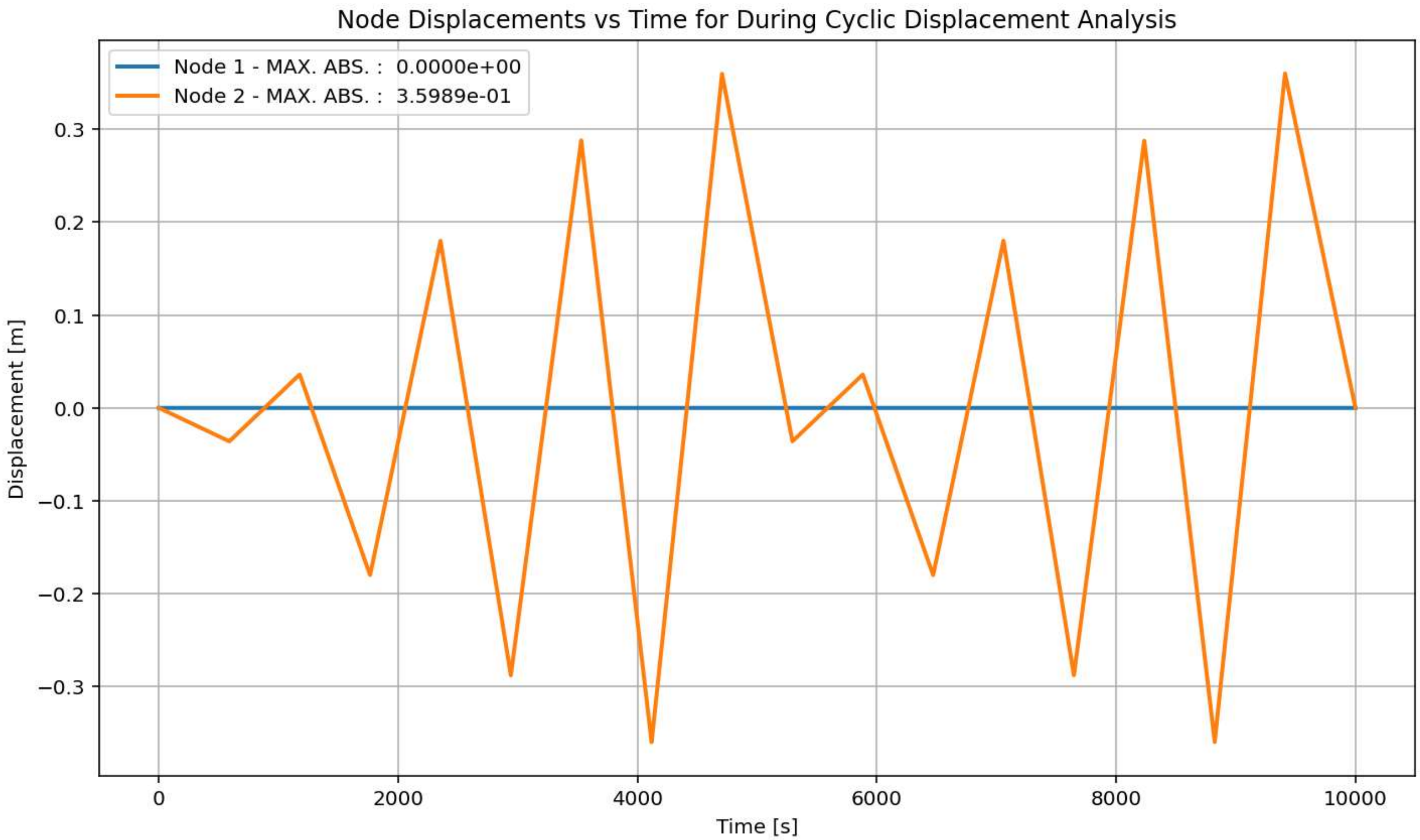


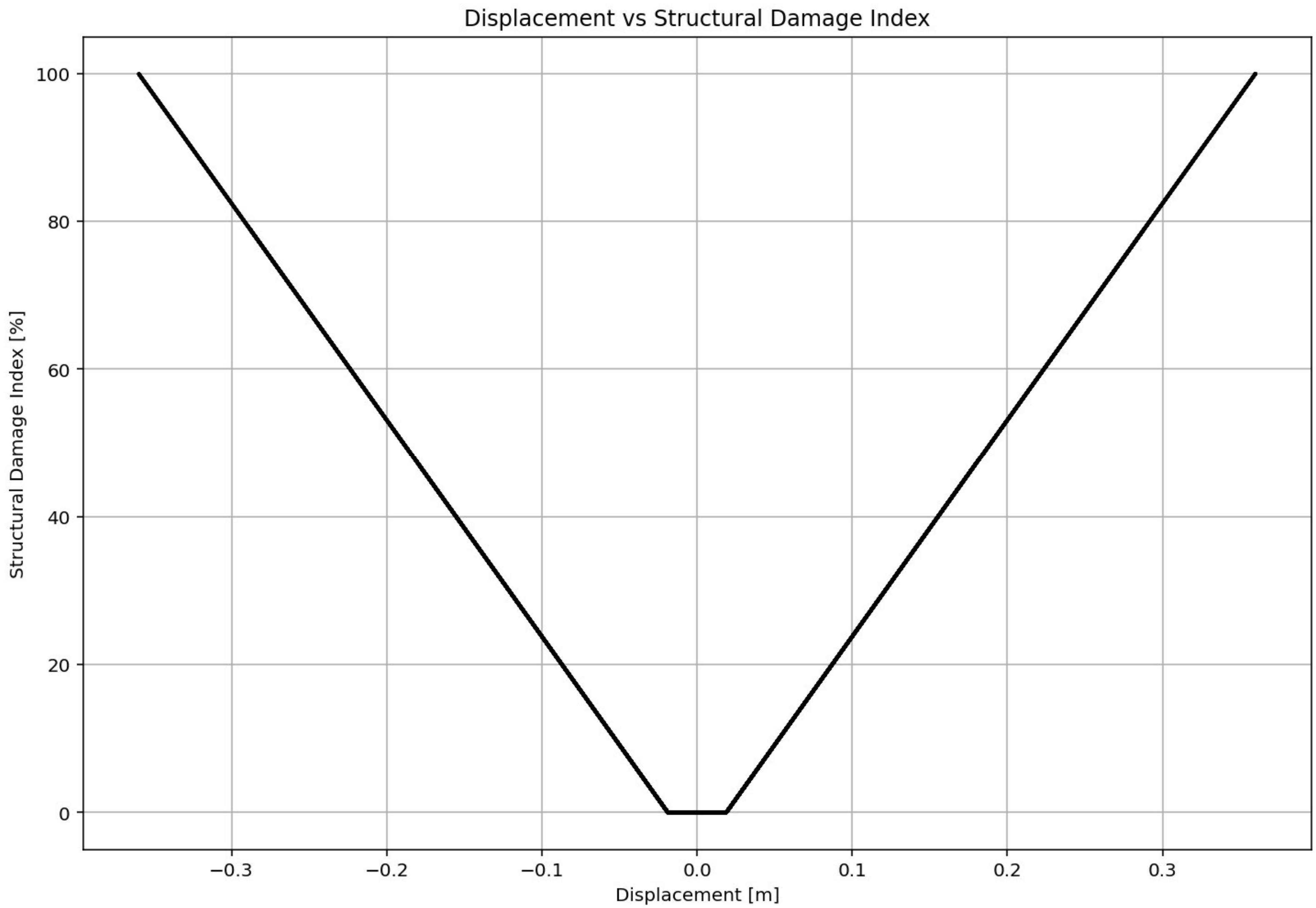

```python

    if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
        IDctrlDOF = 1   # 1: Horizental Dispalcement - 2: Vertical Dispalce
        DMAX = -DSU     # [m] Max. Displacement
        n_points = 10000
        # 4. CYCLIC DISPALCEMENT PROTOCOL
        # Key strain points (same logic as your protocol)
        key_disp = np.array([
            0.0,
            0.1*DMAX,   -0.1*DMAX,
            0.5*DMAX,   -0.5*DMAX,
            0.8*DMAX,   -0.8*DMAX,
            DMAX,       -DMAX,
            0.1*DMAX,   -0.1*DMAX,
            0.5*DMAX,   -0.5*DMAX,
            0.8*DMAX,   -0.8*DMAX,
            DMAX,       -DMAX,
            0.0
        ])

        # Generate 1000-point displacement protocol
        disp_protocol = np.interp(
            np.linspace(0, len(key_disp) - 1, n_points),
            np.arange(len(key_disp)),
            key_disp
        )

        ops.analysis('Static')
        STEP = 0.0
        for target_disp in disp_protocol:
            current_disp = ops.nodeDisp(center_node, 1) # DISPALCEMENT APPL
            dU = target_disp - current_disp
            ops.integrator('DisplacementControl', center_node, IDctrlDOF, d
            OK = ops.analyze(1)
```


Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Base Reaction [N]
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]

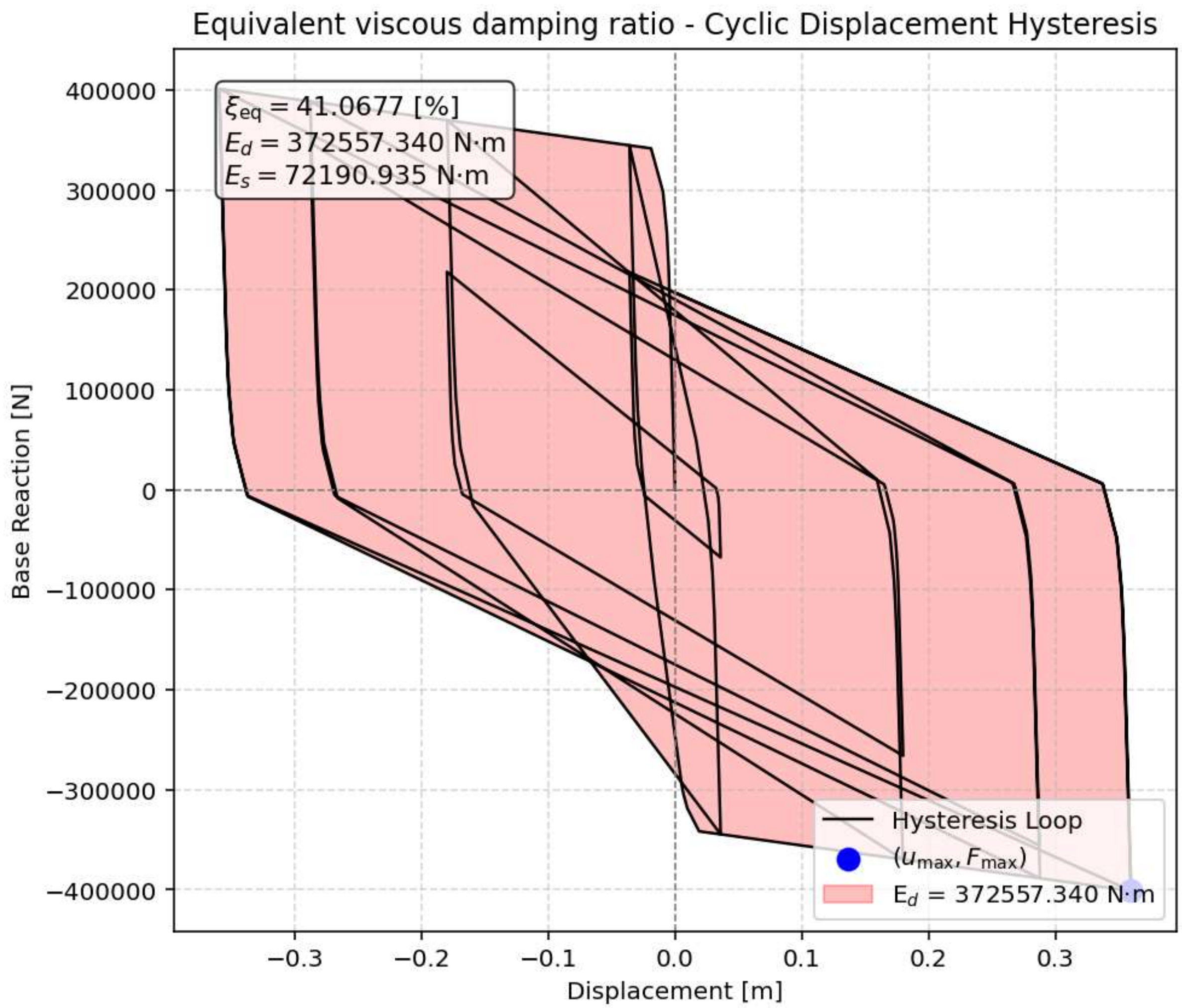

Period of Structure During Cyclic Displacement Analysis
Structural Period [s]
1.0
0.8
0.6
0.4
0.2
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]
PERIOD - MIN VALUES: Min: 0.066 (s) - Mean: 0.547 (s) - Max: 1.063 (s)
PERIOD - MAX VALUES: Min: 0.066 (s) - Mean: 0.547 (s) - Max: 1.063 (s)


elements force vs Time During Cyclic Displacement Analysis
Force [N]
Time [s]
Ele. 1 - MAX. ABS. : 1.0030e+05
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0
2000
4000
6000
8000
10000


Node Displacements vs Time for During Cyclic Displacement Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.5989e-01
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Displacement [m]
0
2000
4000
6000
8000
10000
Time [s]

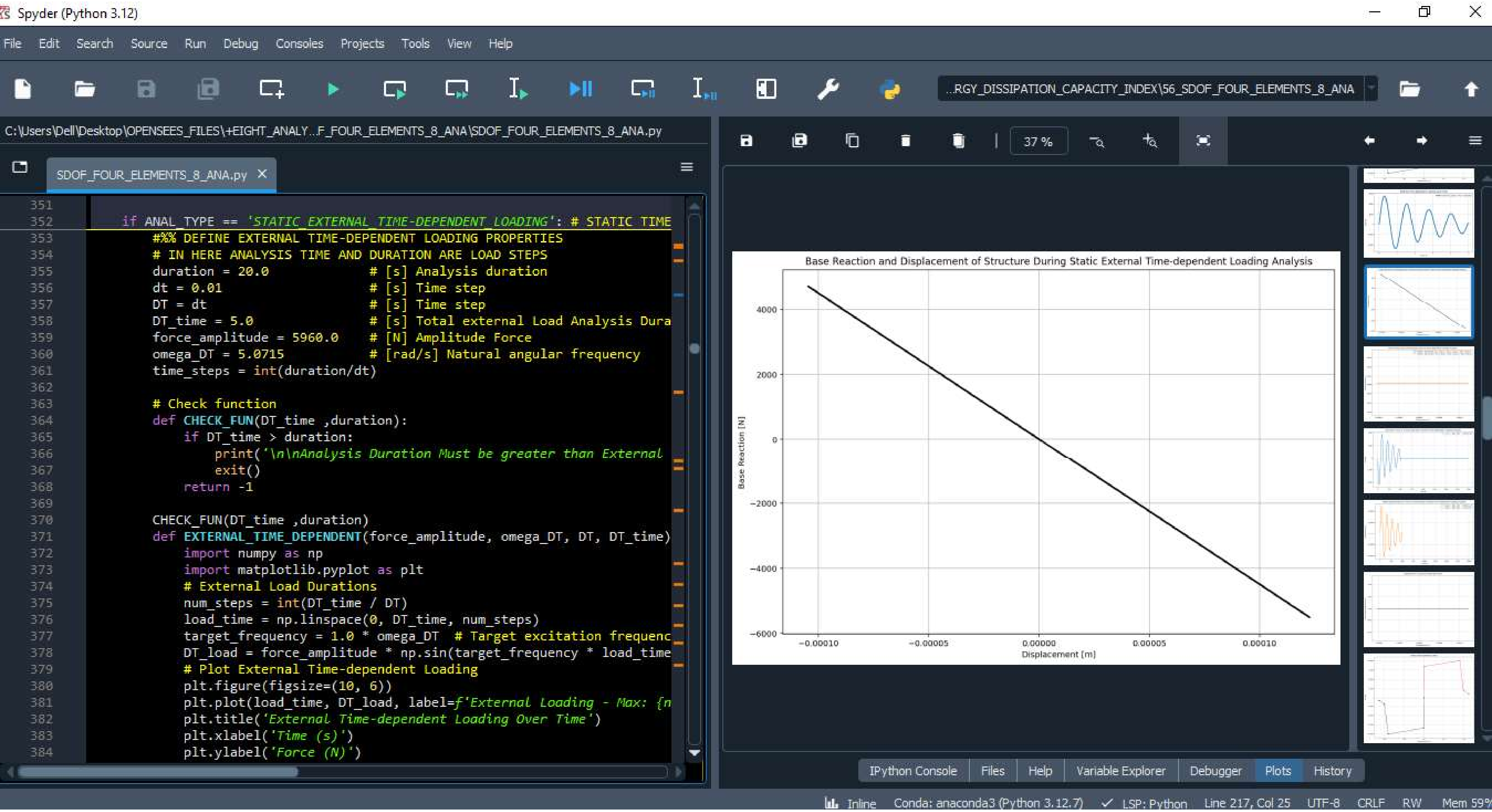


Displacement vs Structural Damage Index
Structural Damage Index [%]
Displacement [m]
0
20
40
60
80
100
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

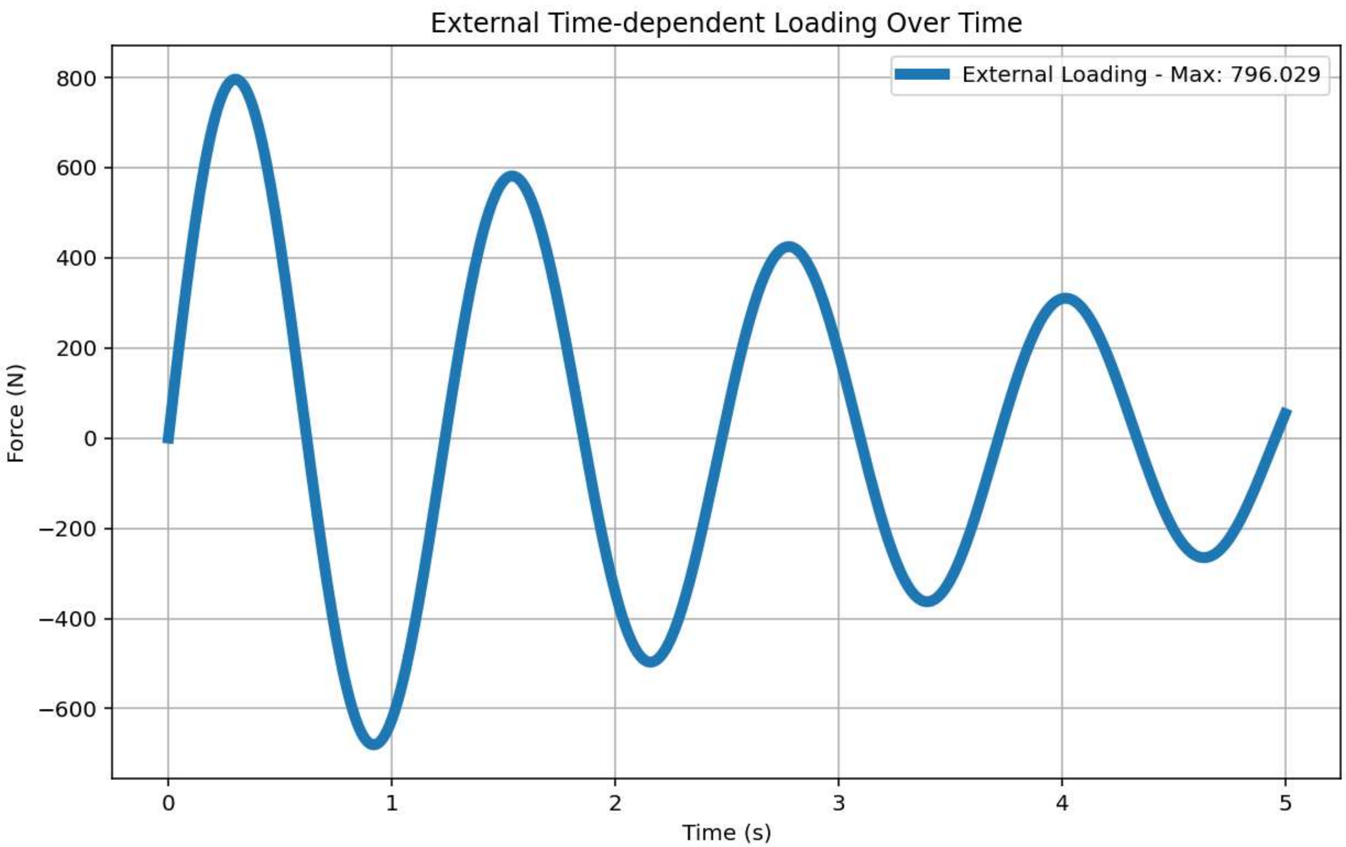


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 41.0677$ [%]
$E_d = 372557.340$ N·m
$E_s = 72190.935$ N·m
Base Reaction [N]
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 372557.340 N·m

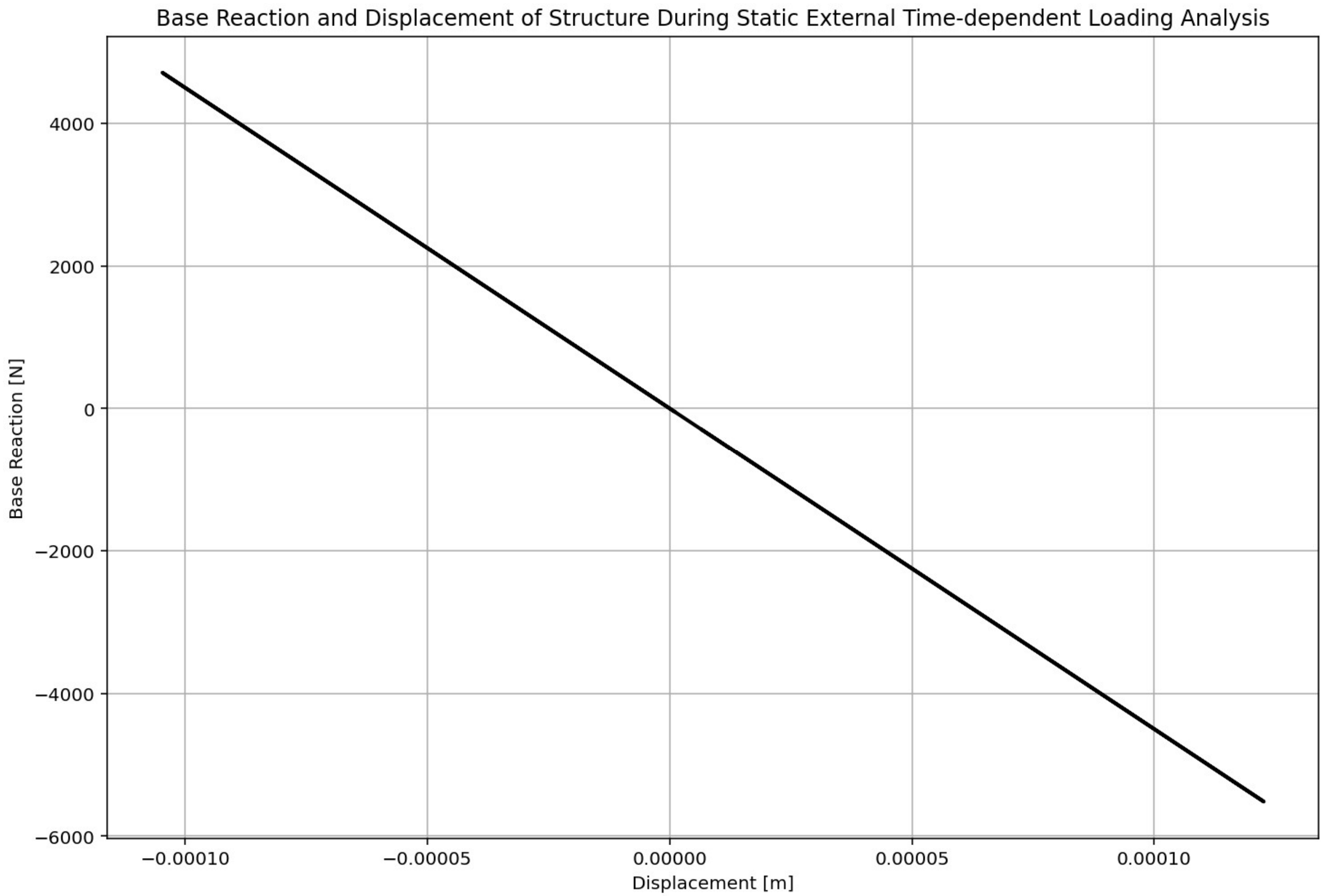
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

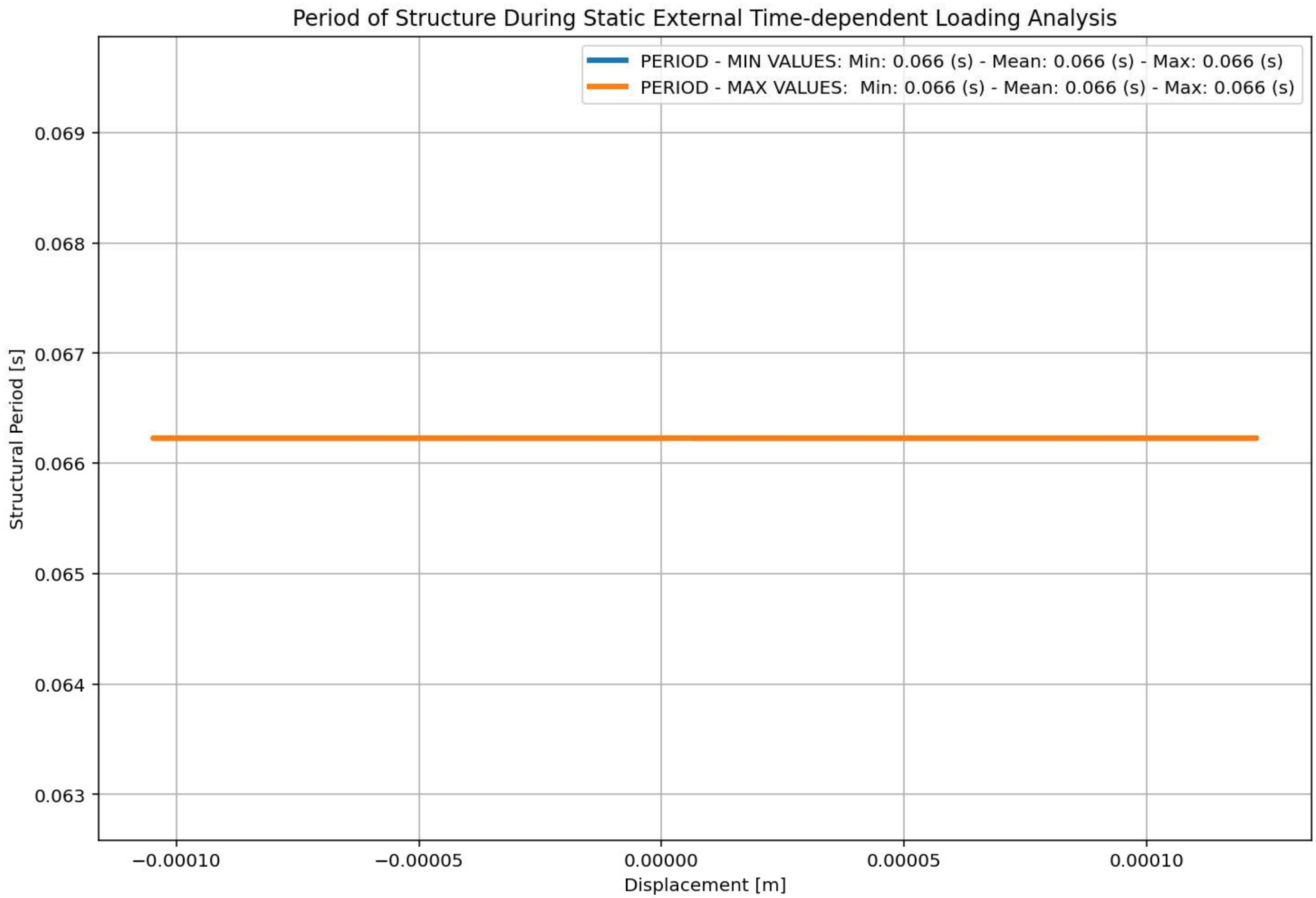


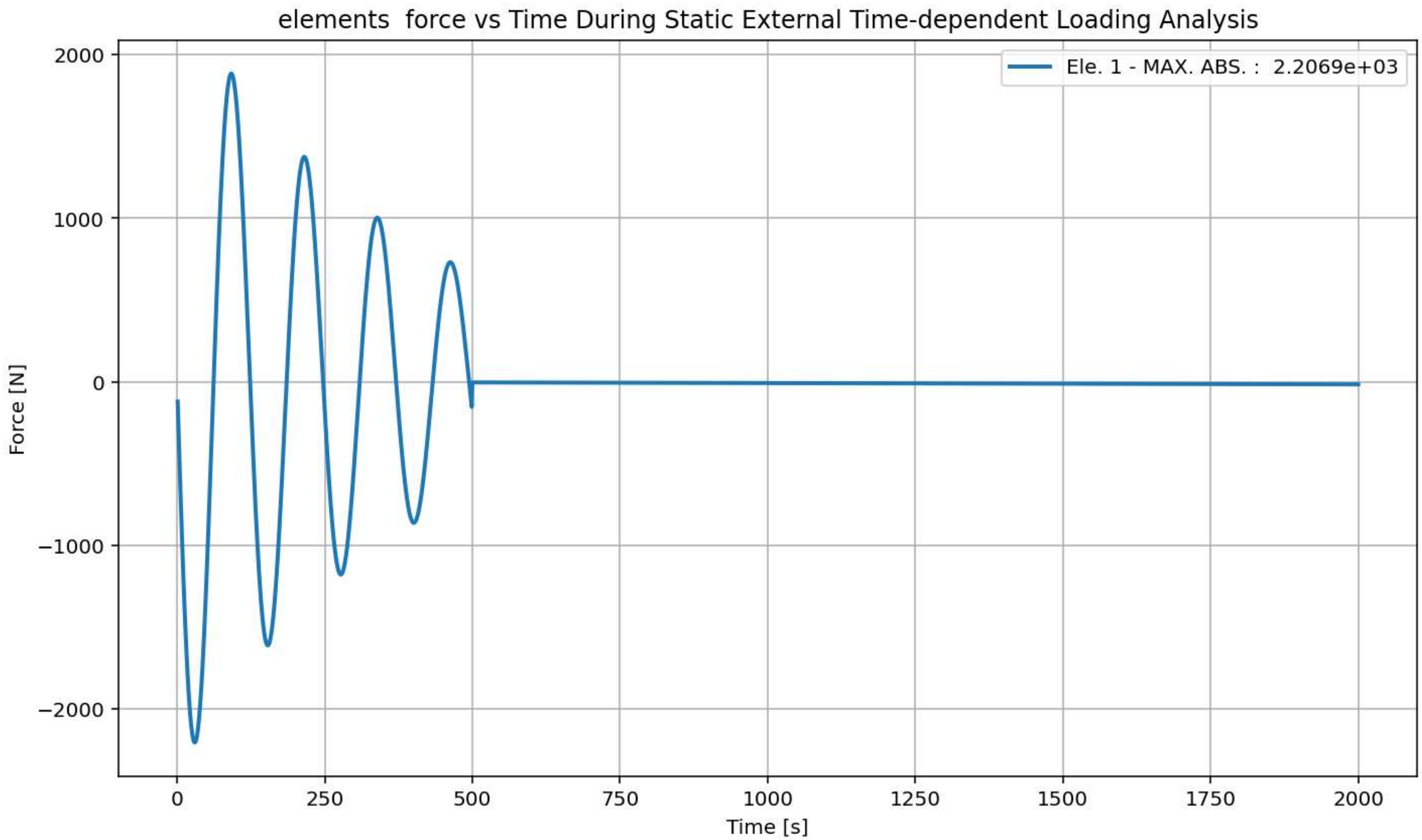
Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
..RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY..F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
if ANAL_TYPE == 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING': # STATIC TIME
#%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
# IN HERE ANALYSIS TIME AND DURATION ARE LOAD STEPS
duration = 20.0 # [s] Analysis duration
dt = 0.01 # [s] Time step
DT = dt # [s] Time step
DT_time = 5.0 # [s] Total external Load Analysis Dura
force_amplitude = 5960.0 # [N] Amplitude Force
omega_DT = 5.0715 # [rad/s] Natural angular frequency
time_steps = int(duration/dt)
# Check function
def CHECK_FUN(DT_time ,duration):
if DT_time > duration:
print('\n\nAnalysis Duration Must be greater than External
exit()
return -1
CHECK_FUN(DT_time ,duration)
def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time)
import numpy as np
import matplotlib.pyplot as plt
# External Load Durations
num_steps = int(DT_time / DT)
load_time = np.linspace(0, DT_time, num_steps)
target_frequency = 1.0 * omega_DT # Target excitation frequenc
DT_load = force_amplitude * np.sin(target_frequency * load_time
# Plot External Time-dependent Loading
plt.figure(figsize=(10, 6))
plt.plot(load_time, DT_load, label=f'External Loading - Max: {n
plt.title('External Time-dependent Loading Over Time')
plt.xlabel('Time (s)')
plt.ylabel('Force (N)')
37 %
Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
Displacement [m]
4000
2000
0
−2000
−4000
−6000
−0.00010
−0.00005
0.00000
0.00005
0.00010
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 217, Col 25 UTF-8 CRLF RW Mem 59%

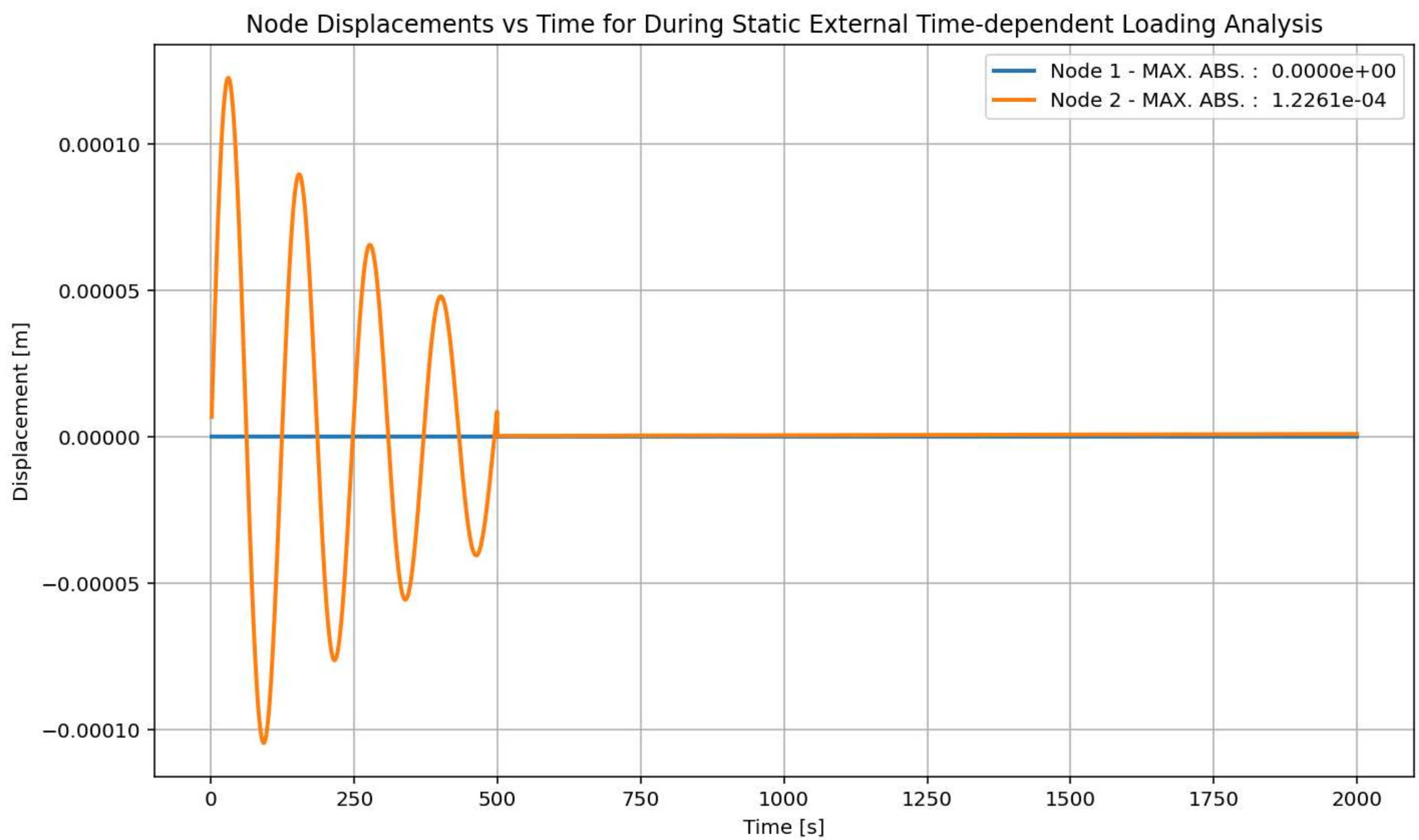


External Time-dependent Loading Over Time
External Loading - Max: 796.029
800
600
400
200
0
−200
−400
−600
Force (N)
0
1
2
3
4
5
Time (s)

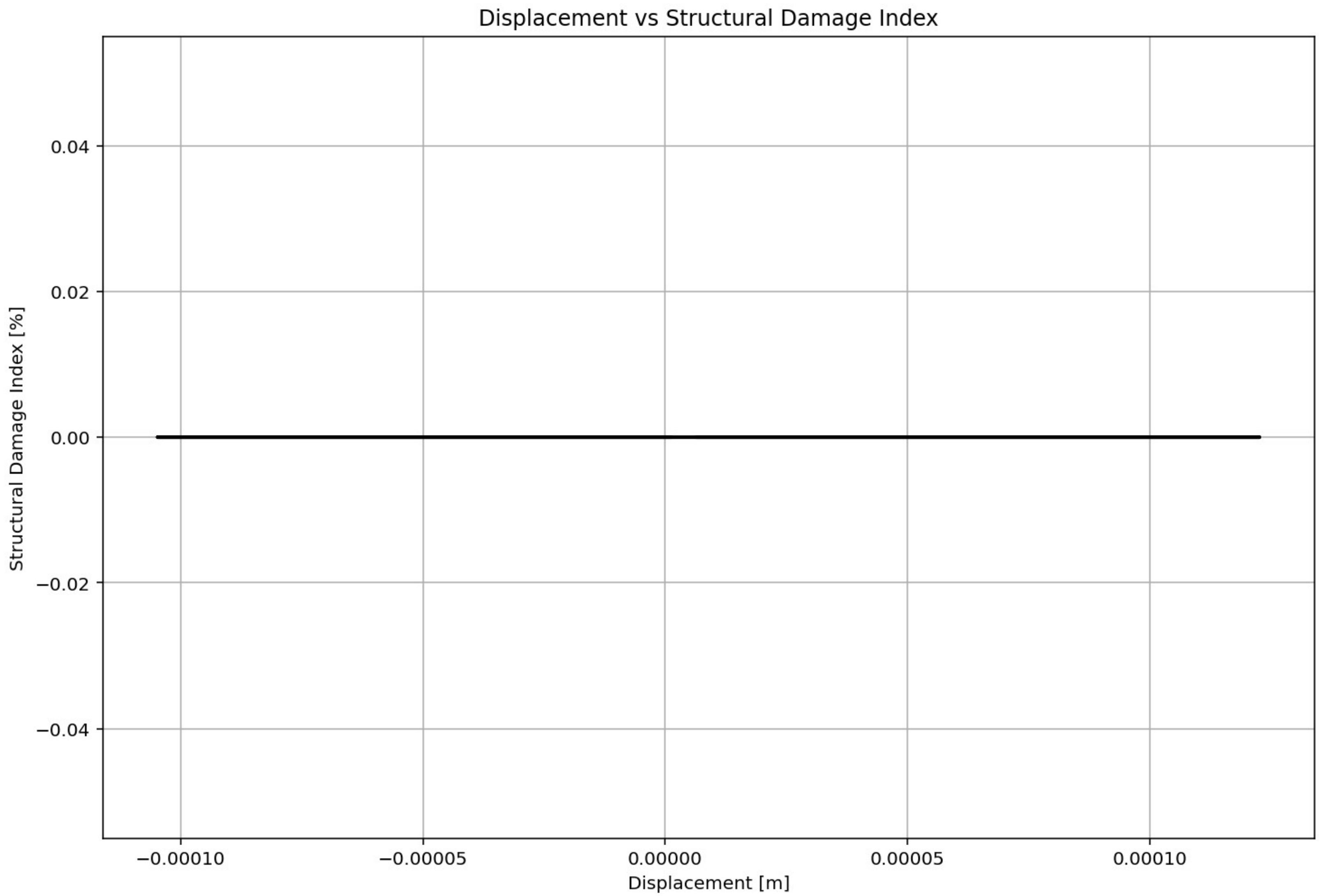

Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
4000
2000
0
−2000
−4000
−6000
Base Reaction [N]
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]


Period of Structure During Static External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
PERIOD - MAX VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
0.069
0.068
0.067
0.066
0.065
0.064
0.063
Structural Period [s]
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]

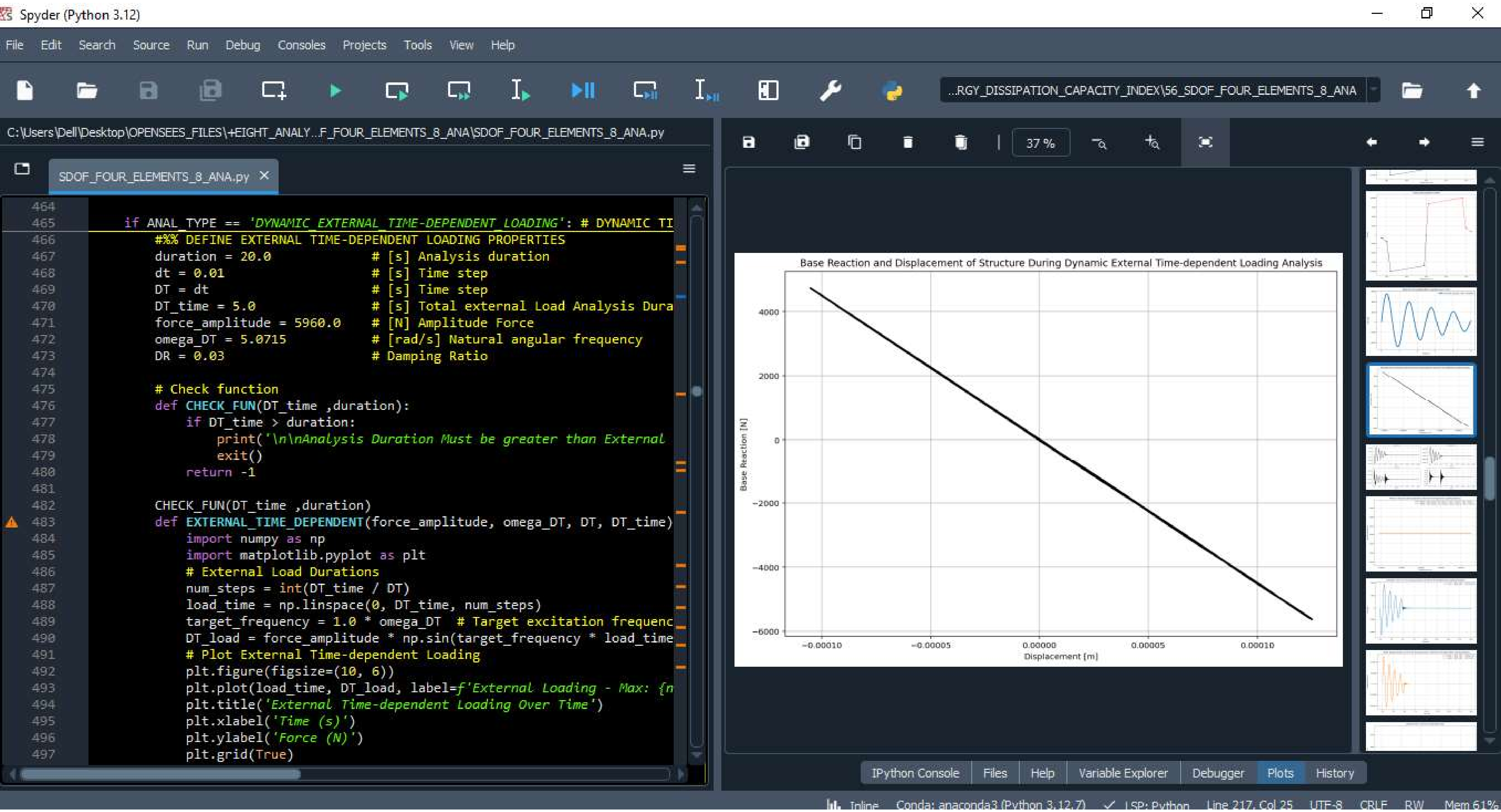

elements force vs Time During Static External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 2.2069e+03
2000
1000
0
−1000
−2000
Force [N]
0
250
500
750
1000
1250
1500
1750
2000
Time [s]

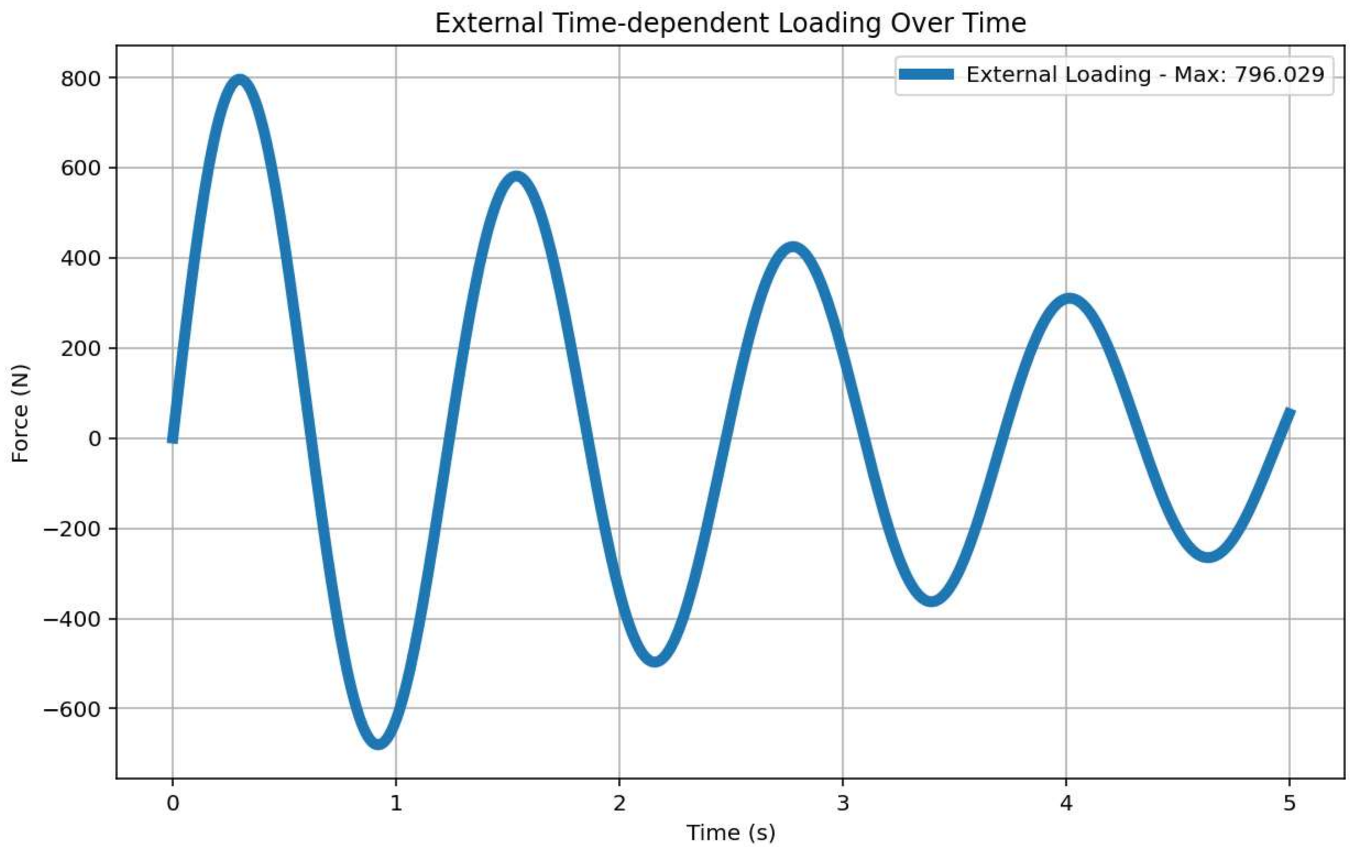


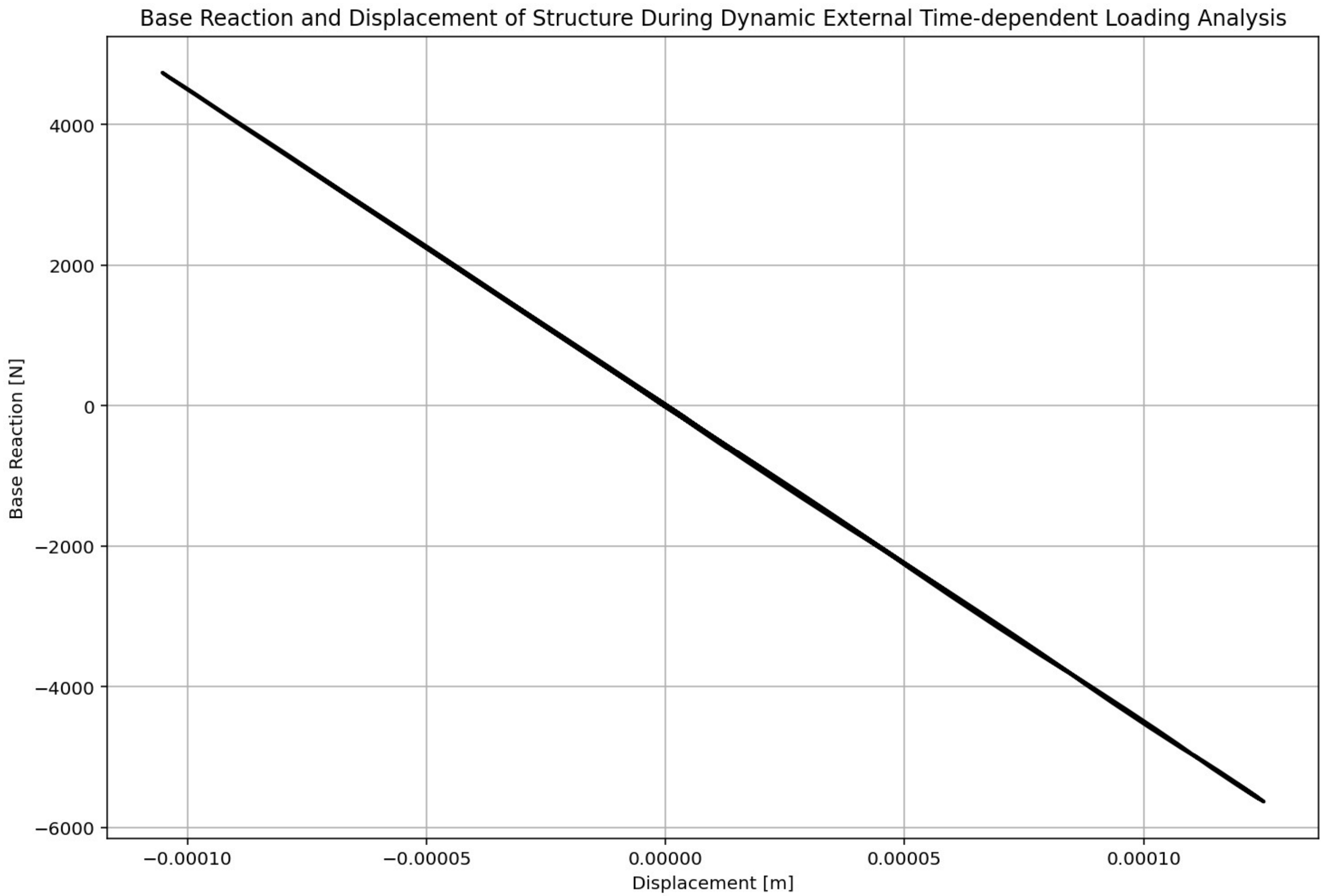
Node Displacements vs Time for During Static External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.2261e-04
Displacement [m]
Time [s]
0.00010
0.00005
0.00000
−0.00005
−0.00010
0
250
500
750
1000
1250
1500
1750
2000

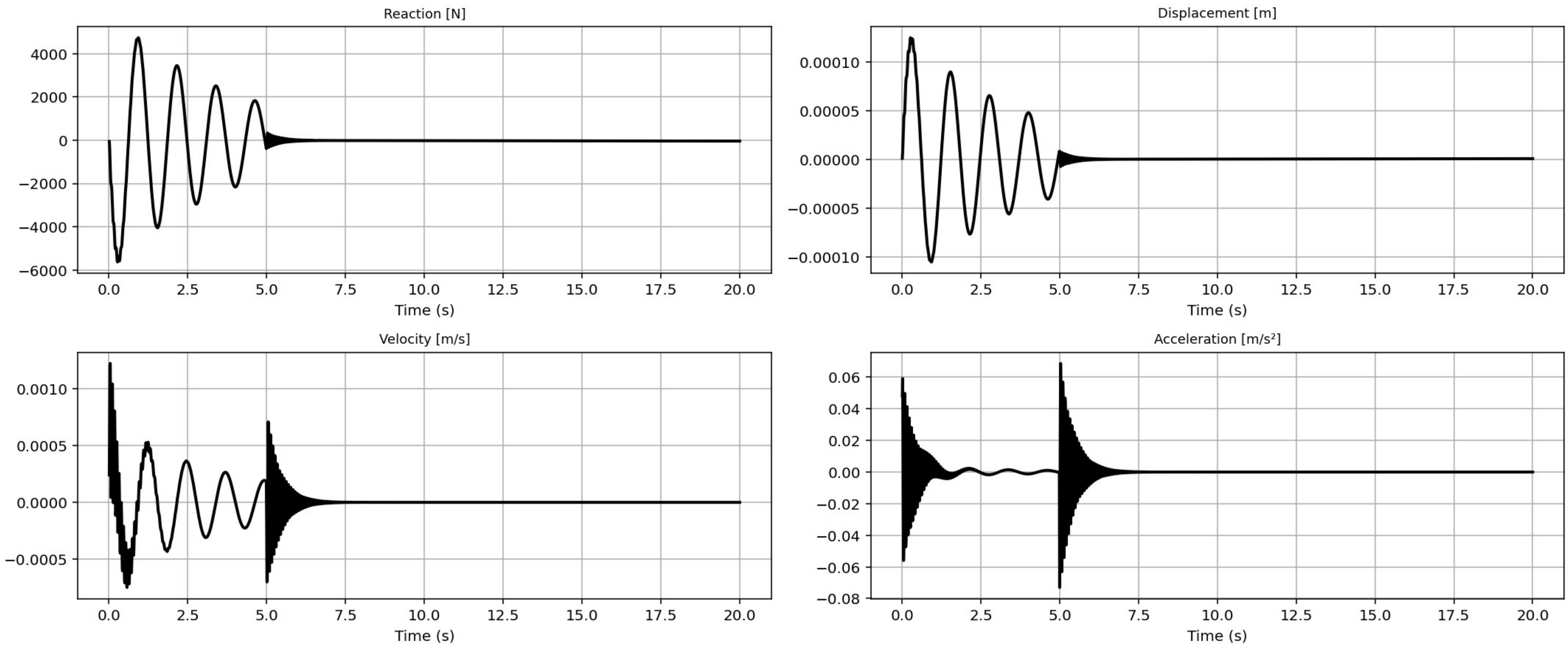

Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]

DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

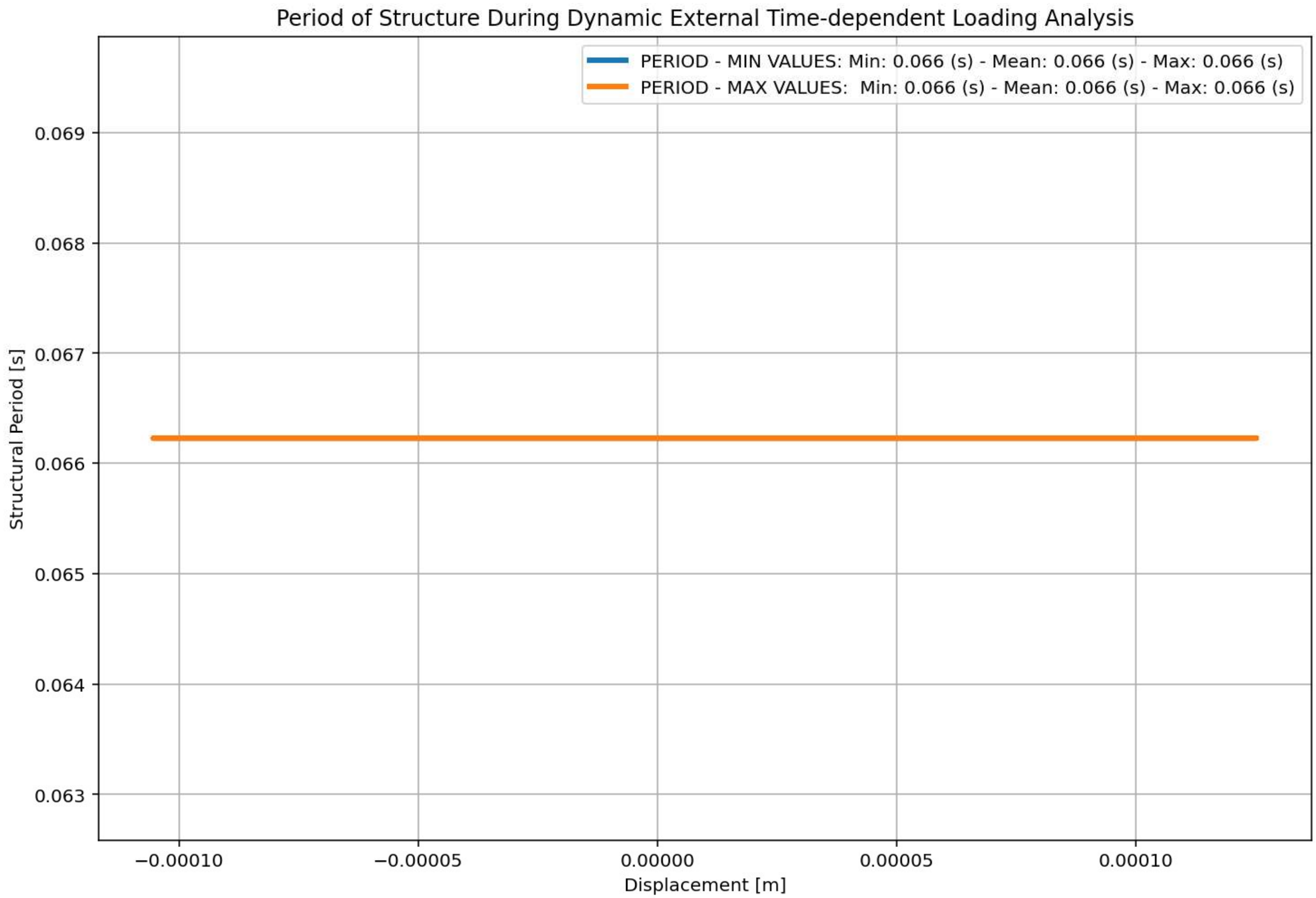

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
    if ANAL_TYPE == 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING': # DYNAMIC TI
        #%% DEFINE EXTERNAL TIME-DEPENDENT LOADING PROPERTIES
        duration = 20.0   # [s] Analysis duration
        dt = 0.01   # [s] Time step
        DT = dt   # [s] Time step
        DT_time = 5.0   # [s] Total external Load Analysis Dura
        force_amplitude = 5960.0   # [N] Amplitude Force
        omega_DT = 5.0715   # [rad/s] Natural angular frequency
        DR = 0.03   # Damping Ratio
        # Check function
        def CHECK_FUN(DT_time ,duration):
            if DT_time > duration:
                print('\n\nAnalysis Duration Must be greater than External
                exit()
            return -1
        CHECK_FUN(DT_time ,duration)
        def EXTERNAL_TIME_DEPENDENT(force_amplitude, omega_DT, DT, DT_time):
            import numpy as np
            import matplotlib.pyplot as plt
            # External Load Durations
            num_steps = int(DT_time / DT)
            load_time = np.linspace(0, DT_time, num_steps)
            target_frequency = 1.0 * omega_DT # Target excitation frequenc
            DT_load = force_amplitude * np.sin(target_frequency * load_time
            # Plot External Time-dependent Loading
            plt.figure(figsize=(10, 6))
            plt.plot(load_time, DT_load, label=f'External Loading - Max: {n
            plt.title('External Time-dependent Loading Over Time')
            plt.xlabel('Time (s)')
            plt.ylabel('Force (N)')
            plt.grid(True)
37 %
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
Displacement [m]
4000
2000
0
−2000
−4000
−6000
−0.00010
−0.00005
0.00000
0.00005
0.00010
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 217, Col 25 UTF-8 CRLF RW Mem 61%

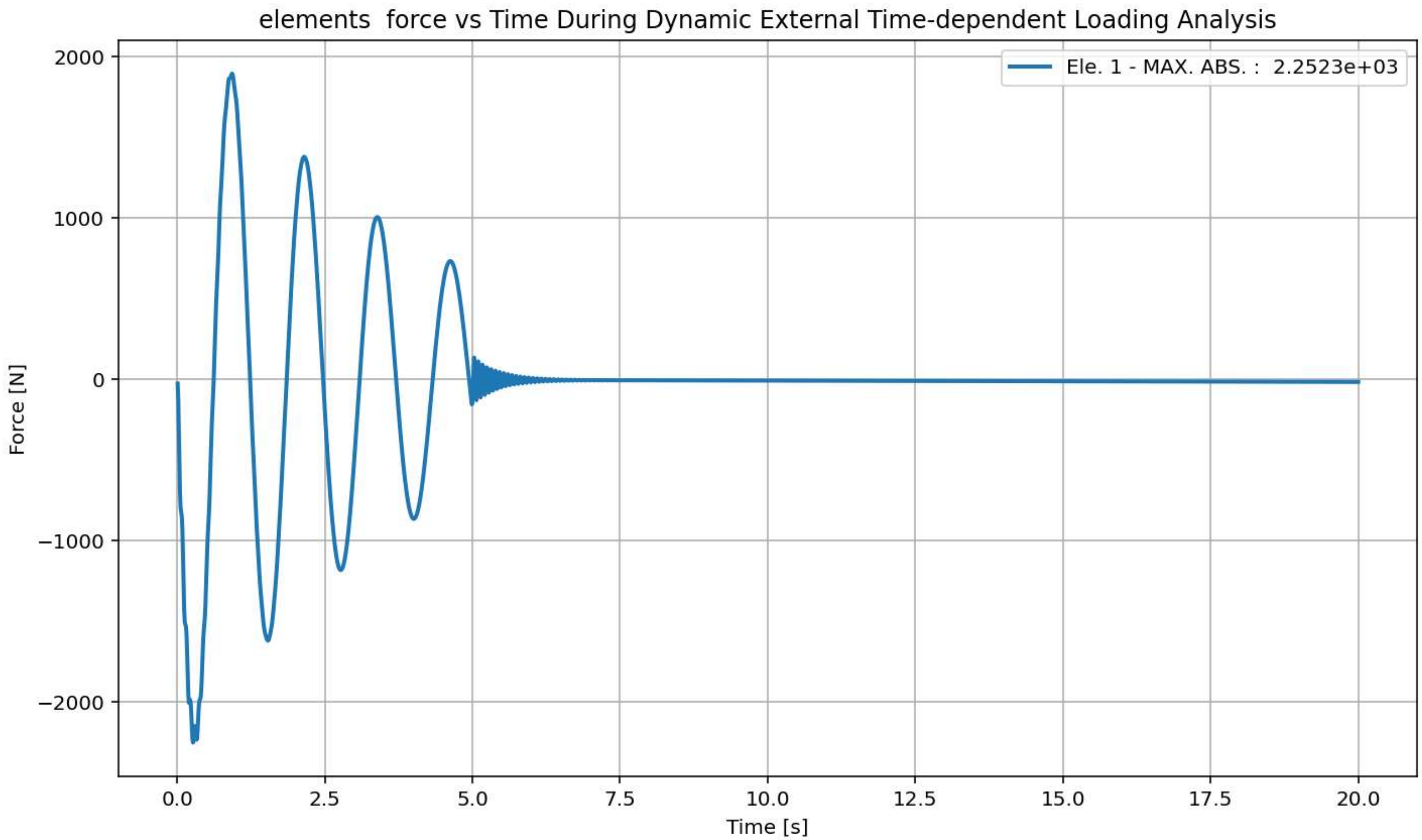

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
Time (s)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5

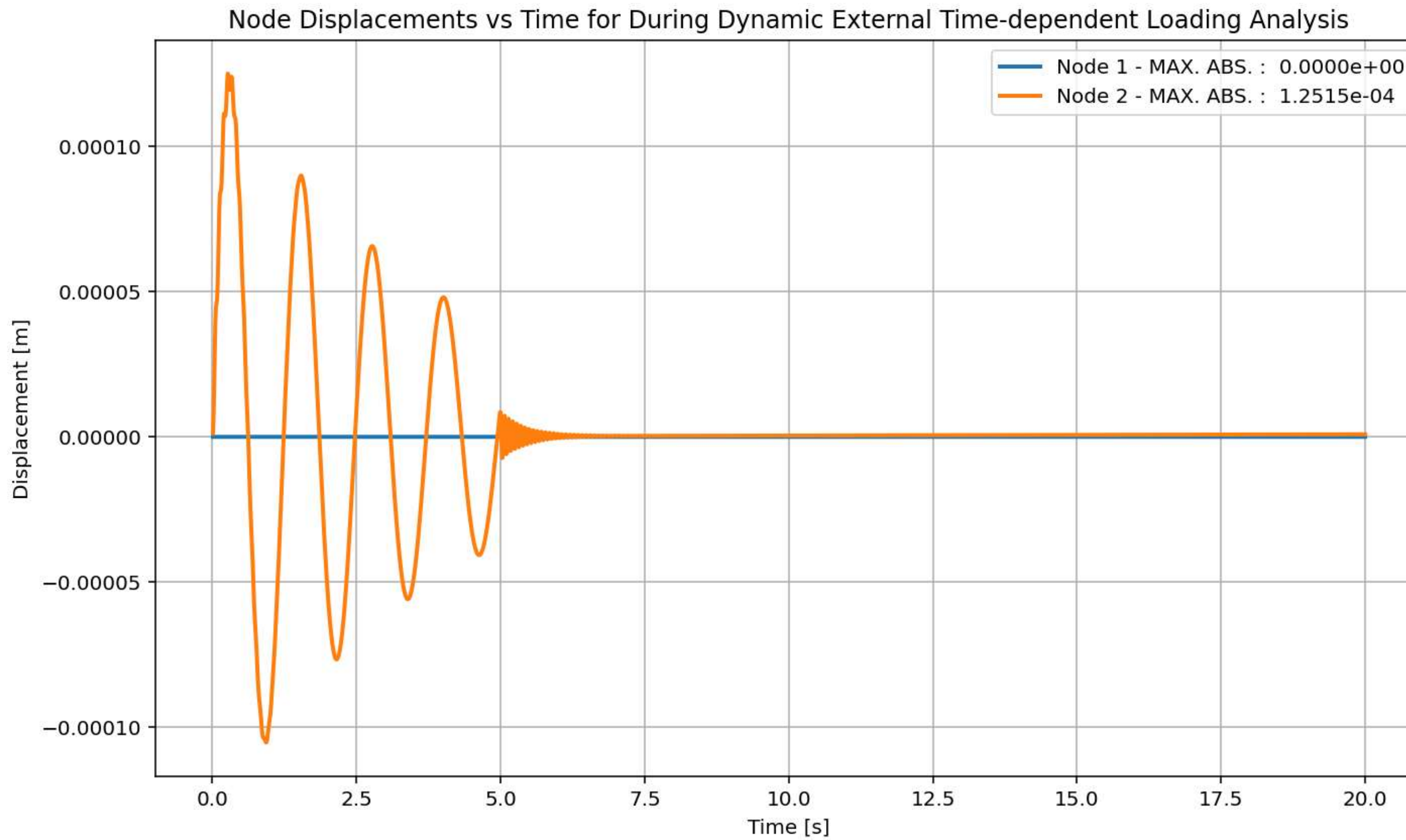

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
4000
2000
0
−2000
−4000
−6000
Base Reaction [N]
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]



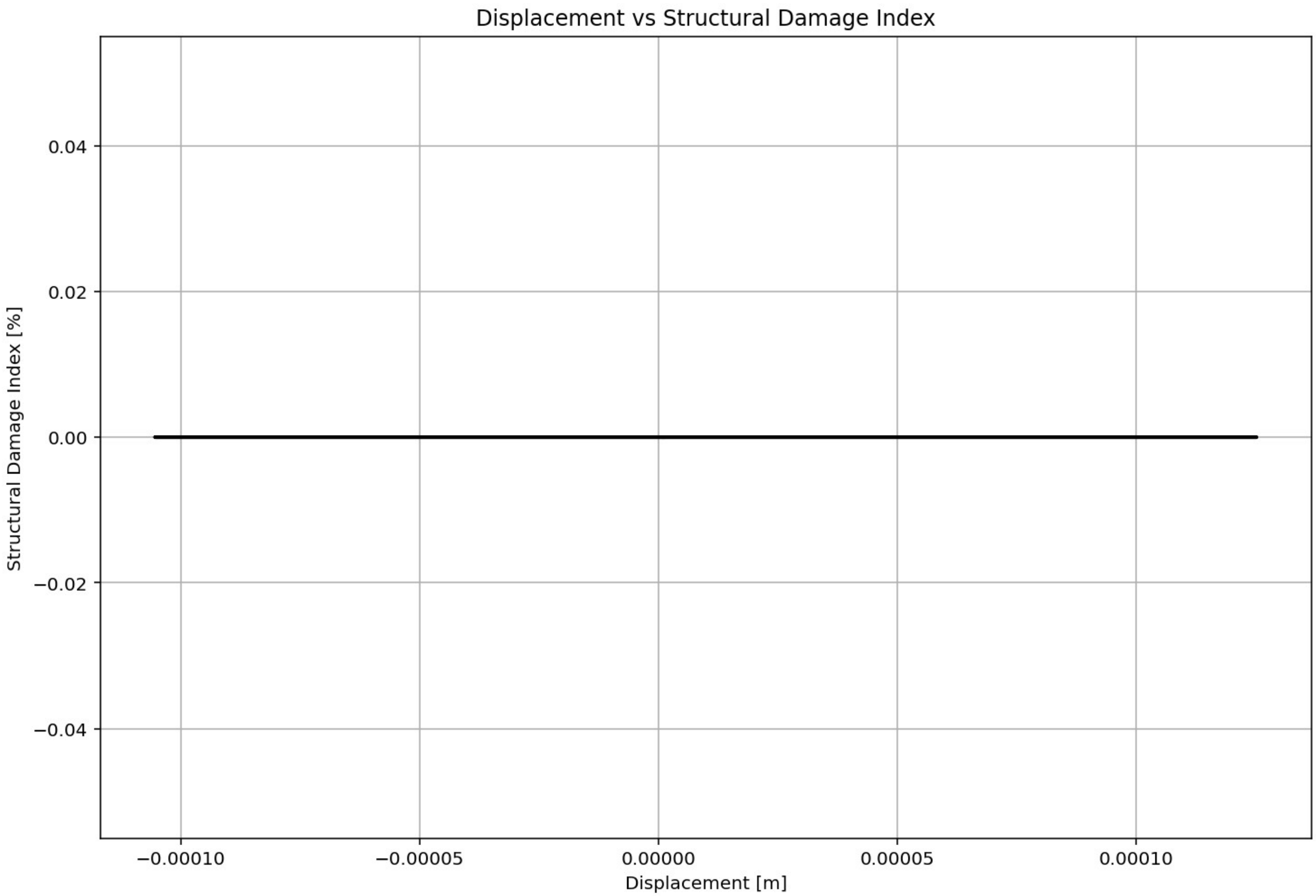
Reaction [N]
Displacement [m]
Velocity [m/s]
Acceleration [m/s²]
Time (s)


Period of Structure During Dynamic External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
PERIOD - MAX VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
0.069
0.068
0.067
0.066
0.065
0.064
0.063
Structural Period [s]
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]


elements force vs Time During Dynamic External Time-dependent Loading Analysis
Ele. 1 - MAX. ABS. : 2.2523e+03
Force [N]
Time [s]
2000
1000
0
−1000
−2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

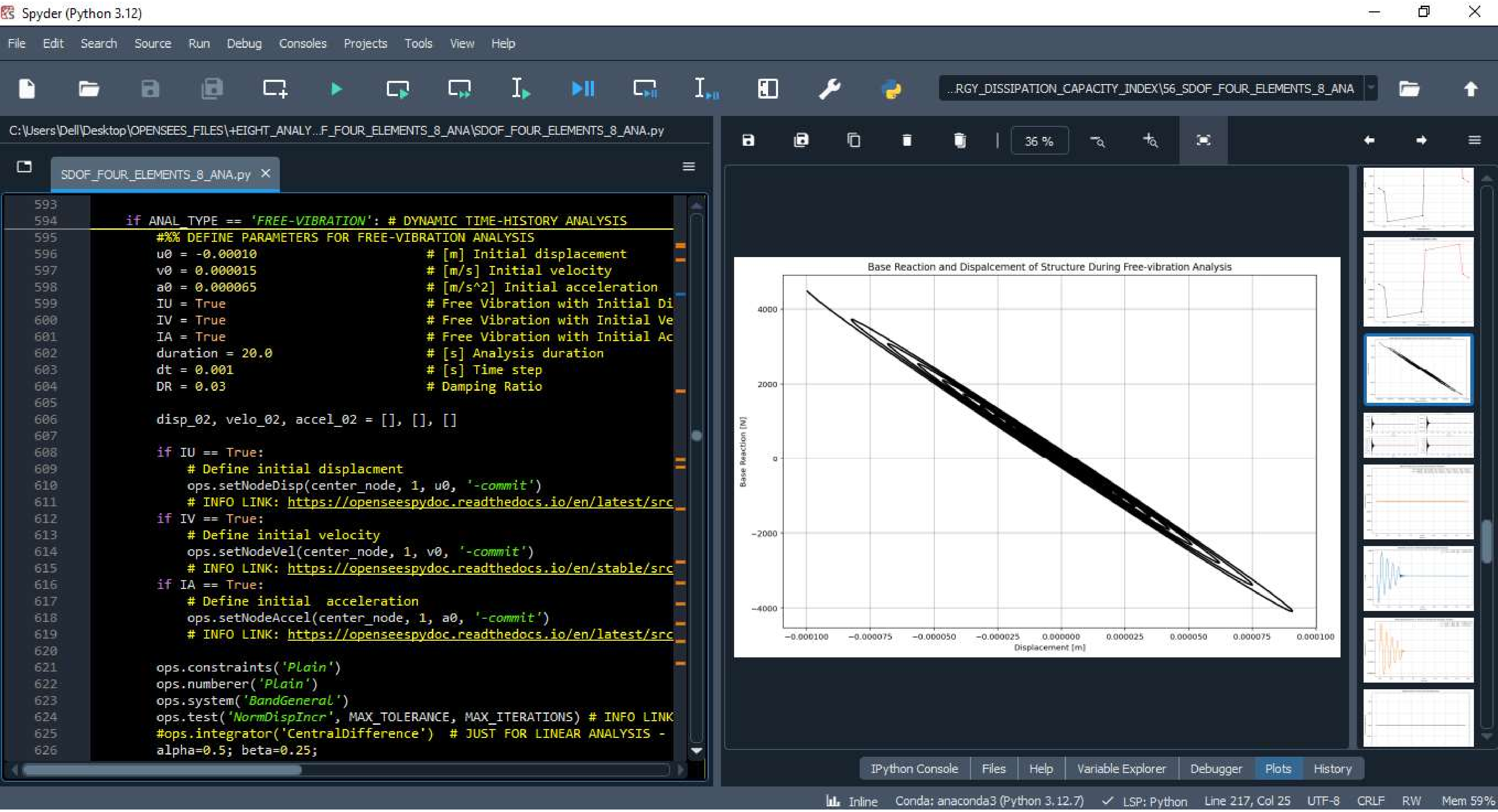


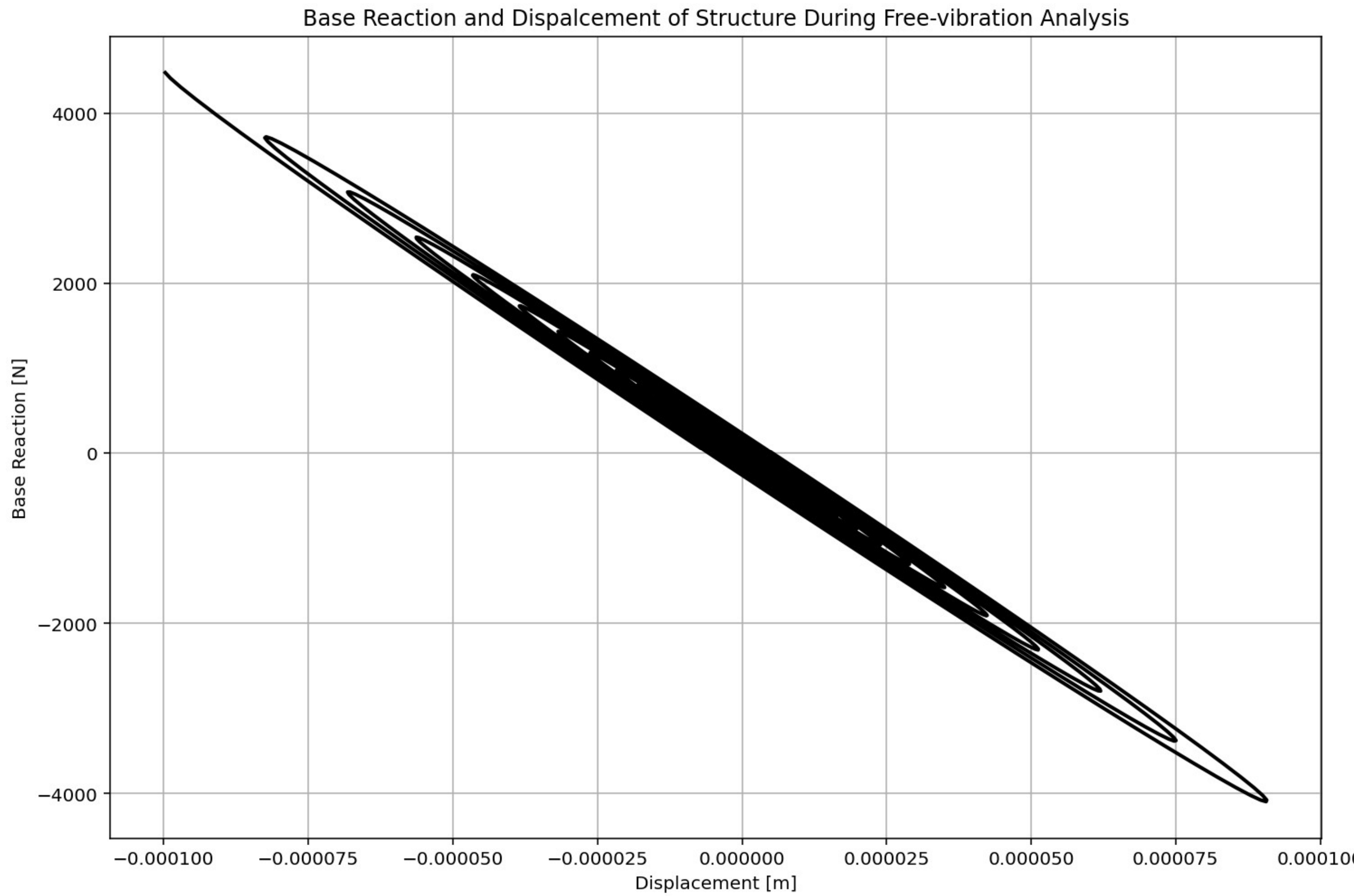
Node Displacements vs Time for During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.2515e-04
0.00010
0.00005
0.00000
−0.00005
−0.00010
Displacement [m]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

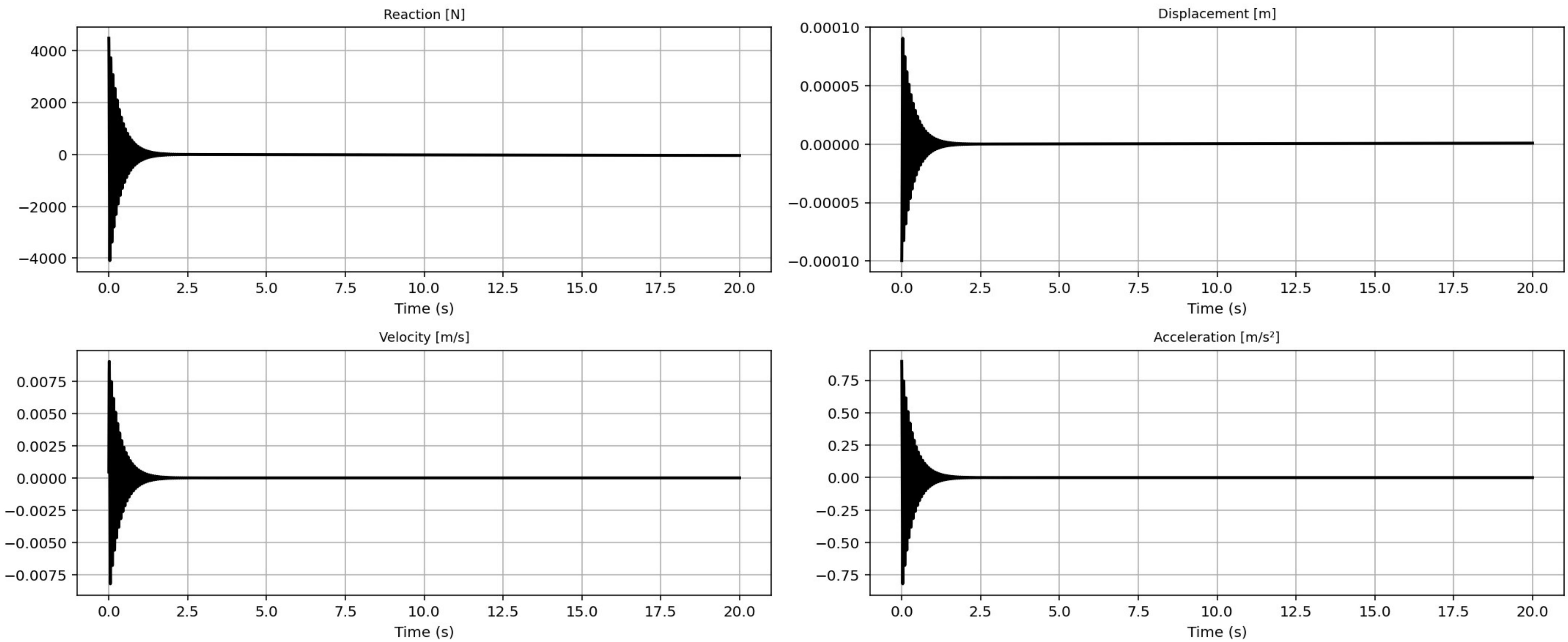


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.00010
−0.00005
0.00000
0.00005
0.00010
Displacement [m]

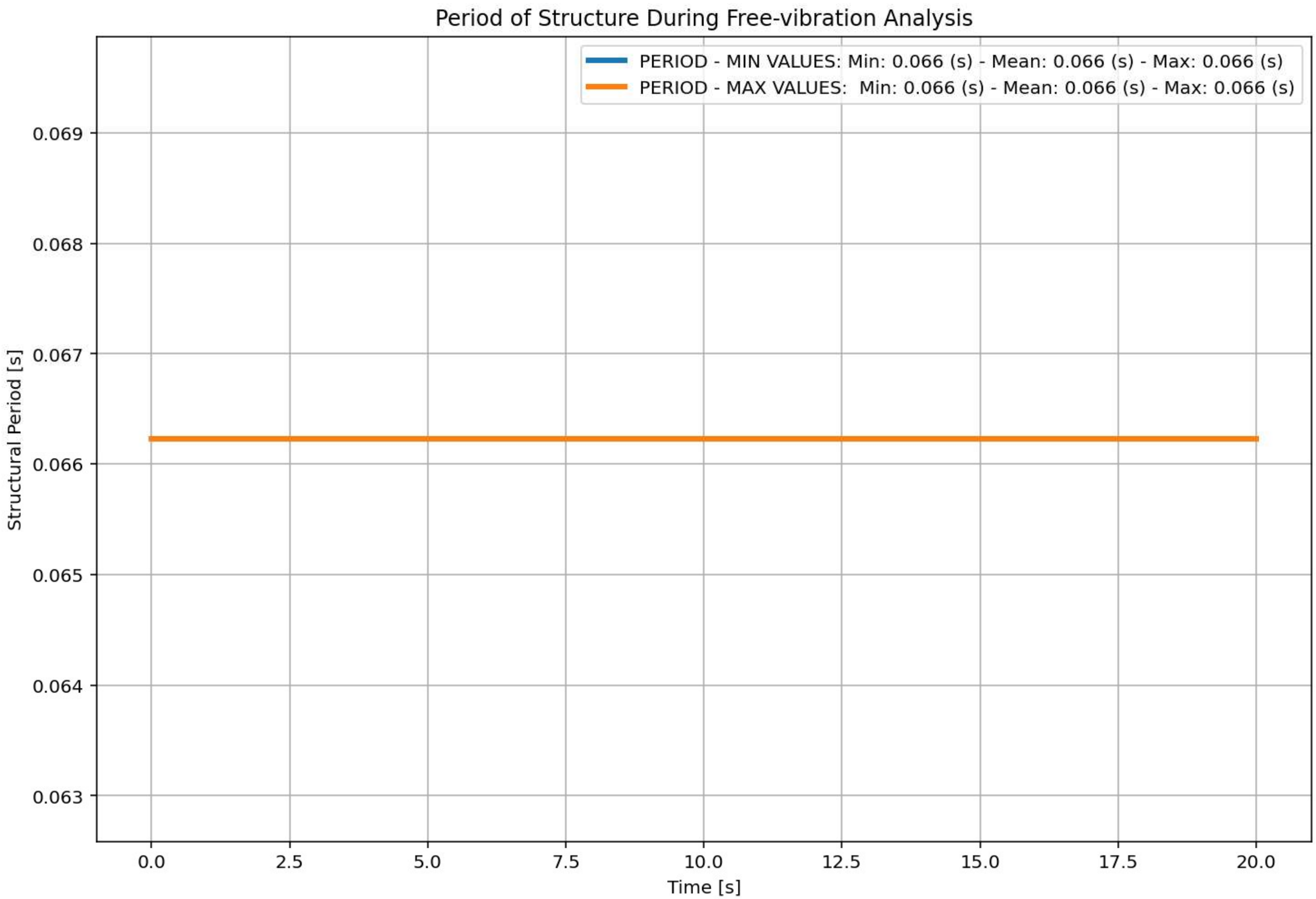
FREE-VIBRATION ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
if ANAL_TYPE == 'FREE-VIBRATION': # DYNAMIC TIME-HISTORY ANALYSIS
#%% DEFINE PARAMETERS FOR FREE-VIBRATION ANALYSIS
u0 = -0.00010 # [m] Initial displacement
v0 = 0.000015 # [m/s] Initial velocity
a0 = 0.000065 # [m/s^2] Initial acceleration
IU = True # Free Vibration with Initial Di
IV = True # Free Vibration with Initial Ve
IA = True # Free Vibration with Initial Ac
duration = 20.0 # [s] Analysis duration
dt = 0.001 # [s] Time step
DR = 0.03 # Damping Ratio
disp_02, velo_02, accel_02 = [], [], []
if IU == True:
# Define initial displacment
ops.setNodeDisp(center_node, 1, u0, '-commit')
# INFO LINK: https://openseespydoc.readthedocs.io/en/latest/src
if IV == True:
# Define initial velocity
ops.setNodeVel(center_node, 1, v0, '-commit')
# INFO LINK: https://openseespydoc.readthedocs.io/en/stable/src
if IA == True:
# Define initial acceleration
ops.setNodeAccel(center_node, 1, a0, '-commit')
# INFO LINK: https://openseespydoc.readthedocs.io/en/latest/src
ops.constraints('Plain')
ops.numberer('Plain')
ops.system('BandGeneral')
ops.test('NormDispIncr', MAX_TOLERANCE, MAX_ITERATIONS) # INFO LINK
#ops.integrator('CentralDifference') # JUST FOR LINEAR ANALYSIS -
alpha=0.5; beta=0.25;
36 %
Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
Displacement [m]
4000
2000
0
−2000
−4000
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 217, Col 25 UTF-8 CRLF RW Mem 59%

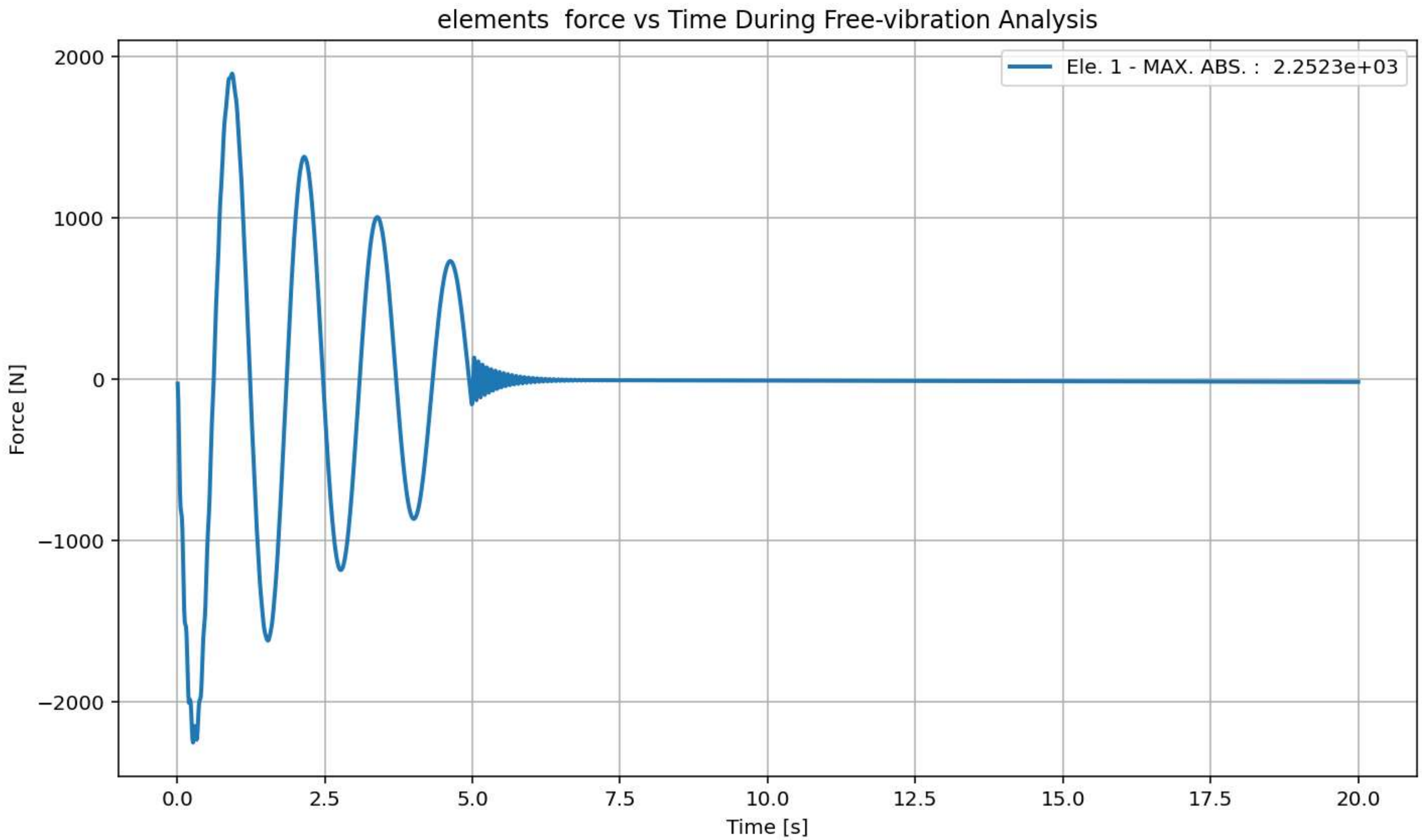

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
4000
2000
0
−2000
−4000
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
Displacement [m]



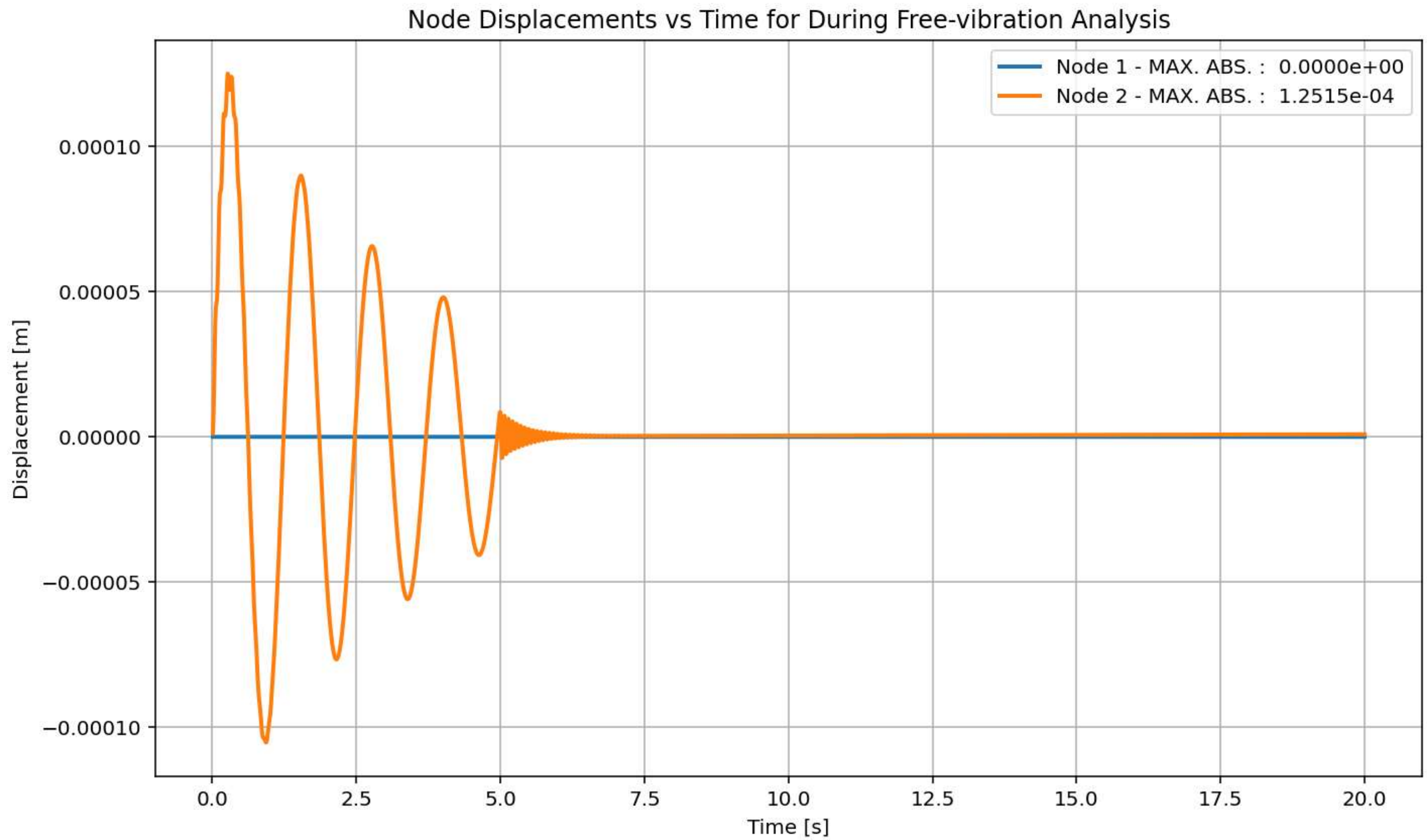
Reaction [N]
4000
2000
0
−2000
−4000
Displacement [m]
0.00010
0.00005
0.00000
−0.00005
−0.00010
Velocity [m/s]
0.0075
0.0050
0.0025
0.0000
−0.0025
−0.0050
−0.0075
Acceleration [m/s²]
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)



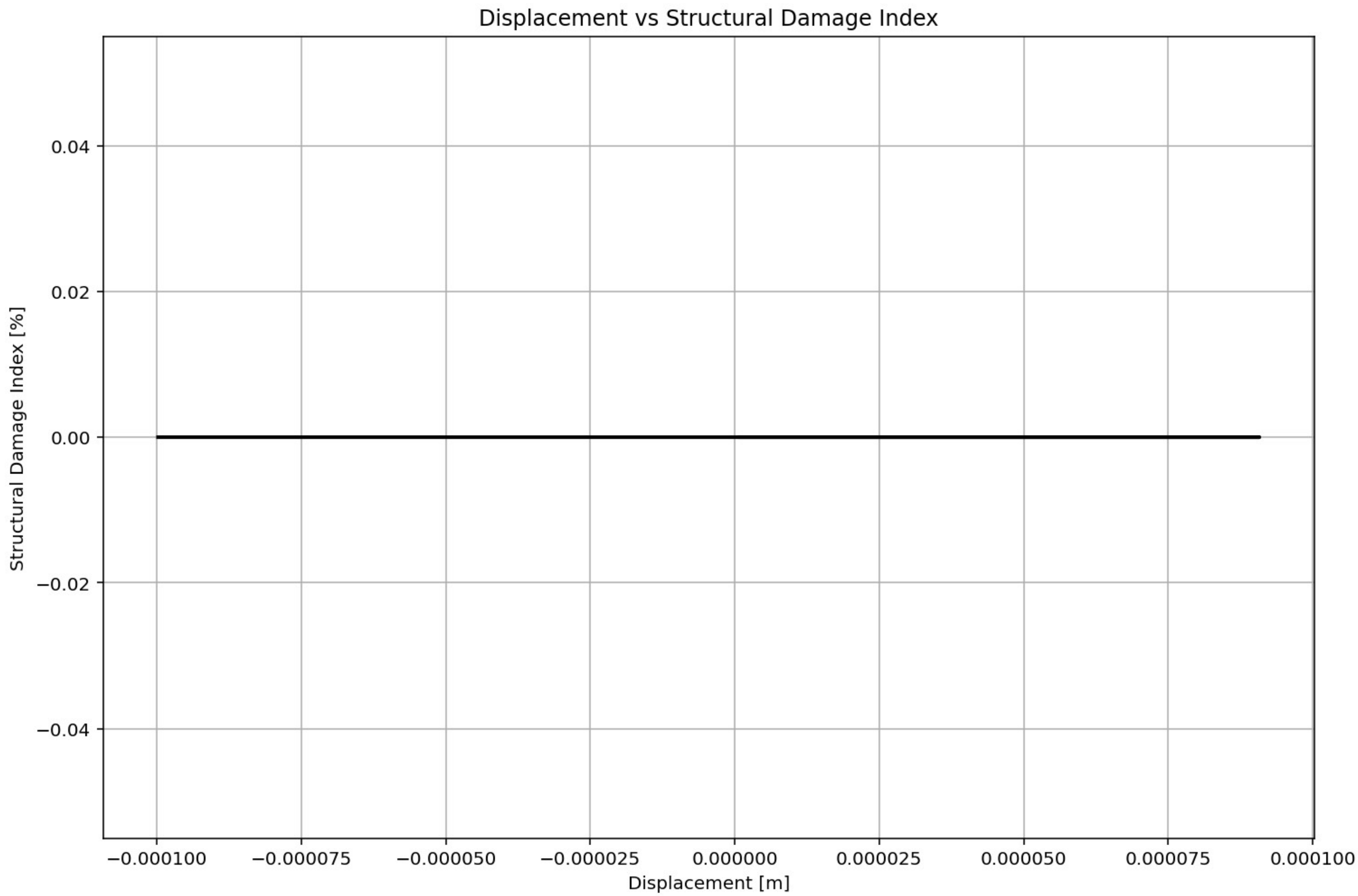
Period of Structure During Free-vibration Analysis
PERIOD - MIN VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
PERIOD - MAX VALUES: Min: 0.066 (s) - Mean: 0.066 (s) - Max: 0.066 (s)
Structural Period [s]
0.069
0.068
0.067
0.066
0.065
0.064
0.063
Time [s]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

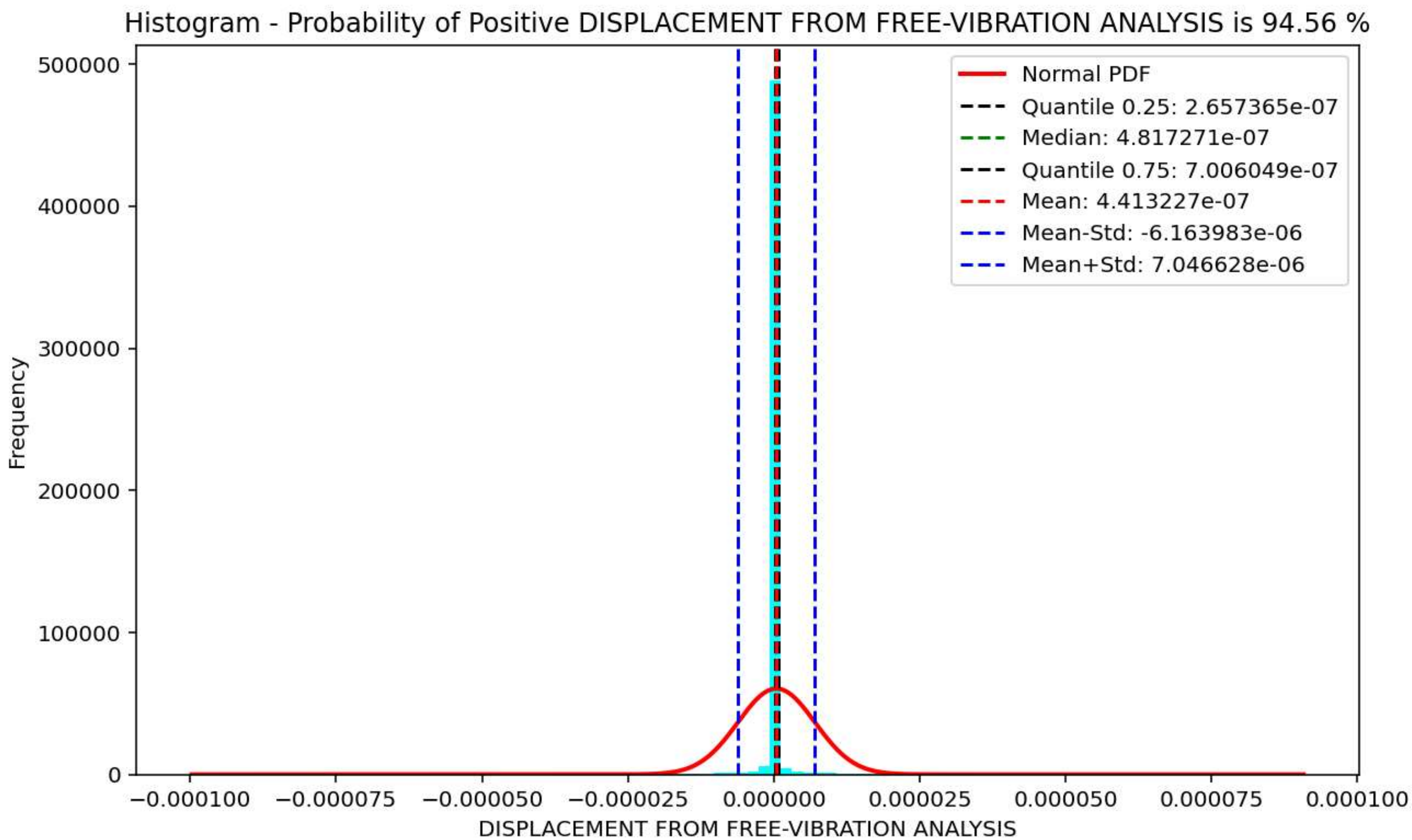


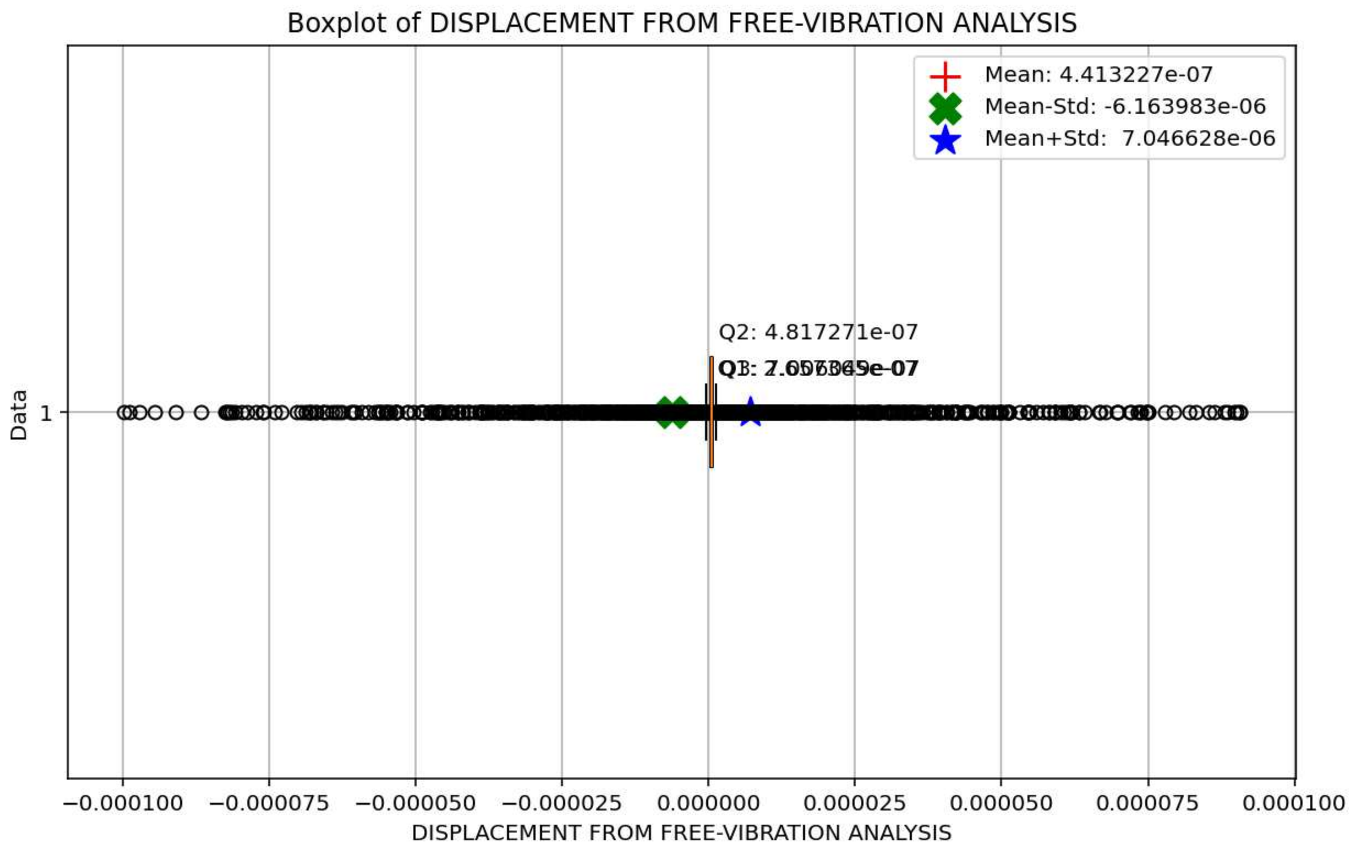
elements force vs Time During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 2.2523e+03
Force [N]
Time [s]
2000
1000
0
−1000
−2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



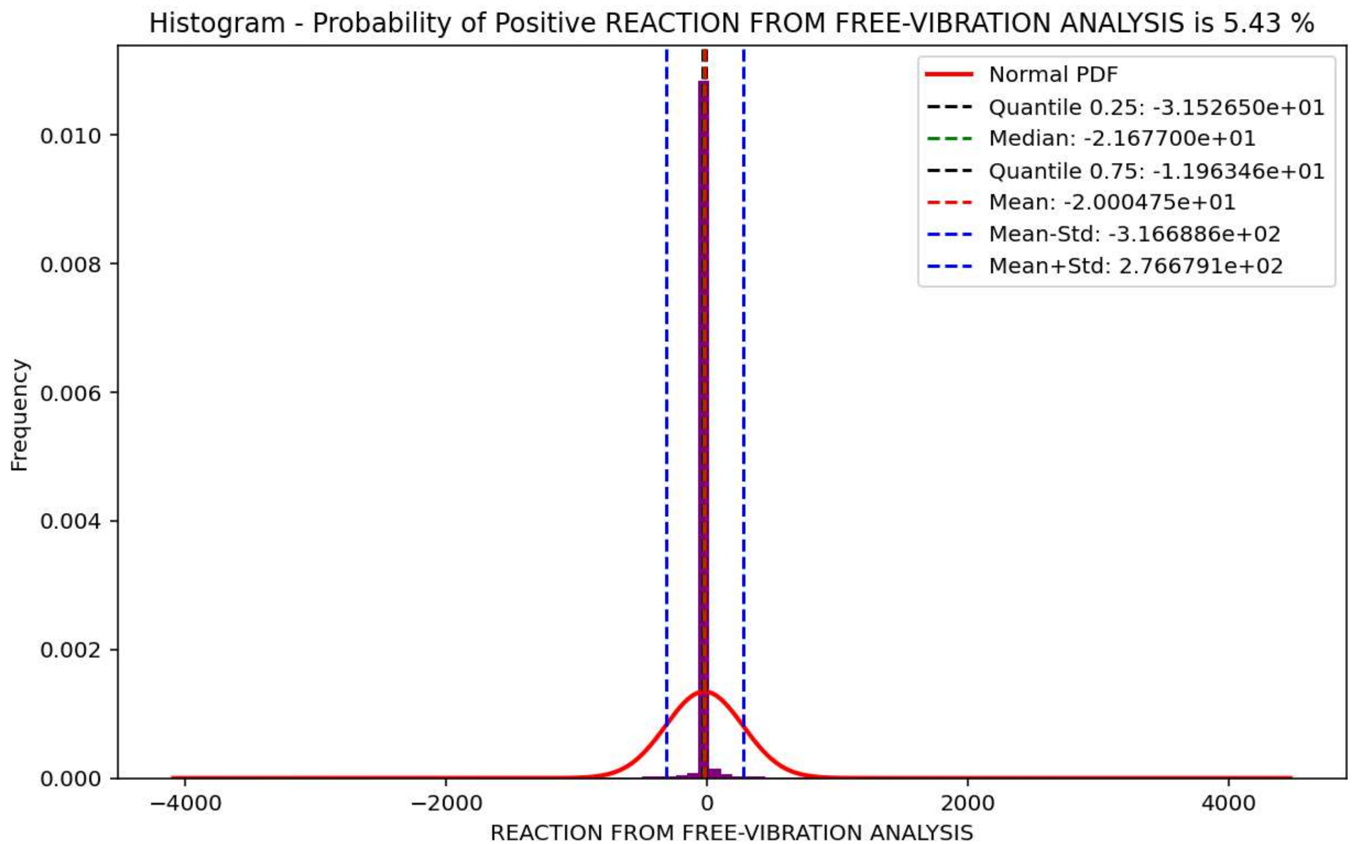
Node Displacements vs Time for During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.2515e-04
0.00010
0.00005
0.00000
−0.00005
−0.00010
Displacement [m]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]


Displacement vs Structural Damage Index
Structural Damage Index [%]
0.04
0.02
0.00
−0.02
−0.04
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
Displacement [m]

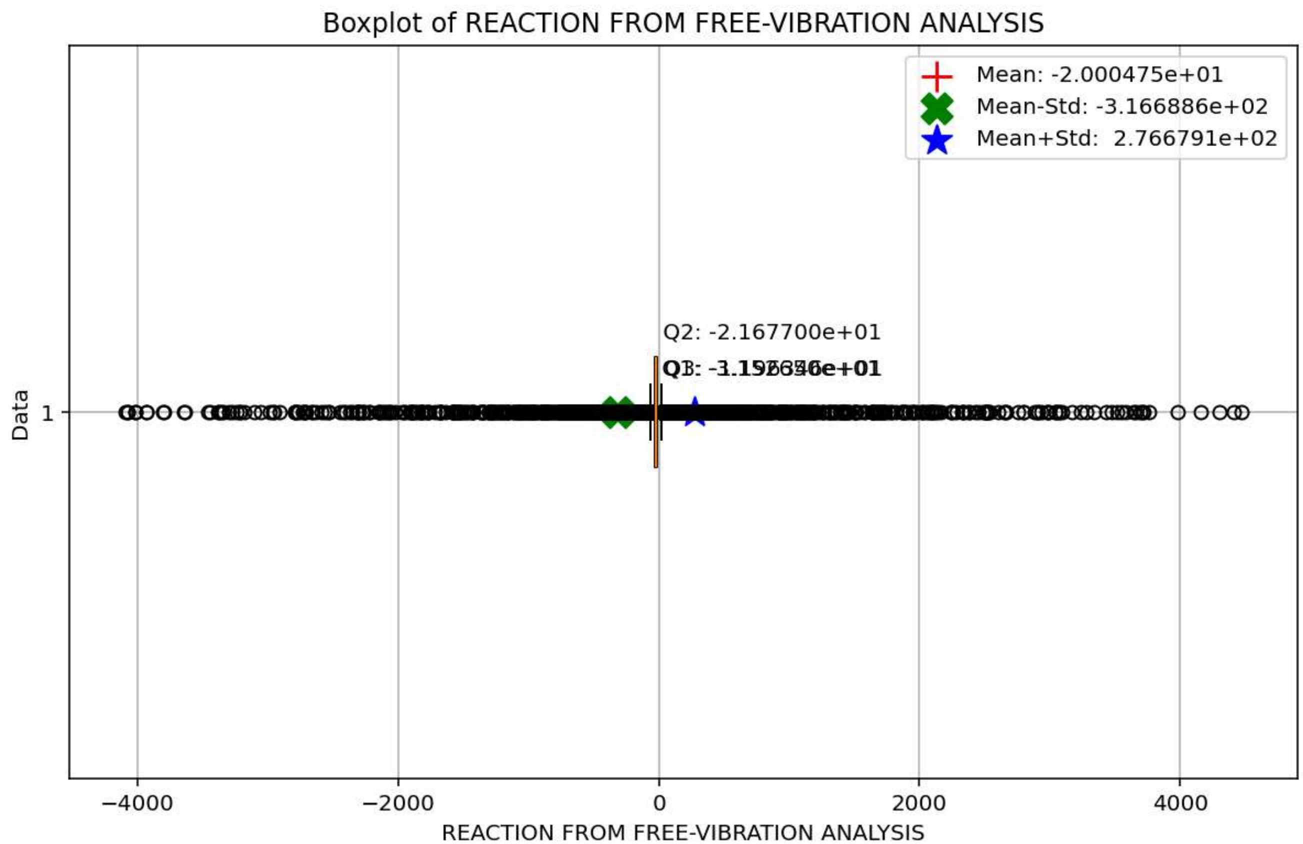

Histogram - Probability of Positive DISPLACEMENT FROM FREE-VIBRATION ANALYSIS is 94.56 %
Normal PDF
Quantile 0.25: 2.657365e-07
Median: 4.817271e-07
Quantile 0.75: 7.006049e-07
Mean: 4.413227e-07
Mean-Std: -6.163983e-06
Mean+Std: 7.046628e-06
Frequency
500000
400000
300000
200000
100000
0
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

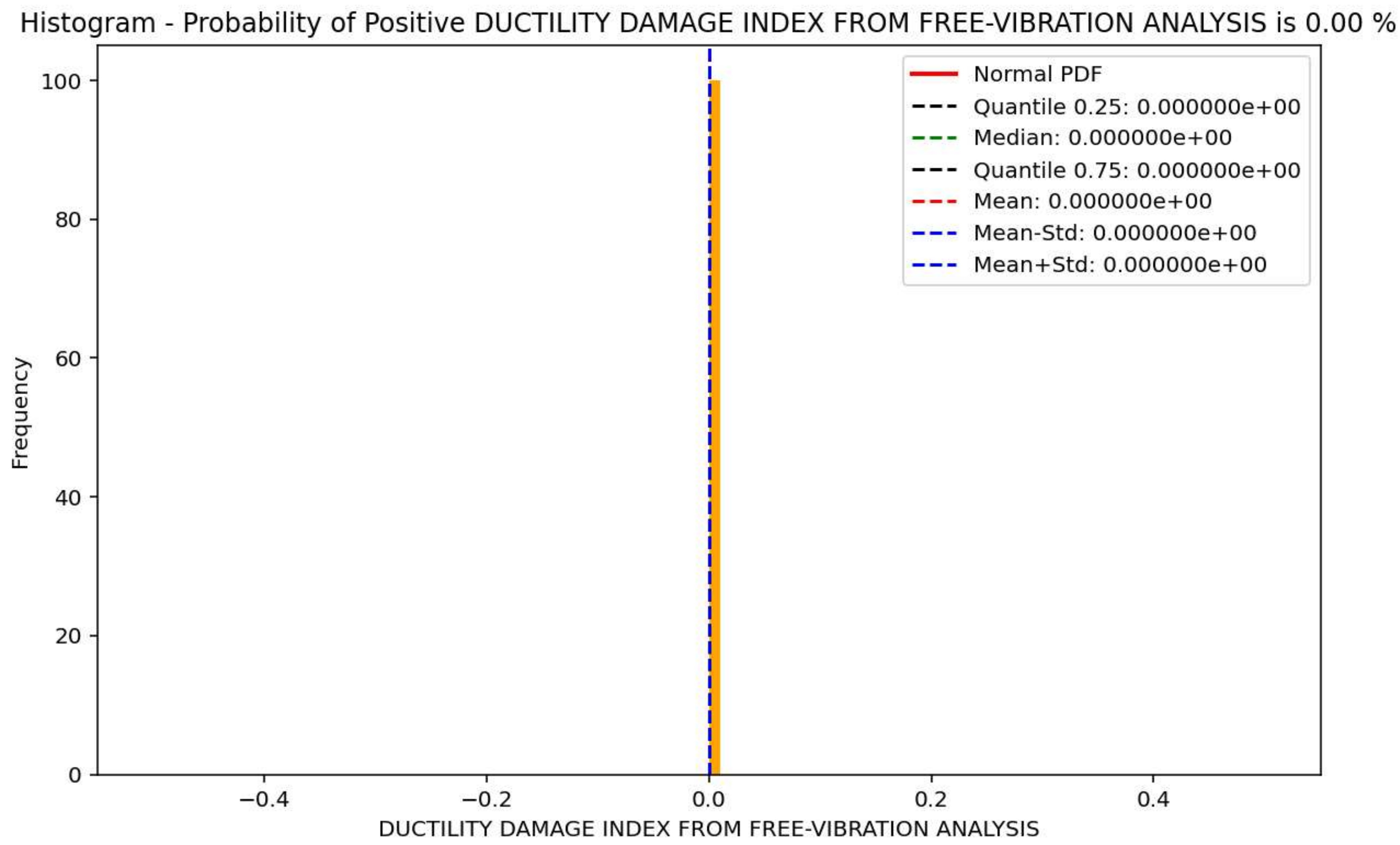


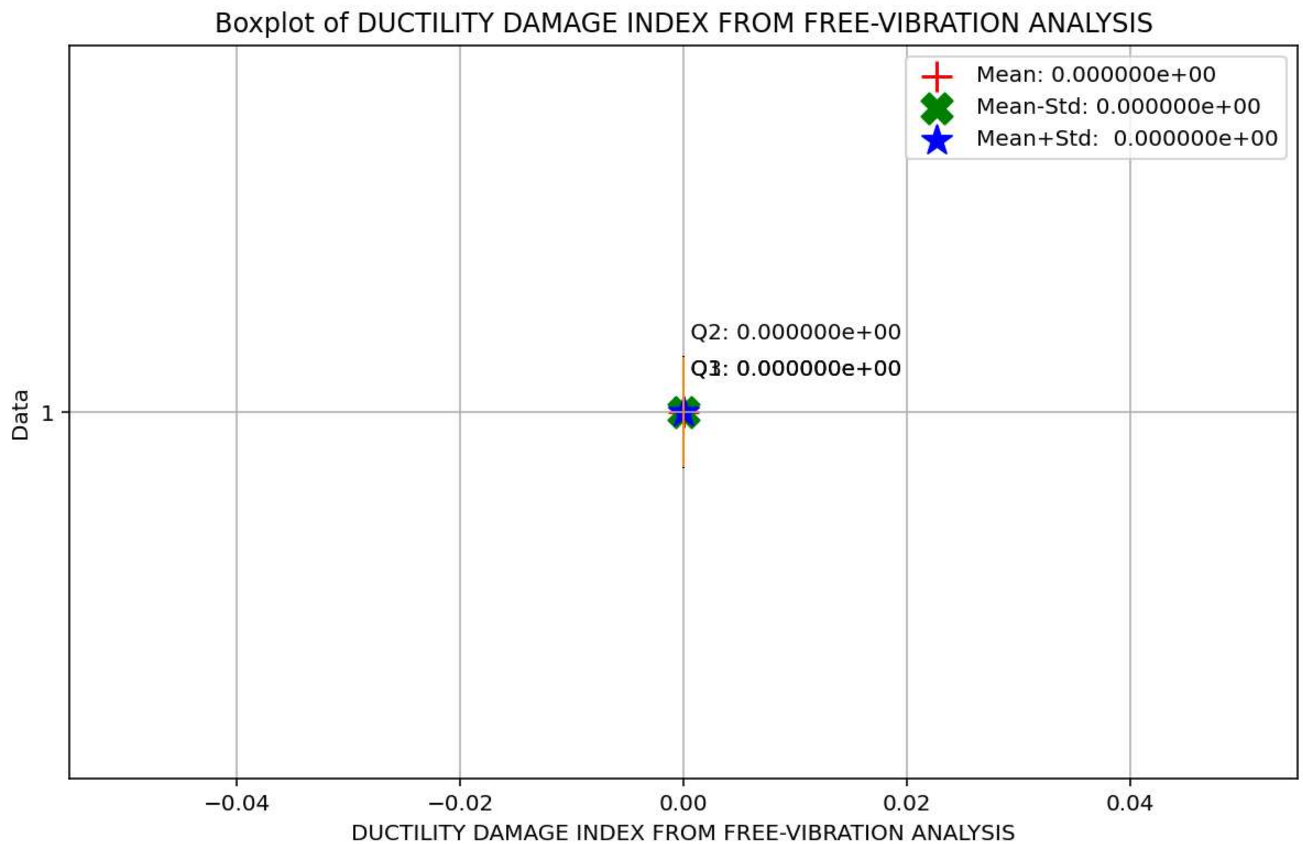
Boxplot of DISPLACEMENT FROM FREE-VIBRATION ANALYSIS
Mean: 4.413227e-07
Mean-Std: -6.163983e-06
Mean+Std: 7.046628e-06
Q2: 4.817271e-07
Data
1
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
DISPLACEMENT FROM FREE-VIBRATION ANALYSIS

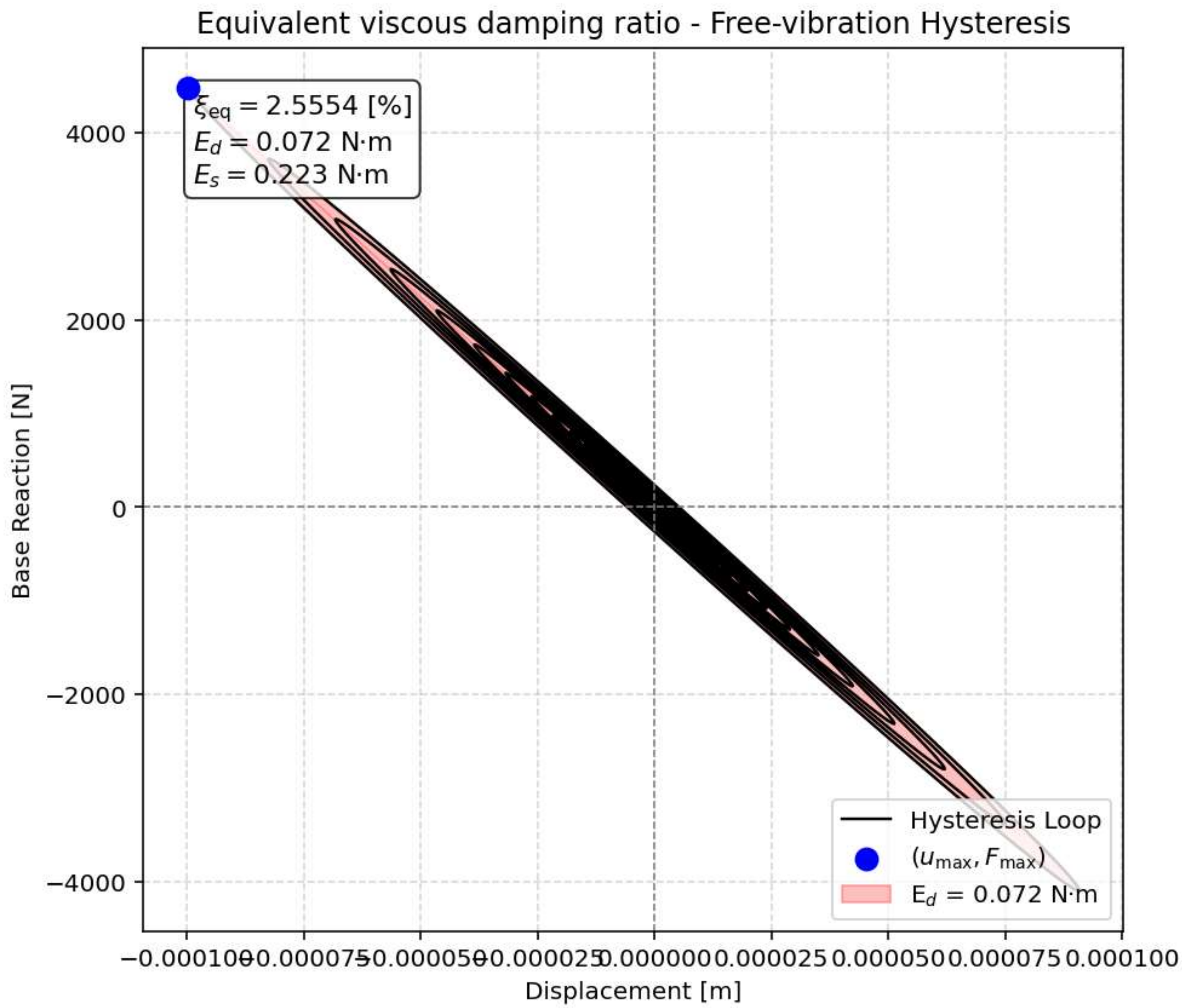


Histogram - Probability of Positive REACTION FROM FREE-VIBRATION ANALYSIS is 5.43 %
Normal PDF
Quantile 0.25: -3.152650e+01
Median: -2.167700e+01
Quantile 0.75: -1.196346e+01
Mean: -2.000475e+01
Mean-Std: -3.166886e+02
Mean+Std: 2.766791e+02
Frequency
0.010
0.008
0.006
0.004
0.002
0.000
−4000
−2000
0
2000
4000
REACTION FROM FREE-VIBRATION ANALYSIS


Boxplot of REACTION FROM FREE-VIBRATION ANALYSIS
Mean: -2.000475e+01
Mean-Std: -3.166886e+02
Mean+Std: 2.766791e+02
Q2: -2.167700e+01
Data
1
−4000
−2000
0
2000
4000
REACTION FROM FREE-VIBRATION ANALYSIS

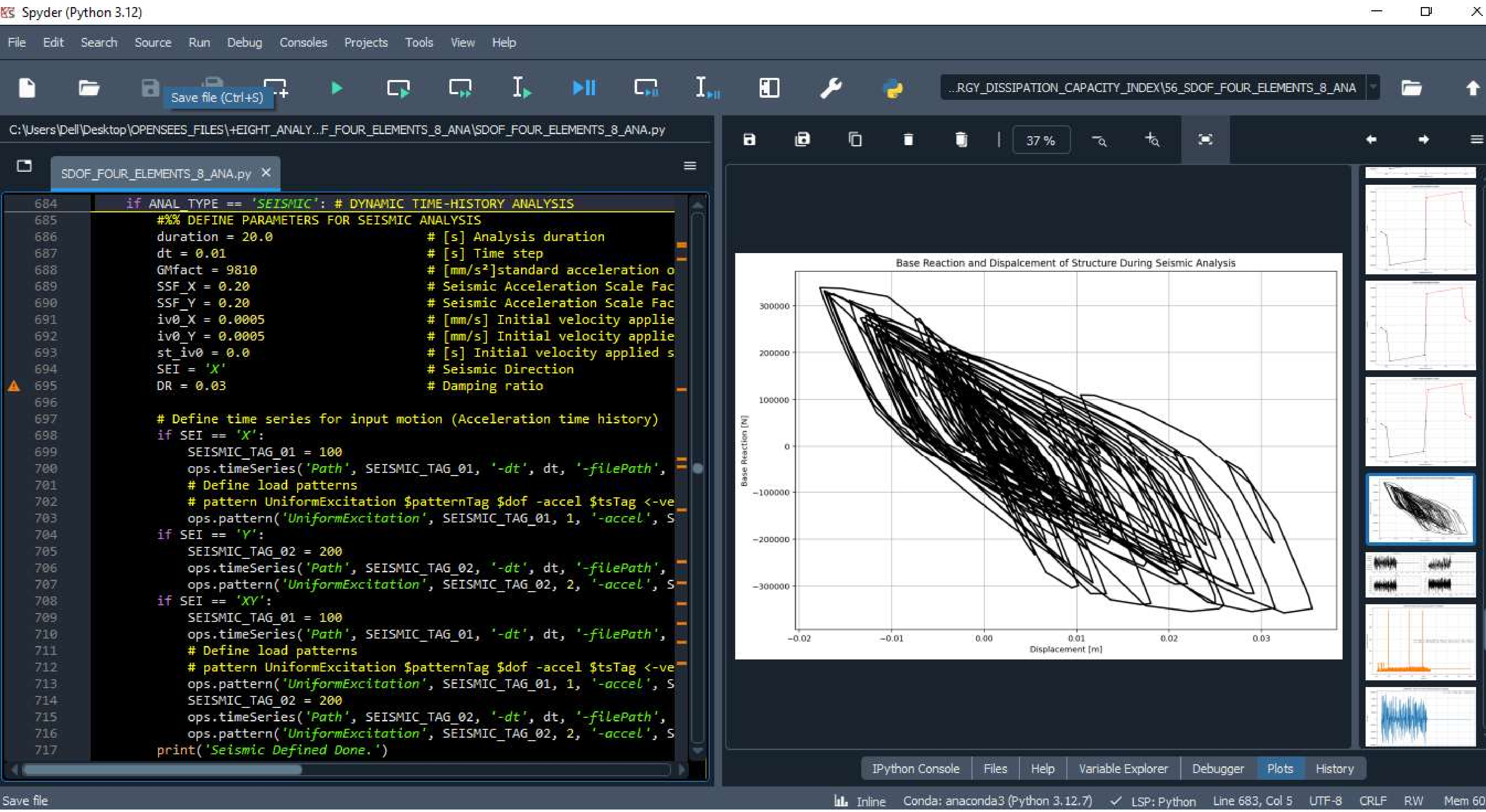

Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS is 0.00 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Frequency
100
80
60
40
20
0
−0.4
−0.2
0.0
0.2
0.4
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS


Boxplot of DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS
Mean: 0.000000e+00
Mean-Std: 0.000000e+00
Mean+Std: 0.000000e+00
Q2: 0.000000e+00
Q3: 0.000000e+00
Q1: 0.000000e+00
Data
1
−0.04
−0.02
0.00
0.02
0.04
DUCTILITY DAMAGE INDEX FROM FREE-VIBRATION ANALYSIS

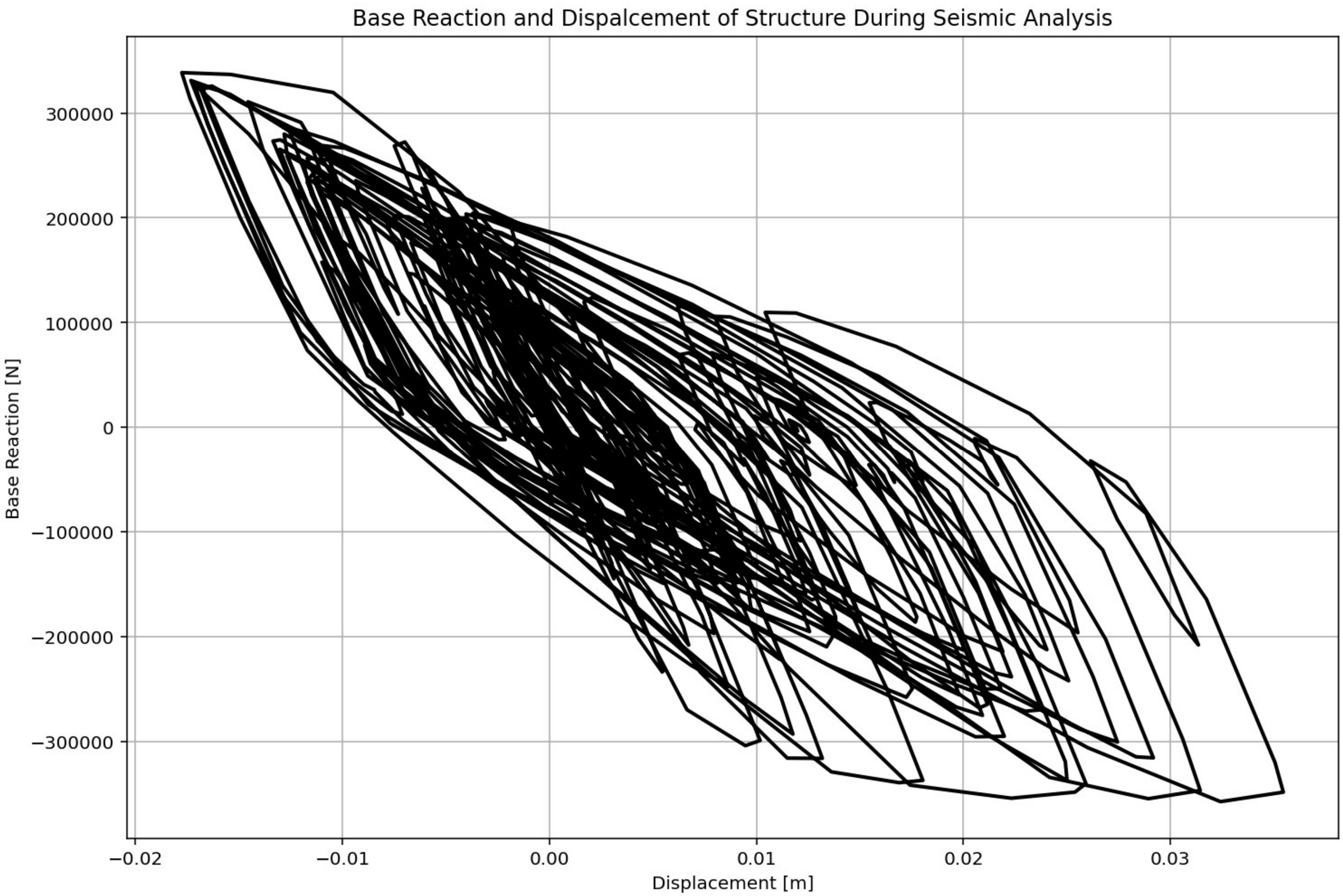


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 2.5554$ [%]
$E_d = 0.072$ N·m
$E_s = 0.223$ N·m
4000
2000
0
−2000
−4000
Base Reaction [N]
−0.000100
−0.000075
−0.000050
−0.000025
0.000000
0.000025
0.000050
0.000075
0.000100
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.072 N·m

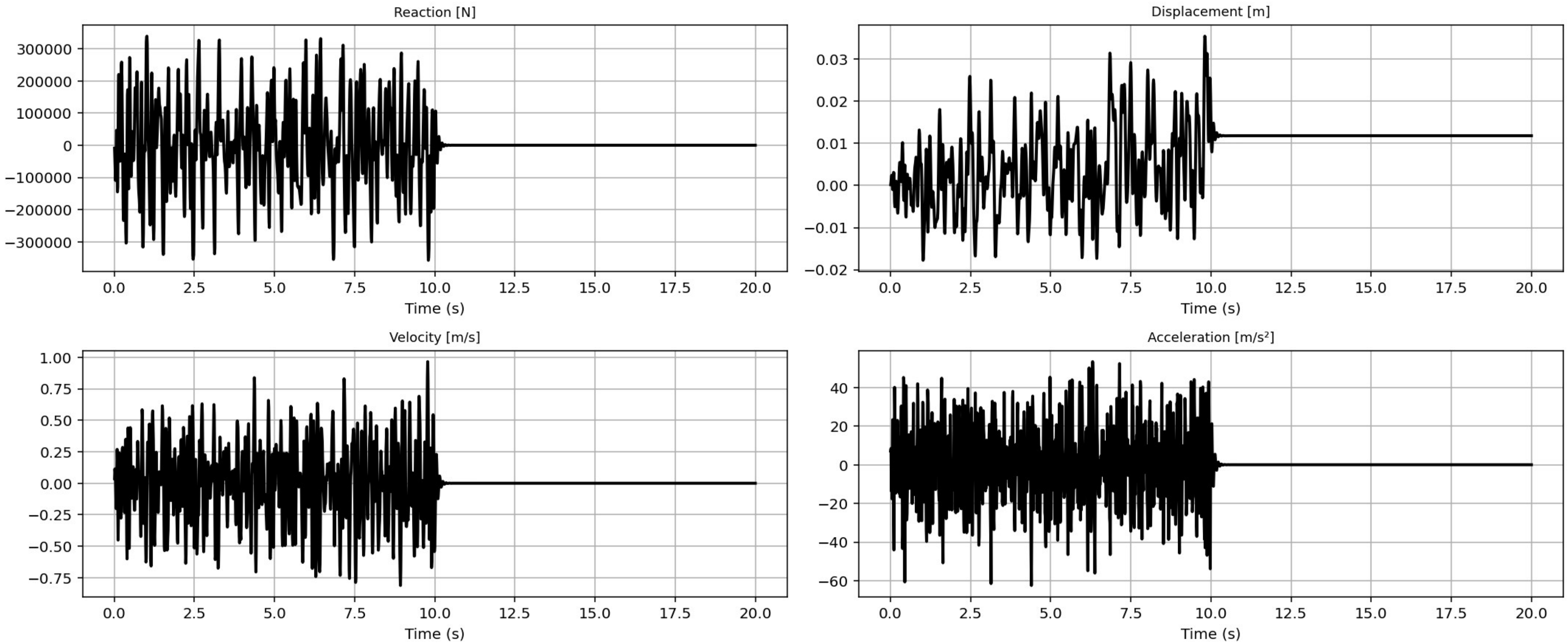
SEISMIC ANALYSIS

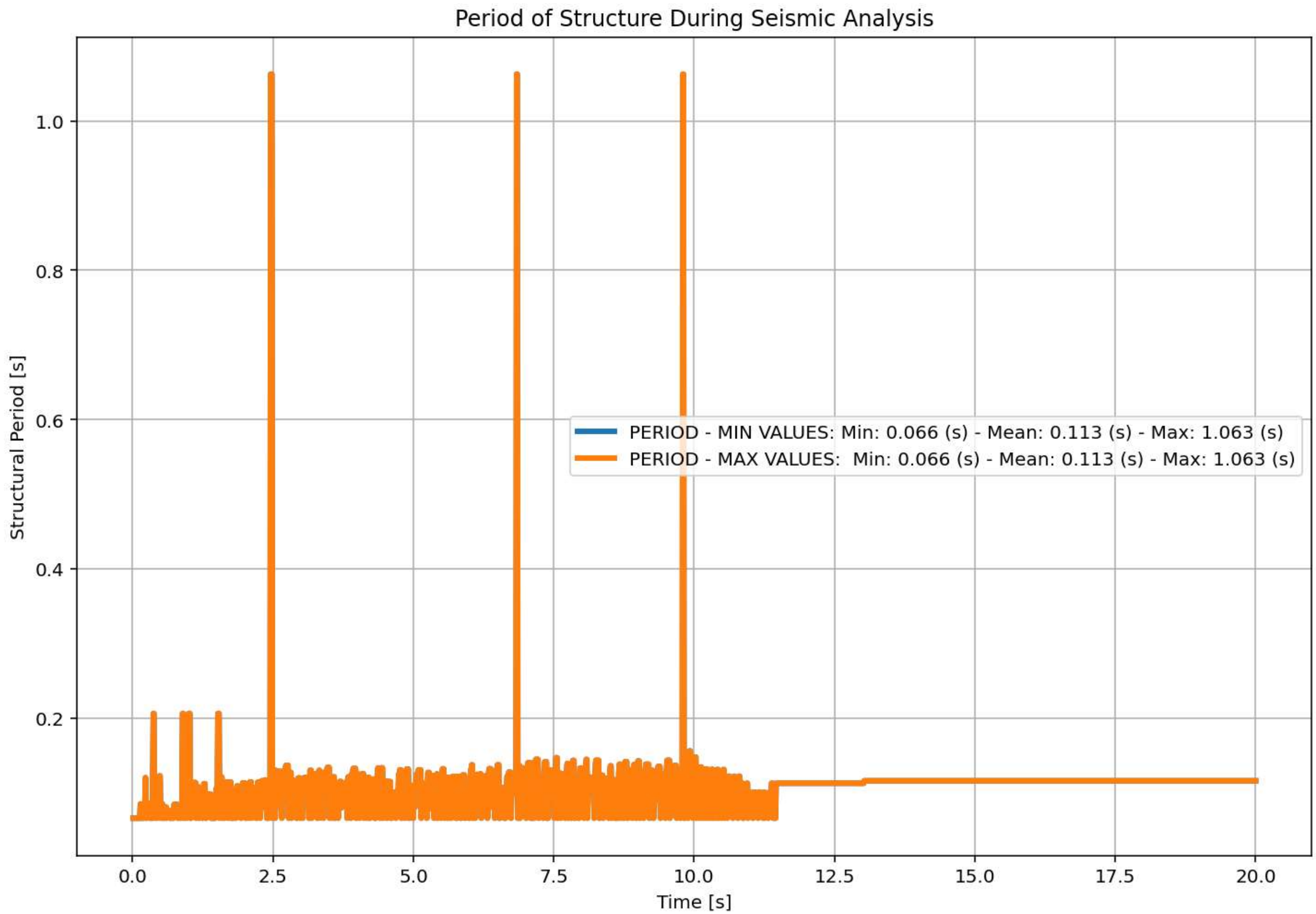

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
Save file (Ctrl+S)
...RGY_DISSIPATION_CAPACITY_INDEX\56_SDOF_FOUR_ELEMENTS_8_ANA
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...F_FOUR_ELEMENTS_8_ANA\SDOF_FOUR_ELEMENTS_8_ANA.py
SDOF_FOUR_ELEMENTS_8_ANA.py
    if ANAL_TYPE == 'SEISMIC': # DYNAMIC TIME-HISTORY ANALYSIS
        #%% DEFINE PARAMETERS FOR SEISMIC ANALYSIS
        duration = 20.0   # [s] Analysis duration
        dt = 0.01   # [s] Time step
        GMfact = 9810   # [mm/s²]standard acceleration o
        SSF_X = 0.20   # Seismic Acceleration Scale Fac
        SSF_Y = 0.20   # Seismic Acceleration Scale Fac
        iv0_X = 0.0005   # [mm/s] Initial velocity applie
        iv0_Y = 0.0005   # [mm/s] Initial velocity applie
        st_iv0 = 0.0   # [s] Initial velocity applied s
        SEI = 'X'   # Seismic Direction
        DR = 0.03   # Damping ratio

        # Define time series for input motion (Acceleration time history)
        if SEI == 'X':
            SEISMIC_TAG_01 = 100
            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
            # Define load patterns
            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
        if SEI == 'Y':
            SEISMIC_TAG_02 = 200
            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
        if SEI == 'XY':
            SEISMIC_TAG_01 = 100
            ops.timeSeries('Path', SEISMIC_TAG_01, '-dt', dt, '-filePath',
            # Define load patterns
            # pattern UniformExcitation $patternTag $dof -accel $tsTag <-ve
            ops.pattern('UniformExcitation', SEISMIC_TAG_01, 1, '-accel', S
            SEISMIC_TAG_02 = 200
            ops.timeSeries('Path', SEISMIC_TAG_02, '-dt', dt, '-filePath',
            ops.pattern('UniformExcitation', SEISMIC_TAG_02, 2, '-accel', S
        print('Seismic Defined Done.')
37 %
Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
300000
200000
100000
0
−100000
−200000
−300000
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]
IPython Console Files Help Variable Explorer Debugger Plots History
Save file
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 683, Col 5 UTF-8 CRLF RW Mem 60%

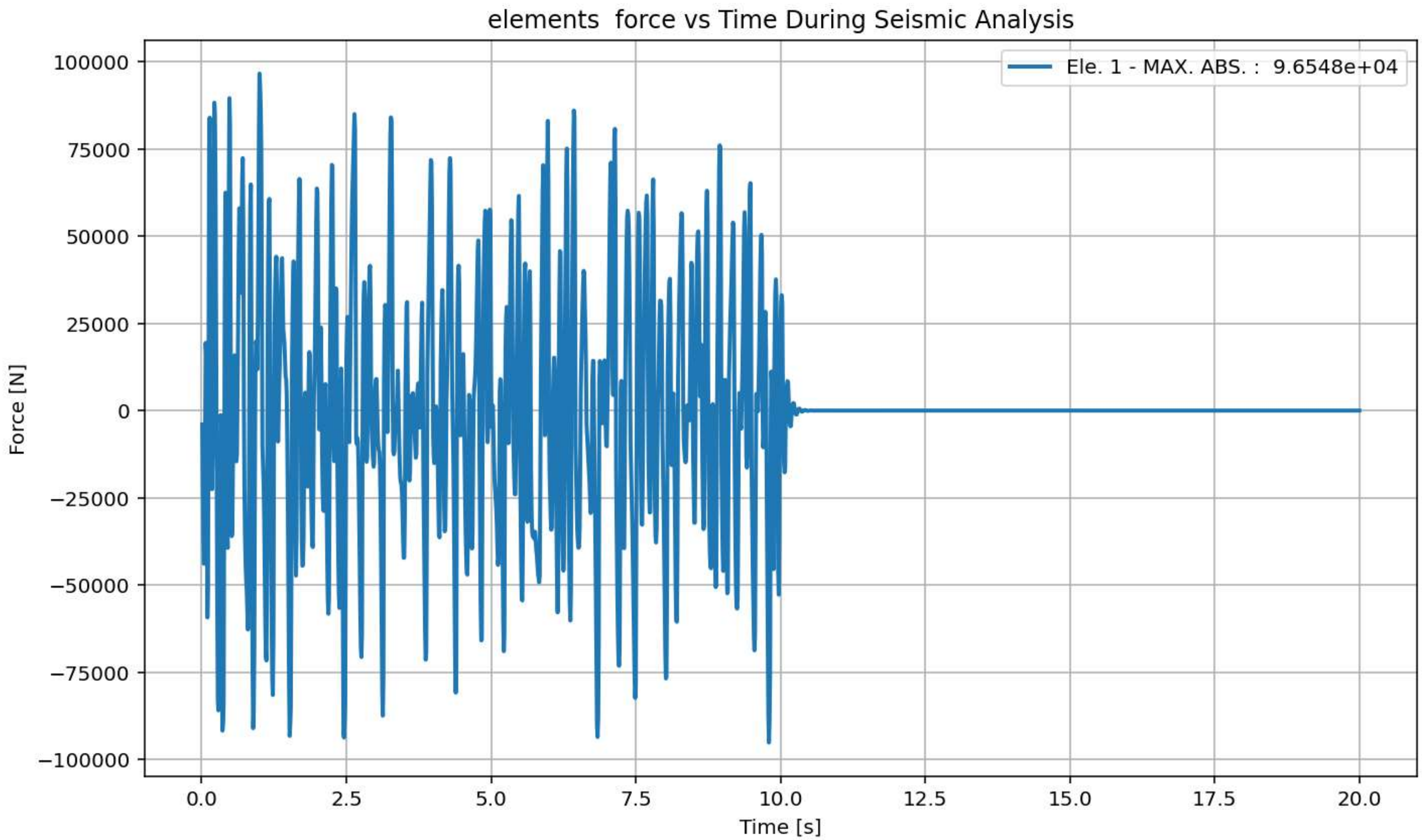


Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
300000
200000
100000
0
−100000
−200000
−300000
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]

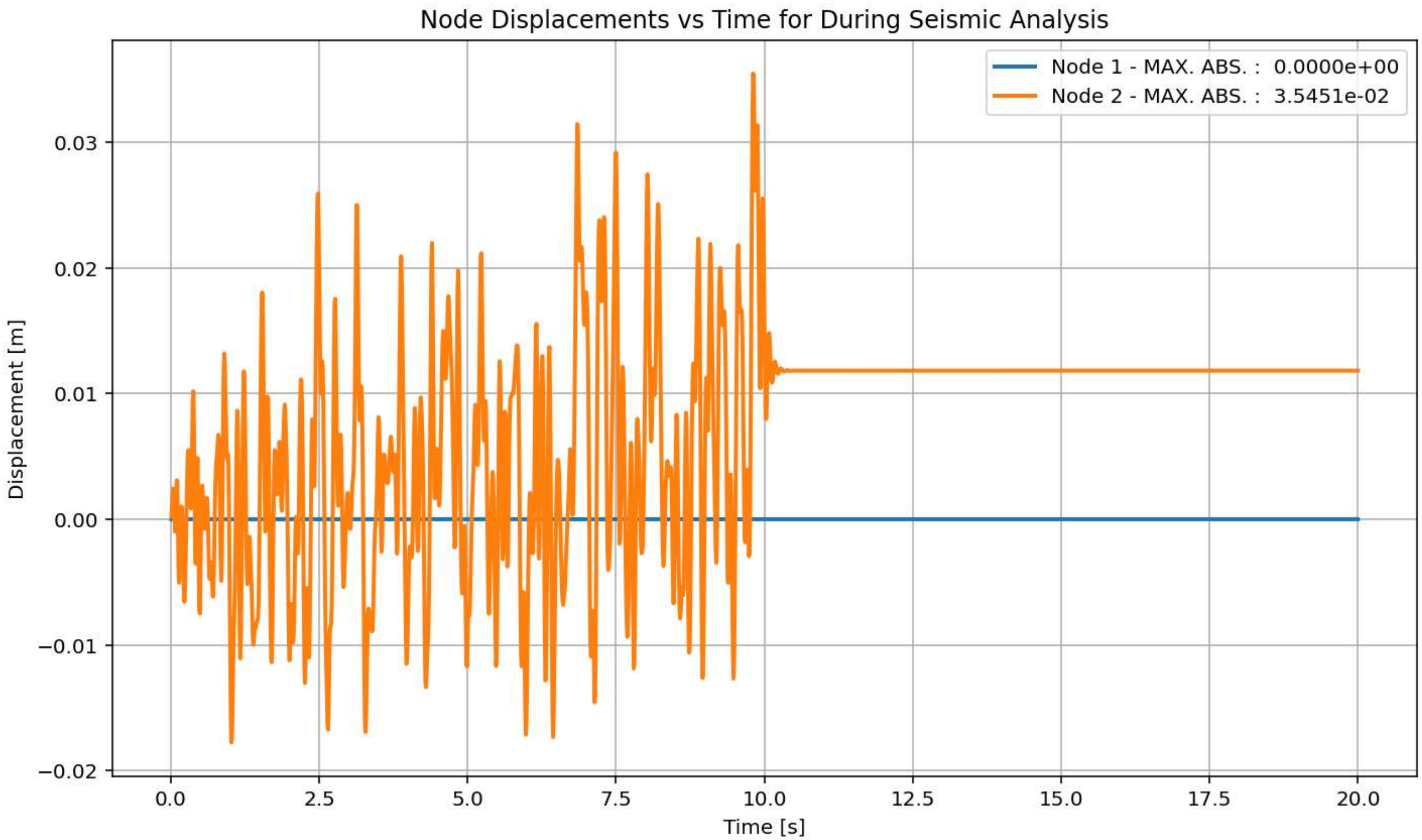

Reaction [N]
300000
200000
100000
0
−100000
−200000
−300000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)
Displacement [m]
0.03
0.02
0.01
0.00
−0.01
−0.02
Time (s)
Velocity [m/s]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Time (s)
Acceleration [m/s²]
40
20
0
−20
−40
−60
Time (s)

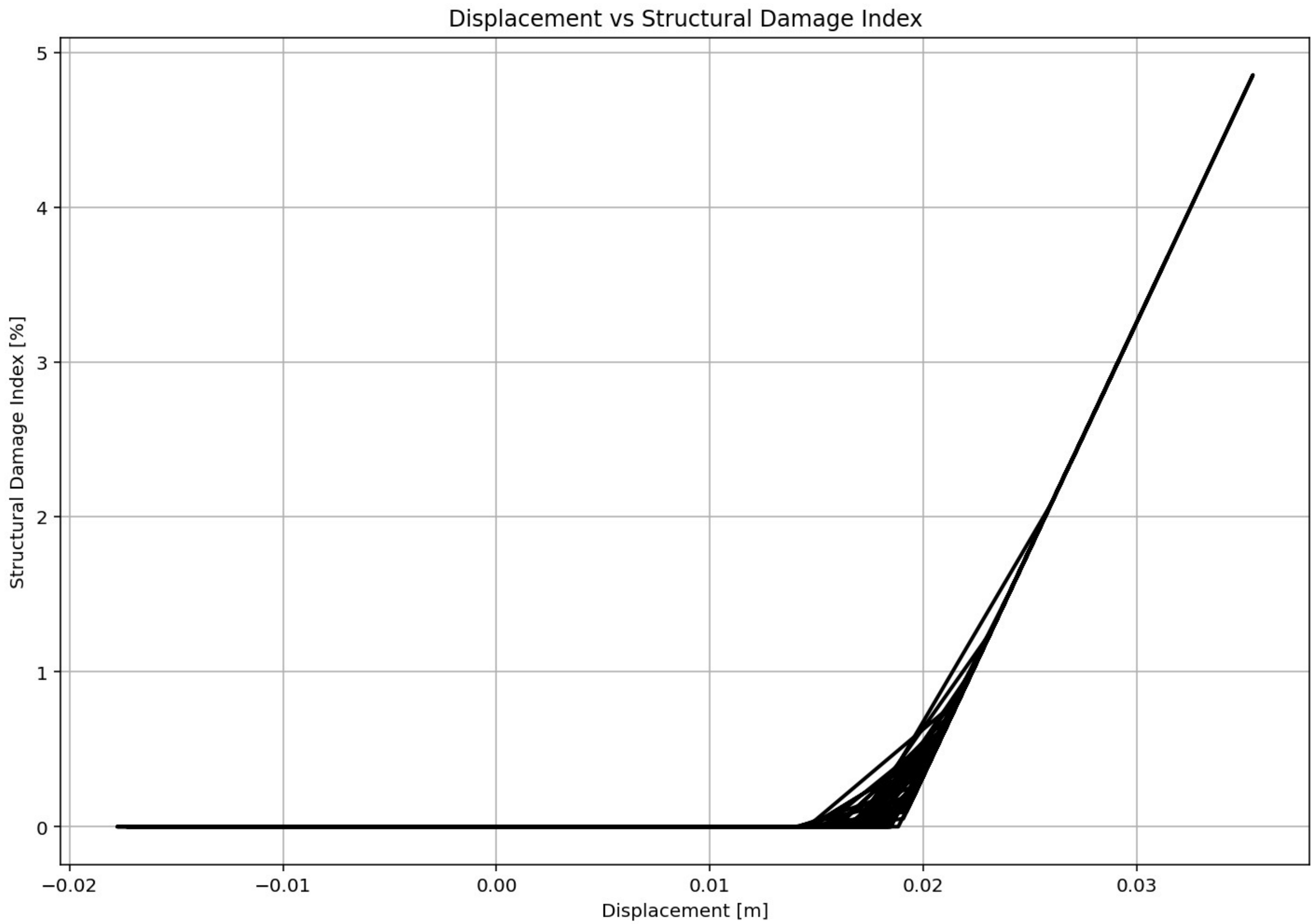


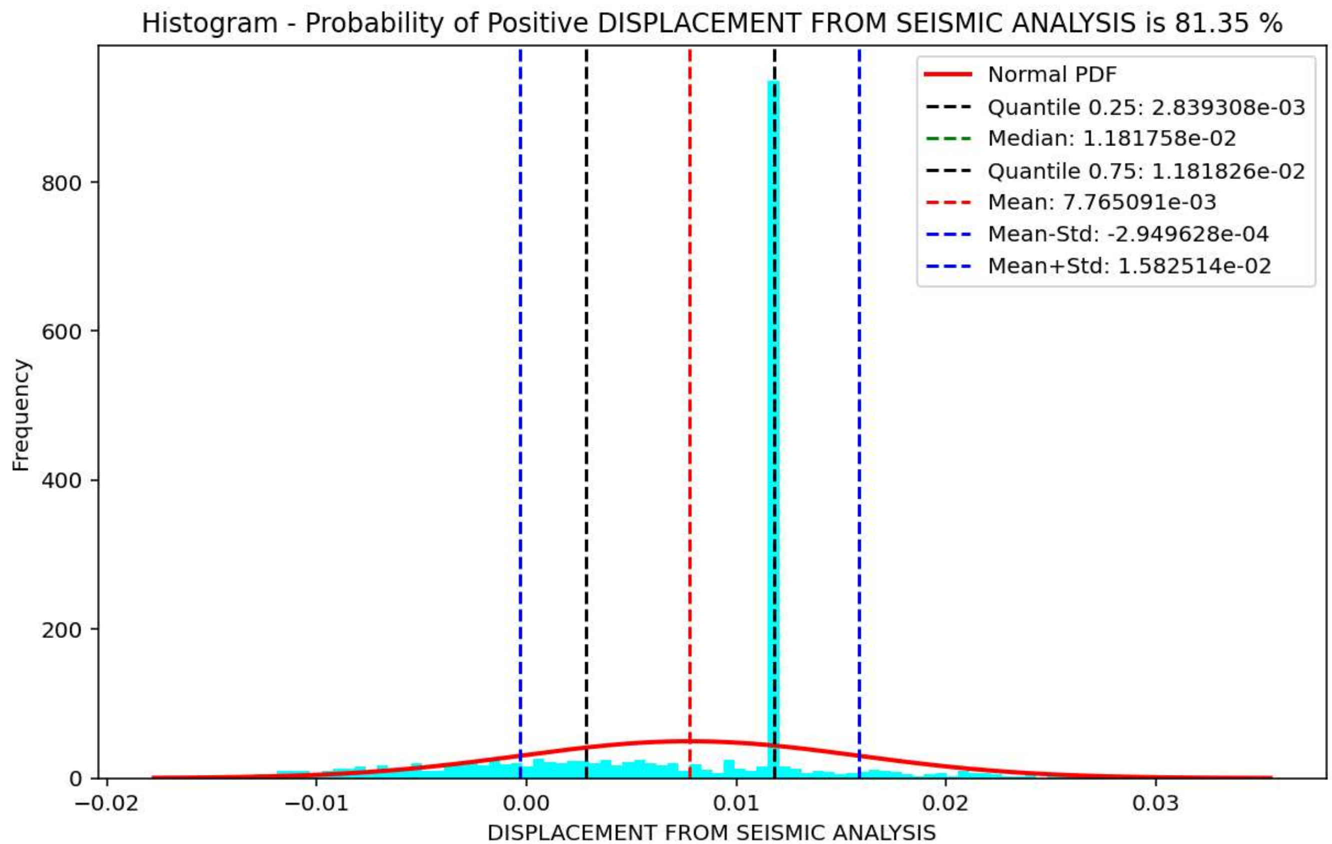
Period of Structure During Seismic Analysis
Structural Period [s]
Time [s]
PERIOD - MIN VALUES: Min: 0.066 (s) - Mean: 0.113 (s) - Max: 1.063 (s)
PERIOD - MAX VALUES: Min: 0.066 (s) - Mean: 0.113 (s) - Max: 1.063 (s)
0.2
0.4
0.6
0.8
1.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



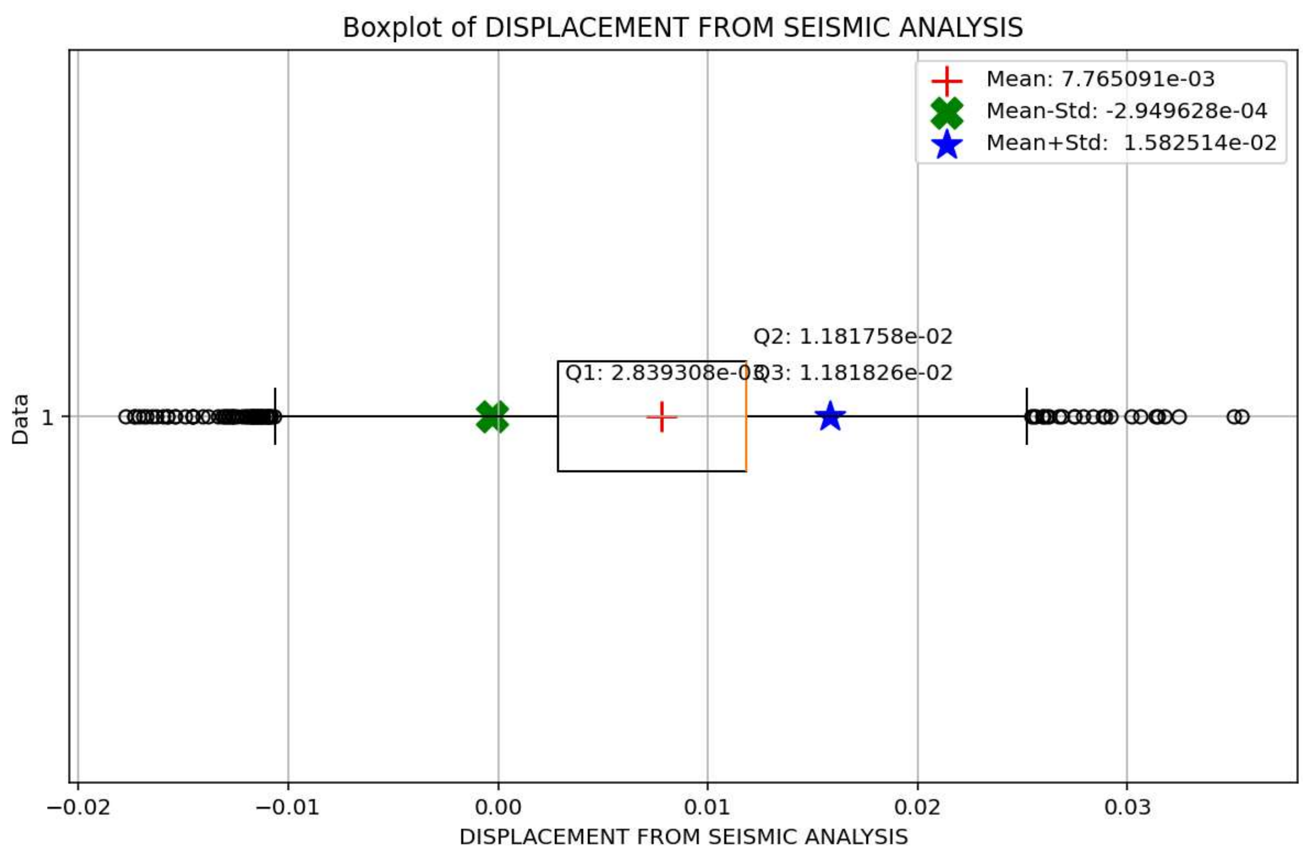
elements force vs Time During Seismic Analysis
Ele. 1 - MAX. ABS. : 9.6548e+04
Force [N]
Time [s]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

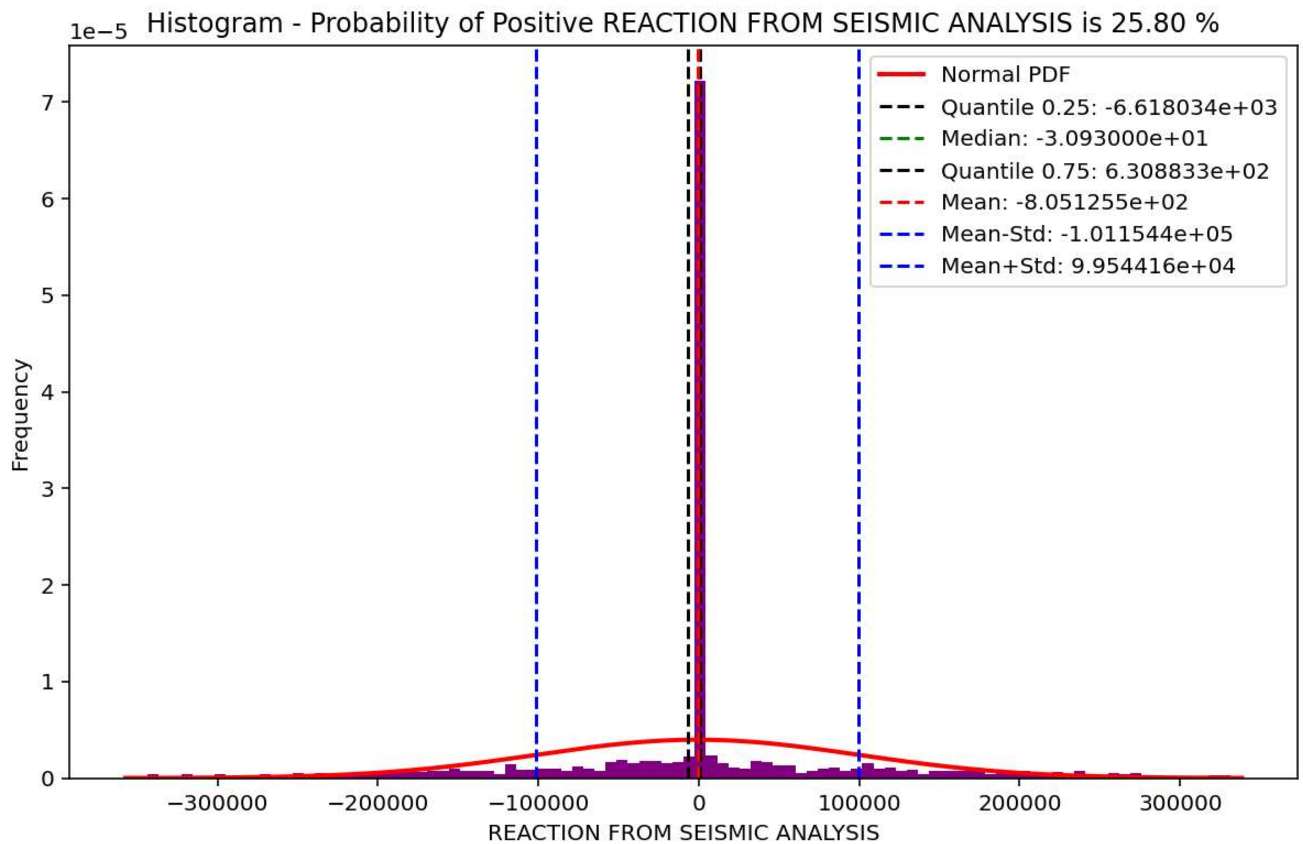

Node Displacements vs Time for During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.5451e-02
0.03
0.02
0.01
0.00
−0.01
−0.02
Displacement [m]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

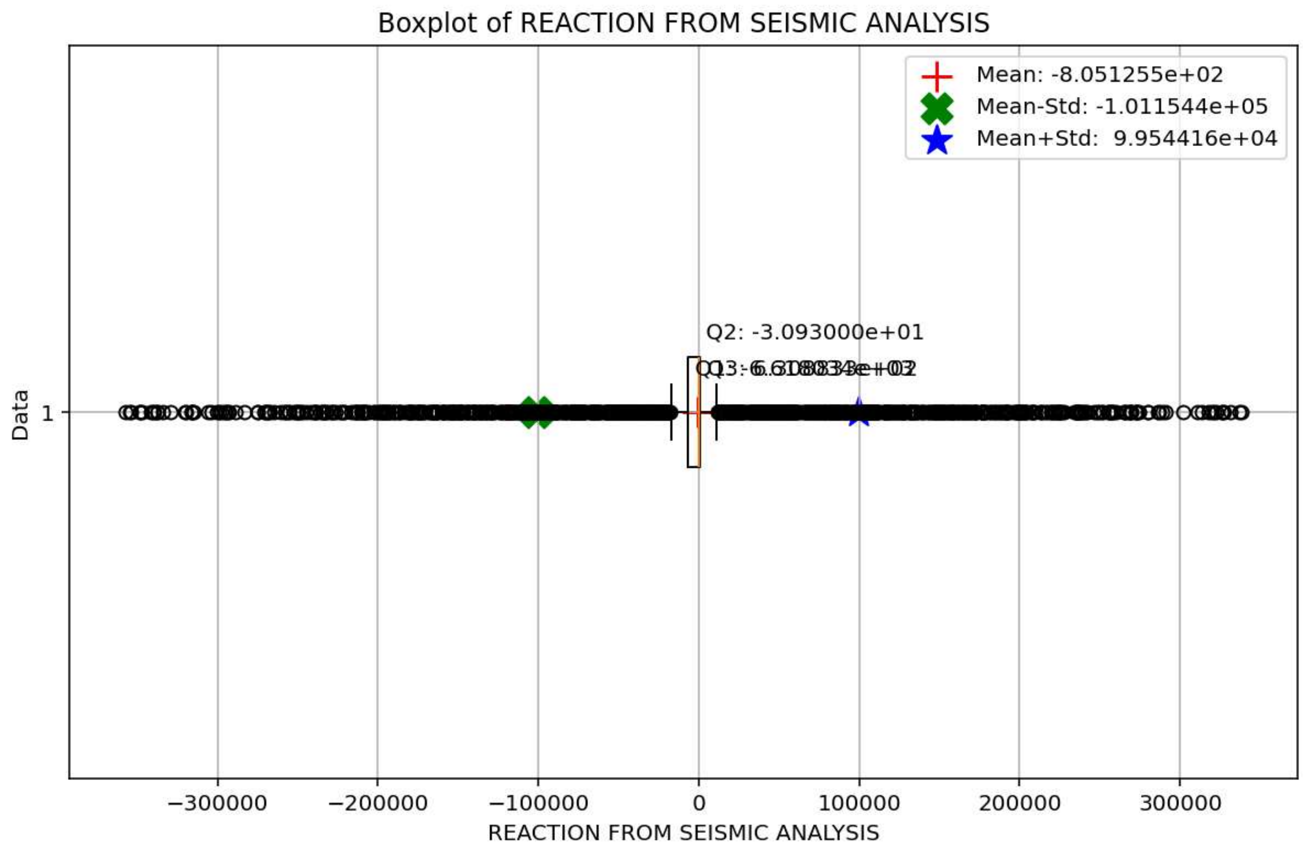

Displacement vs Structural Damage Index
Structural Damage Index [%]
Displacement [m]
5
4
3
2
1
0
−0.02
−0.01
0.00
0.01
0.02
0.03

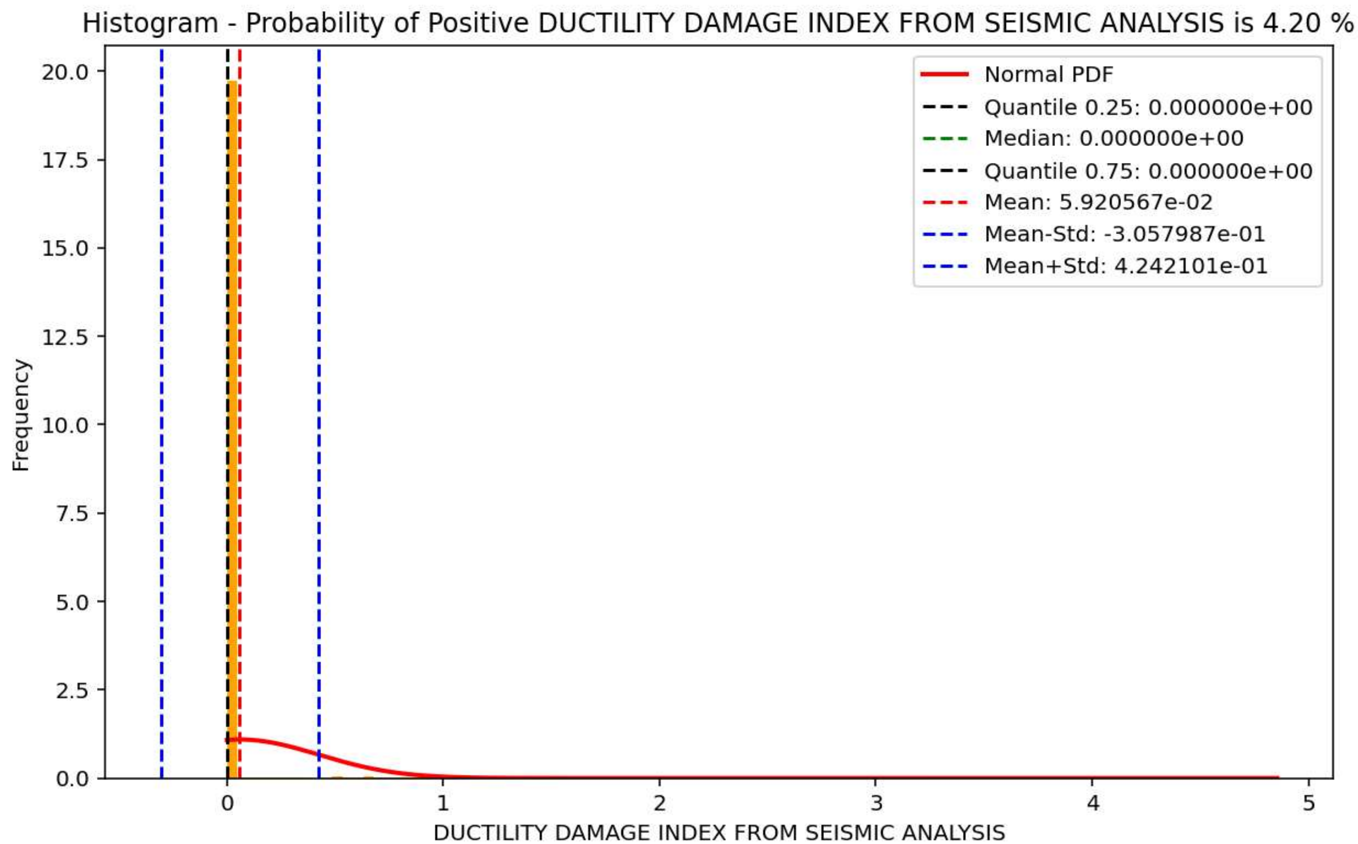


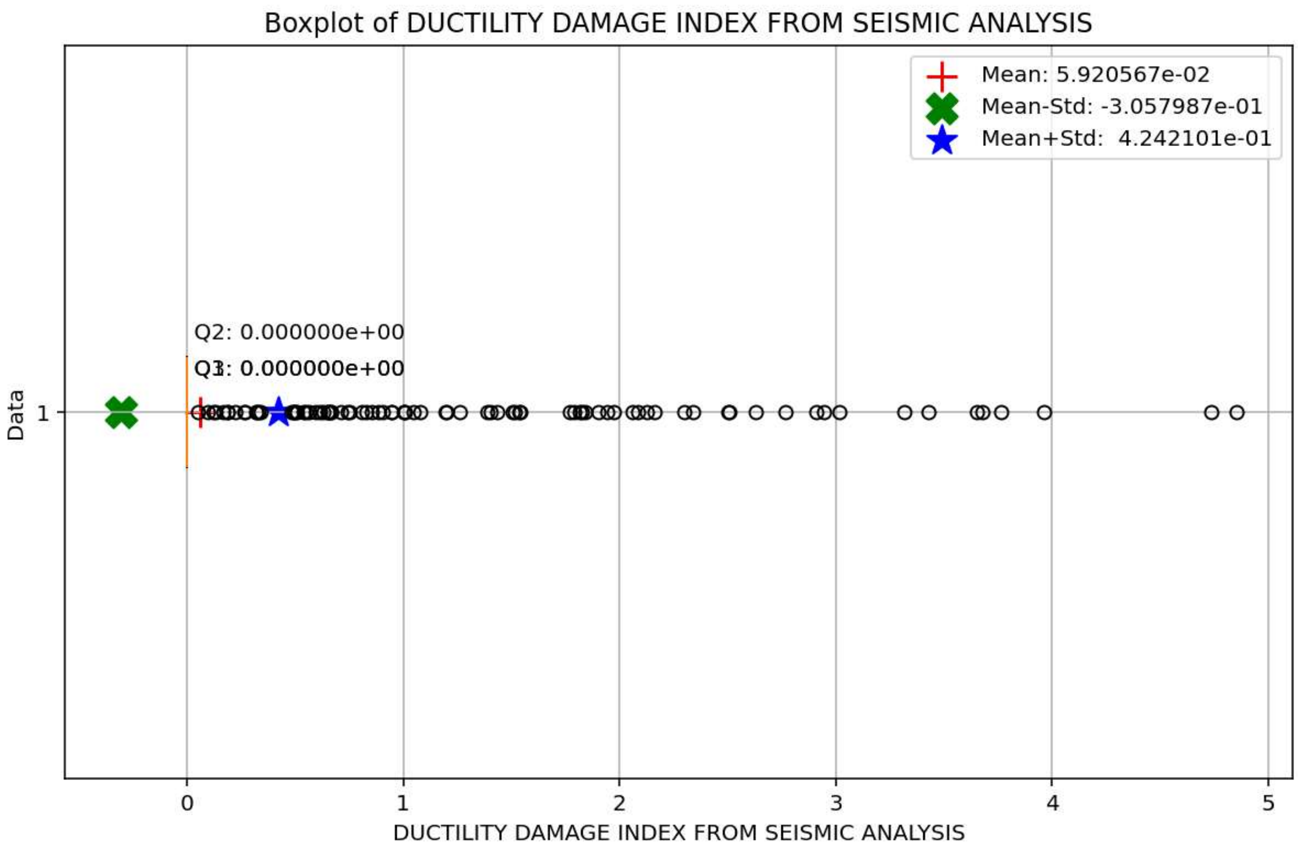
Histogram - Probability of Positive DISPLACEMENT FROM SEISMIC ANALYSIS is 81.35 %
Normal PDF
Quantile 0.25: 2.839308e-03
Median: 1.181758e-02
Quantile 0.75: 1.181826e-02
Mean: 7.765091e-03
Mean-Std: -2.949628e-04
Mean+Std: 1.582514e-02
Frequency
800
600
400
200
0
−0.02
−0.01
0.00
0.01
0.02
0.03
DISPLACEMENT FROM SEISMIC ANALYSIS

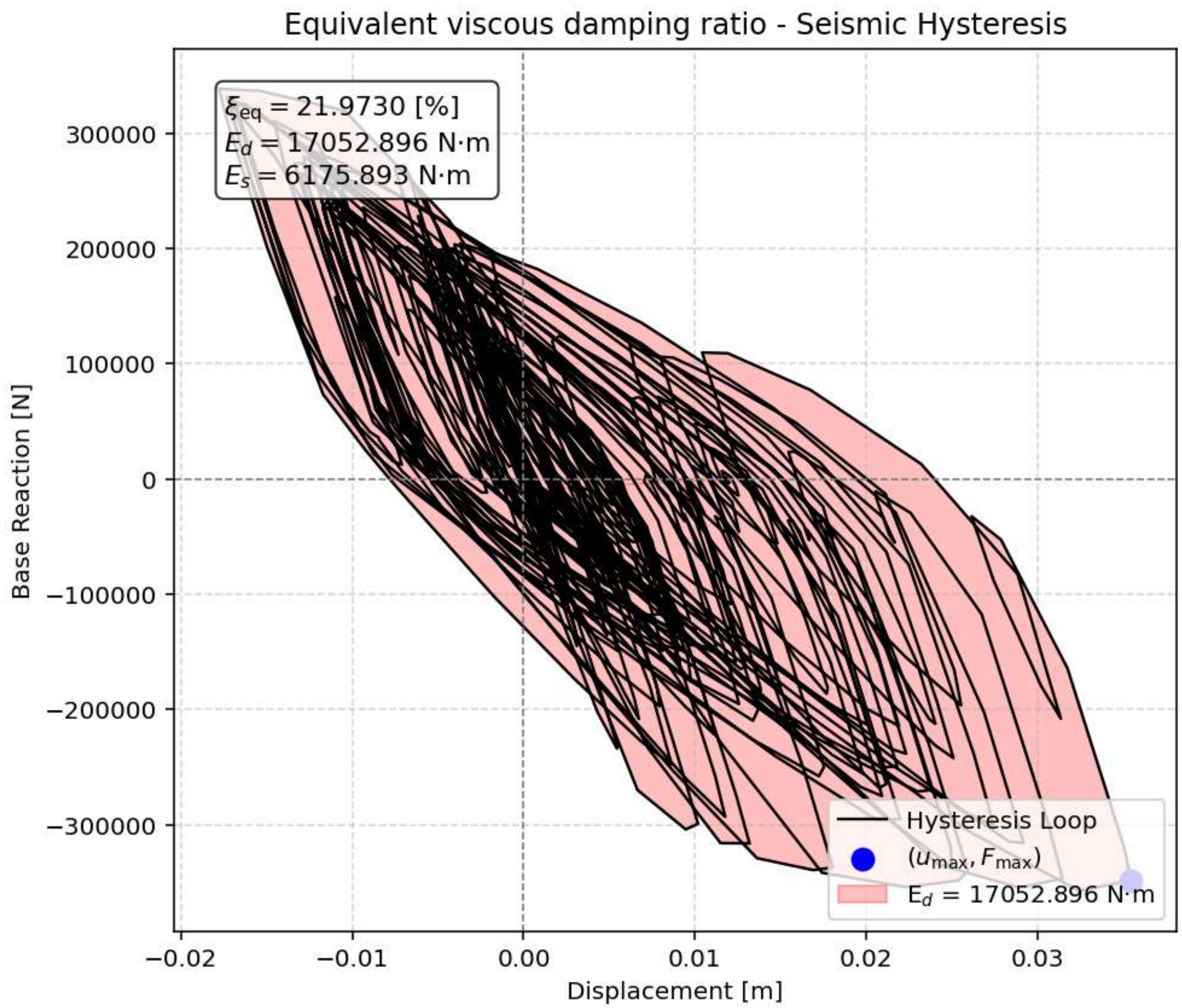

Boxplot of DISPLACEMENT FROM SEISMIC ANALYSIS
Mean: 7.765091e-03
Mean-Std: -2.949628e-04
Mean+Std: 1.582514e-02
Q2: 1.181758e-02
Q1: 2.839308e-03
Q3: 1.181826e-02
Data
1
−0.02
−0.01
0.00
0.01
0.02
0.03
DISPLACEMENT FROM SEISMIC ANALYSIS


Histogram - Probability of Positive REACTION FROM SEISMIC ANALYSIS is 25.80 %
1e−5
Frequency
REACTION FROM SEISMIC ANALYSIS
Normal PDF
Quantile 0.25: -6.618034e+03
Median: -3.093000e+01
Quantile 0.75: 6.308833e+02
Mean: -8.051255e+02
Mean-Std: -1.011544e+05
Mean+Std: 9.954416e+04
−300000
−200000
−100000
0
100000
200000
300000
0
1
2
3
4
5
6
7

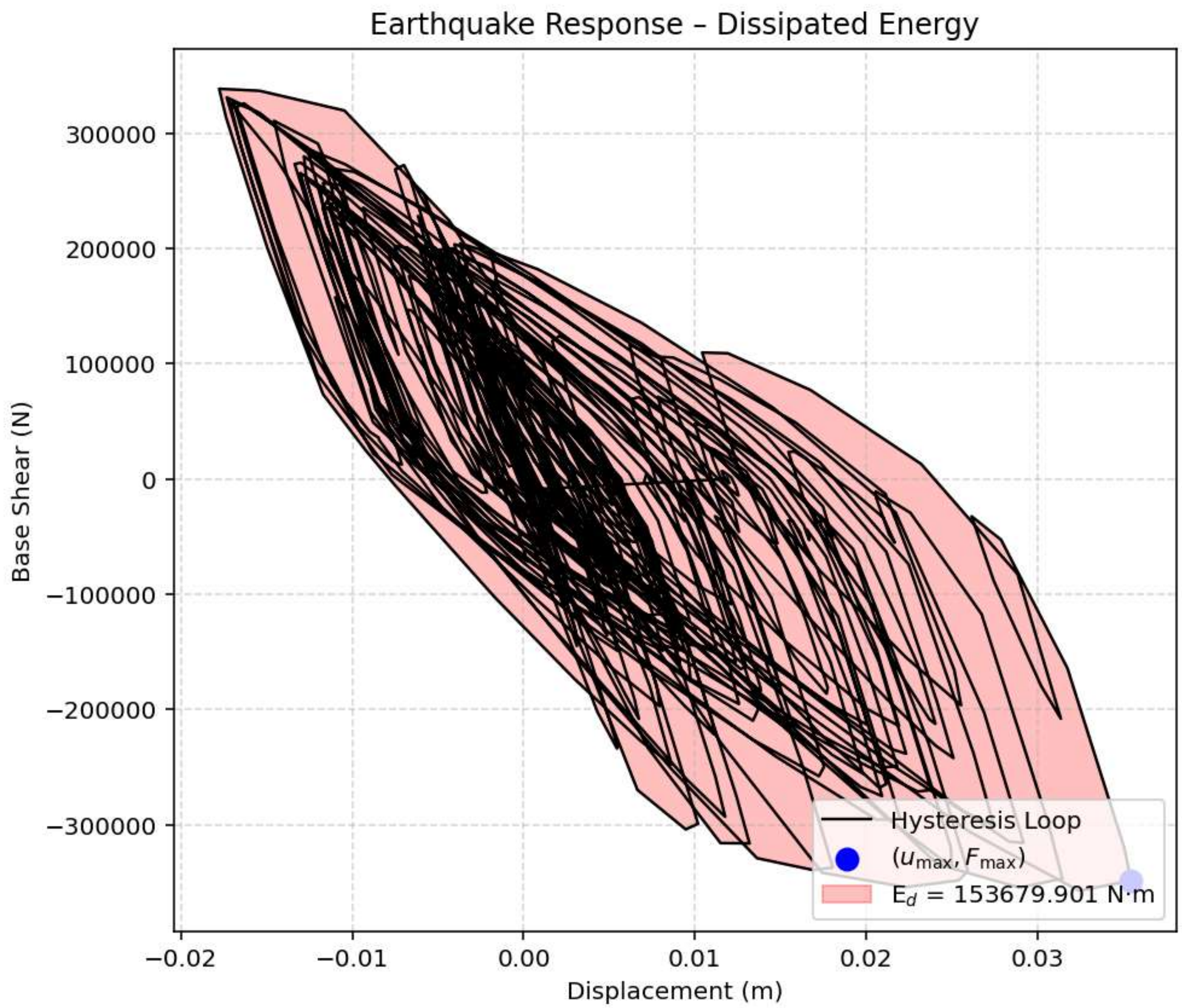

Boxplot of REACTION FROM SEISMIC ANALYSIS
Mean: -8.051255e+02
Mean-Std: -1.011544e+05
Mean+Std: 9.954416e+04
Q2: -3.093000e+01
Data
1
−300000
−200000
−100000
0
100000
200000
300000
REACTION FROM SEISMIC ANALYSIS


Histogram - Probability of Positive DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS is 4.20 %
Normal PDF
Quantile 0.25: 0.000000e+00
Median: 0.000000e+00
Quantile 0.75: 0.000000e+00
Mean: 5.920567e-02
Mean-Std: -3.057987e-01
Mean+Std: 4.242101e-01
Frequency
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0
1
2
3
4
5
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS

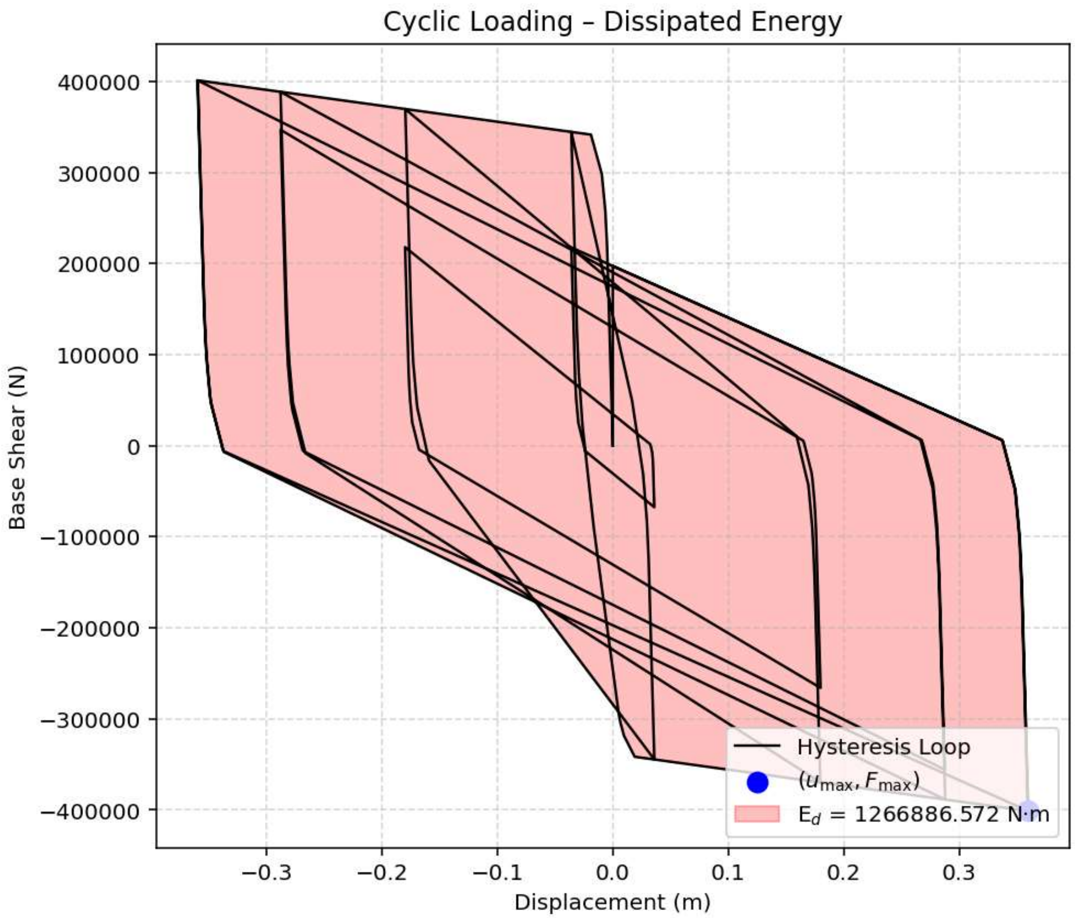

Boxplot of DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS
Mean: 5.920567e-02
Mean-Std: -3.057987e-01
Mean+Std: 4.242101e-01
Q2: 0.000000e+00
Q1: 0.000000e+00
Q3: 0.000000e+00
Data
1
0
1
2
3
4
5
DUCTILITY DAMAGE INDEX FROM SEISMIC ANALYSIS



Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 21.9730$ [%]
$E_d = 17052.896$ N·m
$E_s = 6175.893$ N·m
300000
200000
100000
0
−100000
−200000
−300000
Base Reaction [N]
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 17052.896 N·m

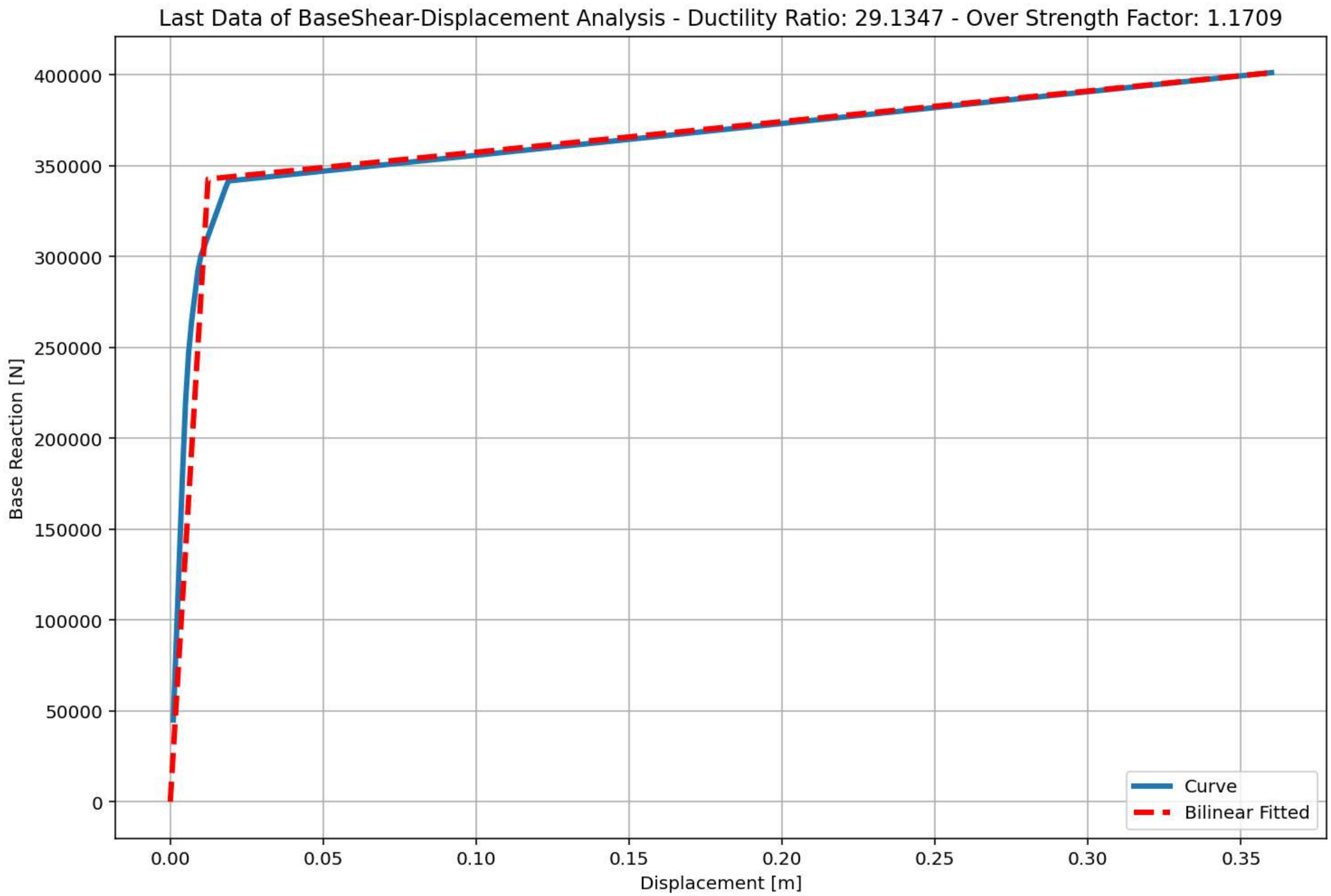


Earthquake Response – Dissipated Energy
300000
200000
100000
0
−100000
−200000
−300000
Base Shear (N)
−0.02
−0.01
0.00
0.01
0.02
0.03
Displacement (m)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 153679.901 N·m


Cyclic Loading – Dissipated Energy
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
Base Shear (N)
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement (m)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 1266886.572 N·m

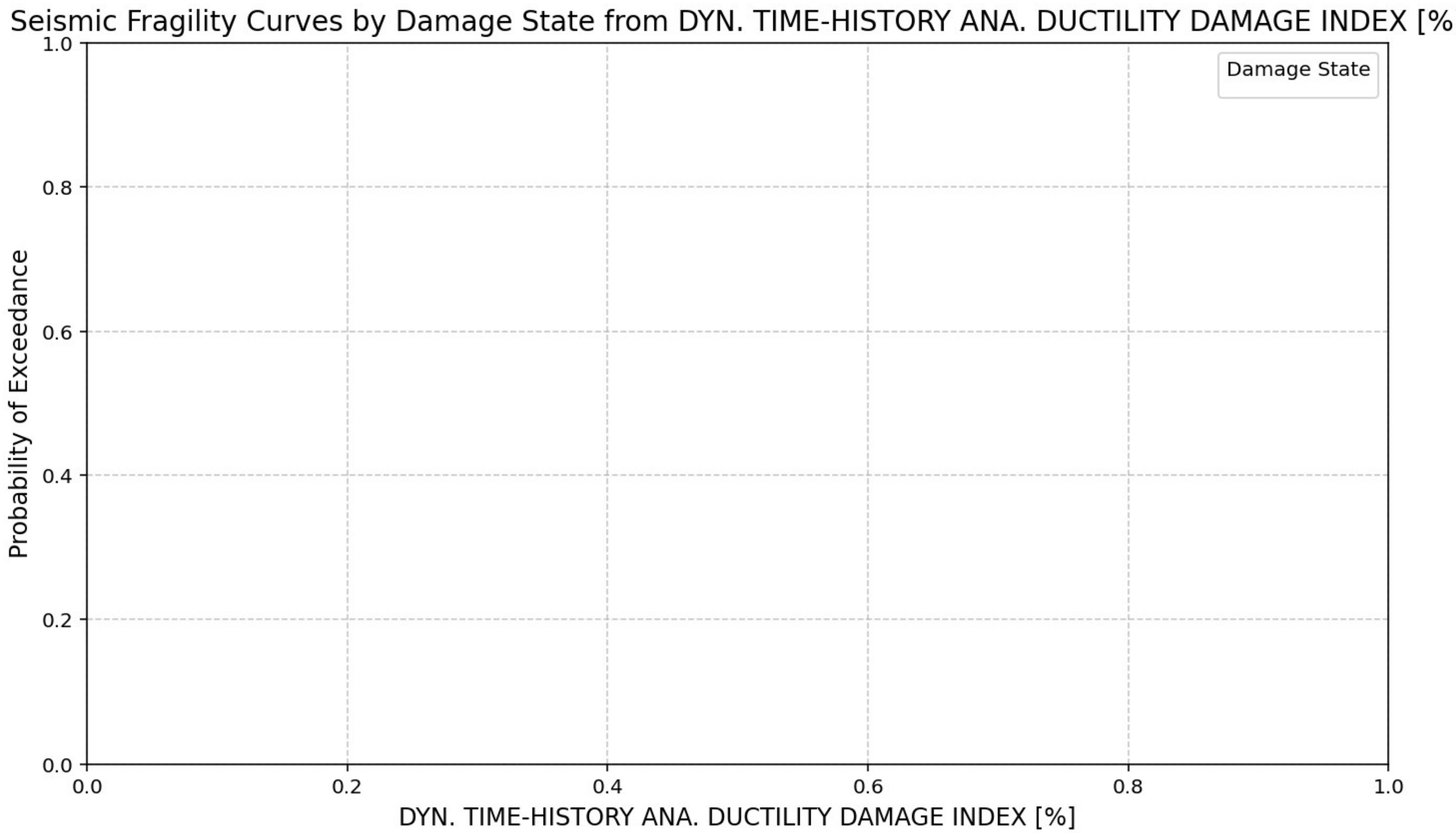
Dissipated Energy from Earthquake= 153679.90 N·m
Dissipated Energy from Cyclic Displacement= 1266886.57 N·m

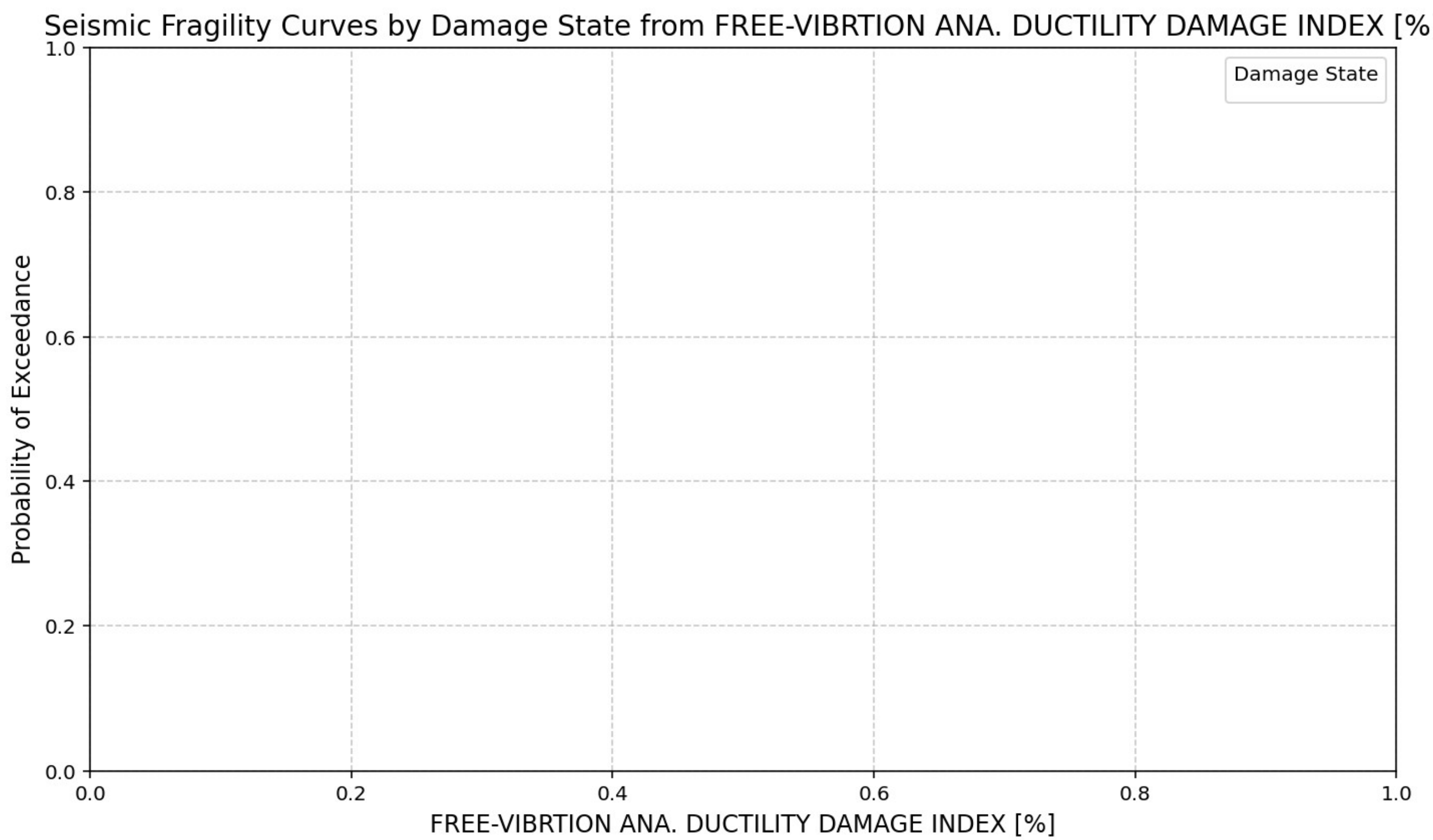
DISSIPATED ENERGY CAPACITY INDEX: 12.131 [%]
ZONE 2: MINOR DAMAGE



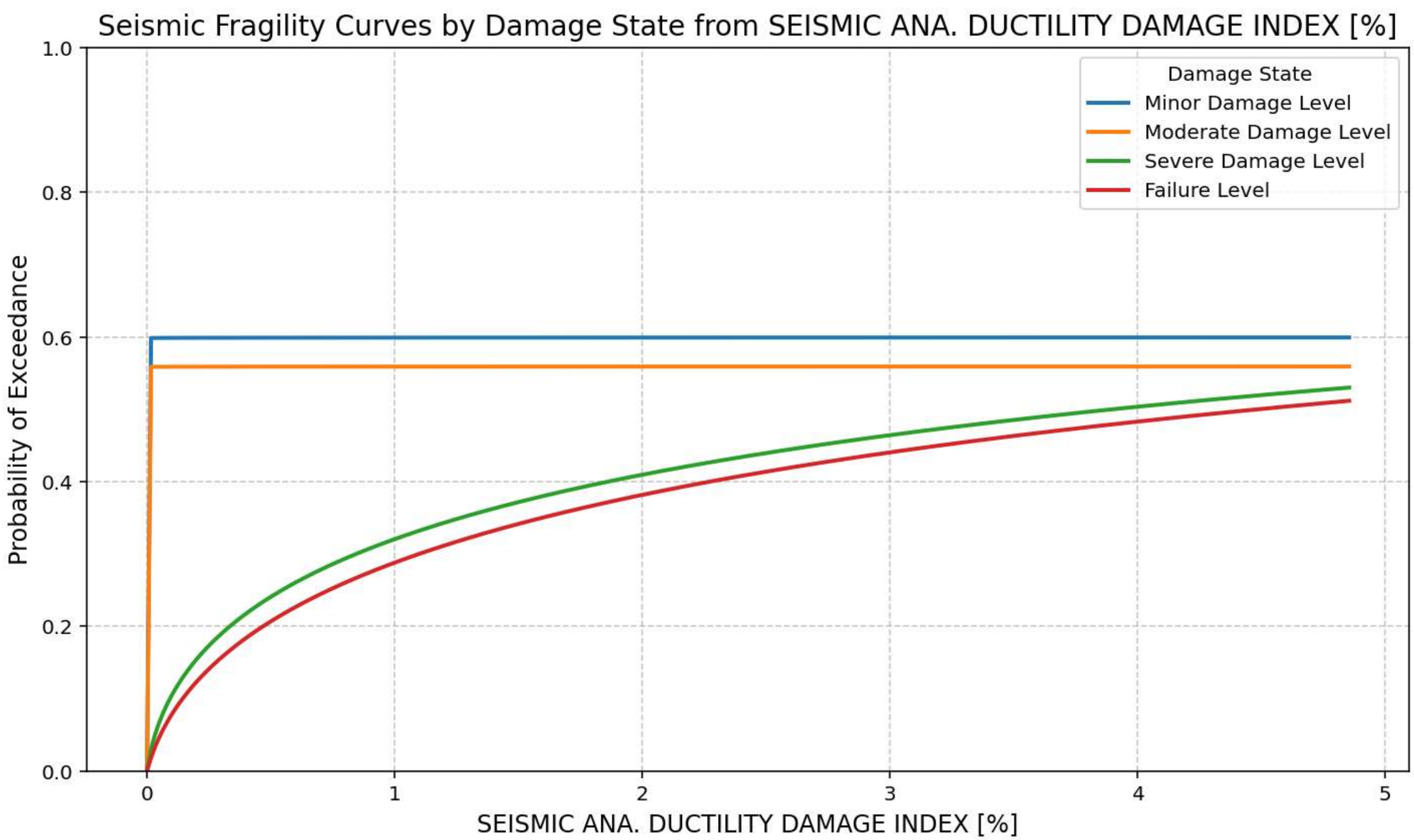
Last Data of BaseShear-Displacement Analysis - Ductility Ratio: 29.1347 - Over Strength Factor: 1.1709
Base Reaction [N]
Displacement [m]
0
50000
100000
150000
200000
250000
300000
350000
400000
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.88855314e-02 8.49848911e+04]
 [3.49000000e-01 9.91148908e+04]]
+==========================+
+----------------------------------------------------+
 Structure Elastic Stiffness :     4500000.00
 Structure Plastic Stiffness :     283996.82
 Structure Tangent Stiffness :     42803.33
 Structure Ductility Ratio :      18.48
 Structure Over Strength Factor:   1.17
+----------------------------------------------------+
Over Strength Coefficient (Ω0):      1.1709
Displacement Ductility Ratio (μ):    29.1347
Ductility Coefficient (Rμ):        29.1347
Structural Behavior Coefficient (R): 34.1124
Structural Ductility Damage Index in X Direction: 6.6433 (%)
```


Seismic Fragility Curves by Damage State from DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
0.0
0.2
0.4
0.6
0.8
1.0
DYN. TIME-HISTORY ANA. DUCTILITY DAMAGE INDEX [%]



Seismic Fragility Curves by Damage State from FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Probability of Exceedance
FREE-VIBRTION ANA. DUCTILITY DAMAGE INDEX [%]
0.0
0.2
0.4
0.6
0.8
1.0



Seismic Fragility Curves by Damage State from SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]
Damage State
Minor Damage Level
Moderate Damage Level
Severe Damage Level
Failure Level
Probability of Exceedance
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
SEISMIC ANA. DUCTILITY DAMAGE INDEX [%]